# Supplementary Material for "Strong Colorings of Orthogonality Hypergraphs as Noncontextual Eigenvalue Assignments"

Karl Svozil [a)]
*Institute for Theoretical Physics, TU Wien, Wiedner Hauptstraße 8-10/136, 1040 Vienna, Austria*

(Dated: November 5, 2026)

This supplementary material provides the complete vector and context enumerations, two-valued state tables, normalized strong coloring tables, odd-cycle nonextendability certificates, chromatic signature and separation profile tables, parent-embedding dictionaries, and supplementary proofs and finite verification certificates supporting the main text. Section S1 lists the explicit unnormalized coordinates and context tables for the completed Yu–Oh hypergraph $H_{\mathrm{YO}}^{+}$ and its faithful restrictions $H_{17,10}$ and $H_{21,13}$, the three- and four-dimensional logical gadgets, the 37-ray complex Hesse hypergraph $H_{37}$, the 26-ray suspension $H_{\mathrm{YO}}^{(4)}$ and 65-ray canonical linear completion $H_{65}$, the Cabello completions $C_{36,18}$ and $C_{45,21}$, and the specified finite KS parents (49-ray Schütte–Bub, 87-ray MUB-inventory union, 40- and 72-ray Peres unions, and 68-ray three-ray extensions). Section S2 gives the complete two-valued state tables, extendable-state classifications, normalized coloring tables, odd-cycle certificates, and separation profiles across both regimes. Section S3 presents the supplementary proofs and exact reductions, including Isaacs dot-product snarks and quantitative $\Delta_3$ robustness bounds for $G_{32}$, the 540,672-candidate reduction for the minimum 18-state order-determining family on $L(H_{37})$, the canonical completion selection rule for $H_{65}$, the forced-zero and ray-criticality certificates for the four-dimensional Cabello completions, transverse criticality deletion colorings, and historical Kochen–Specker gadget certificates.

[a)] Electronic mail: karl.svozil@tuwien.ac.at; http://tph.tuwien.ac.at/~svozil

## READING GUIDE AND VERIFICATION SCOPE

This supplement collects complete vector and context inventories (S1), state and coloring enumerations (S2), and extended proofs and certificates (S3). Labels are local to each configuration. "Exact palette" means $\chi(H) = d$, with no additional density or geometric assumption: the outcomes admit a noncontextual assignment of $d$ eigenvalues, that is, every complete experiment can be measured by a maximal observable with one common spectrum of $d$ eigenvalues, with each outcome carrying the same eigenvalue in every context. Colorings are such eigenvalue assignments, and extendable states are the truth assignments "the outcome with eigenvalue $\lambda_a$ occurs" that they produce.

The five-state and four-state $H_{17,10}$ mixtures and both-direction gadget witnesses use the complete state families. Every selected support satisfies every context equation, and rational arithmetic verifies every Born marginal and the attained nonextendable weights $1/9$ and $1/3$. The ordinary implication $10 \Rightarrow 24$ on $H_{17,10}$ gives a false inclusion between distinct rays and hence a failure of order determination.

The companion electronic data include all coordinates, state enumerations (including the compressed 393,216-state $H_{65}$ inventory), verification programs, and the exhaustive certificates for the eighteen-state Hesse minimum and the 65,535-subset Yu–Oh classification. Checking an eighteen-state family establishes an upper bound; the matching lower bound requires the exhaustive certificate in Sec. S3.3. The directories `part-a/data`, `part-a/verification`, `part-b/data`, and `part-b/verification` contain the underlying data and verifiers. Reports and source-integrity hashes are in `audit/`; Sec. S3.7 describes the joint verification entry point and its scope.

## S1. EXPLICIT UNNORMALIZED VECTORS AND CONTEXTS

All numerical vertex labels in this section are local to the named configuration and match the figures and theorems of the main text.

### S1.1. Completed Yu–Oh and the Petersen-derived contexts

The three-dimensional vectors below give all projective rays used in Sec. III C; signs are chosen to make the first nonzero integer coordinate positive. The 26-ray suspension appends zero and adds $e_4$. The abstract $G_{32}, H_9, H_8$ context lists in Eq. (10) and Theorem 11 specify their incidences completely. The geometric obstructions exclude a corresponding injective three-dimensional vector enumeration for these abstract examples.

TABLE S1: All 25 unnormalized completed Yu–Oh vectors.

| $i$ | $r_i$ | $i$ | $r_i$ |
|---|---|---|---|
| 1 | $(0,1,1)$ | 14 | $(2,1,1)$ |
| 2 | $(1,0,0)$ | 15 | $(1,2,-1)$ |
| 3 | $(0,1,-1)$ | 16 | $(1,-2,1)$ |
| 4 | $(1,1,0)$ | 17 | $(1,1,-2)$ |
| 5 | $(0,0,1)$ | 18 | $(1,-1,-2)$ |
| 6 | $(1,-1,0)$ | 19 | $(2,1,-1)$ |
| 7 | $(1,0,1)$ | 20 | $(2,-1,1)$ |
| 8 | $(0,1,0)$ | 21 | $(1,-2,-1)$ |
| 9 | $(1,0,-1)$ | 22 | $(1,1,2)$ |
| 10 | $(1,-1,1)$ | 23 | $(1,2,1)$ |
| 11 | $(1,-1,-1)$ | 24 | $(1,-1,2)$ |
| 12 | $(1,1,1)$ | 25 | $(2,-1,-1)$ |
| 13 | $(1,1,-1)$ | | |

### S1.2. Faithful Yu–Oh restrictions $H_{17,10}$ and $H_{21,13}$ and logical gadgets in dimensions three and four

Tables S2 and S3 record the restriction membership of the 25 parent rays and 16 parent contexts of $H_{\mathrm{YO}}^{+}$ in $H_{17,10}$ and $H_{21,13}$. Tables S4–S14 provide the explicit unnormalized integer coordinates and context lists for the faithful real 3-dimensional gadgets $P_5$, $B_{13,7}$, $D_{16,9}$, $I_{16,9}$, and $T_{19,11}$ and the faithful real 4-dimensional gadgets $W^{(4)}_{11,5}$, $W^{(4)}_{16,6}$, $B^{(4)}_{14,7}$, $L^{(4)}_{18,7}$, $D^{(4)}_{17,9}$, and $L^{(4)}_{22,9}$. Note that for the standalone gadgets $P_5$, $D_{16,9}$, $I_{16,9}$, and $T_{19,11}$, the vertex numbering

is local to each gadget panel; in $L^{(4)}_{18,7}$ [Fig. 3(d)], $r_{13} = (1, 0, 1, 1)$ and $r_{14} = (2, 0, -1, -1)$ are chosen so that $r_{13} \not\perp r_{15}$, whereas in $L^{(4)}_{22,9}$ [Fig. 3(f)], $r_{13} = (1, 0, 4, 4)$ and $r_{14} = (8, 0, -1, -1)$ are used so that $r_{13} \perp r_{15}$ in the added context $\{13, 20, 15, 19\}$.

TABLE S2: All 25 unnormalized parent vectors; membership of the two geometric examples.

| $i$ | $r_i$ | $H_{17,10}$ | $H_{21,13}$ | $i$ | $r_i$ | $H_{17,10}$ | $H_{21,13}$ |
|---|---|---|---|---|---|---|---|
| 1 | $(0, 1, 1)$ | yes | yes | 14 | $(2, 1, 1)$ | yes | yes |
| 2 | $(1, 0, 0)$ | yes | yes | 15 | $(1, 2, -1)$ | yes | yes |
| 3 | $(0, 1, -1)$ | yes | yes | 16 | $(1, -2, 1)$ | – | – |
| 4 | $(1, 1, 0)$ | yes | yes | 17 | $(1, 1, -2)$ | – | – |
| 5 | $(0, 0, 1)$ | yes | yes | 18 | $(1, -1, -2)$ | yes | yes |
| 6 | $(1, -1, 0)$ | yes | yes | 19 | $(2, 1, -1)$ | yes | yes |
| 7 | $(1, 0, 1)$ | yes | yes | 20 | $(2, -1, 1)$ | – | yes |
| 8 | $(0, 1, 0)$ | yes | yes | 21 | $(1, -2, -1)$ | – | yes |
| 9 | $(1, 0, -1)$ | yes | yes | 22 | $(1, 1, 2)$ | – | yes |
| 10 | $(1, -1, 1)$ | yes | yes | 23 | $(1, 2, 1)$ | yes | yes |
| 11 | $(1, -1, -1)$ | yes | yes | 24 | $(1, -1, 2)$ | yes | yes |
| 12 | $(1, 1, 1)$ | – | – | 25 | $(2, -1, -1)$ | – | – |
| 13 | $(1, 1, -1)$ | – | yes | | | | |

TABLE S3: The 16 parent contexts and the two selected restrictions.

| $j$ | context | $H_{17,10}$ | $H_{21,13}$ |
|---|---|---|---|
| 1 | $\{1, 2, 3\}$ | yes | yes |
| 2 | $\{7, 8, 9\}$ | yes | yes |
| 3 | $\{4, 5, 6\}$ | yes | yes |
| 4 | $\{2, 8, 5\}$ | yes | yes |
| 5 | $\{3, 12, 25\}$ | – | – |
| 6 | $\{9, 12, 16\}$ | – | – |
| 7 | $\{6, 12, 17\}$ | – | – |
| 8 | $\{3, 11, 14\}$ | yes | yes |
| 9 | $\{7, 11, 15\}$ | yes | yes |
| 10 | $\{4, 11, 24\}$ | yes | yes |
| 11 | $\{9, 10, 23\}$ | yes | yes |
| 12 | $\{1, 10, 19\}$ | yes | yes |
| 13 | $\{4, 10, 18\}$ | yes | yes |
| 14 | $\{1, 13, 20\}$ | – | yes |
| 15 | $\{7, 13, 21\}$ | – | yes |
| 16 | $\{6, 13, 22\}$ | – | yes |

### S1.3. 3D panel (a): $P_5$

TABLE S4: Unnormalized vectors and all contexts of $P_5$.

| $i$ | $r_i$ | $j$ | Context |
|---|---|---|---|
| 1 | $(1, 0, 0)$ | 1 | $\{1, 6, 2\}$ |
| 2 | $(0, 1, 1)$ | 2 | $\{2, 7, 3\}$ |
| 3 | $(1, 1, -1)$ | 3 | $\{3, 8, 4\}$ |
| 4 | $(1, -2, -1)$ | 4 | $\{4, 9, 5\}$ |
| 5 | $(0, 1, -2)$ | 5 | $\{5, 10, 1\}$ |
| 6 | $(0, 1, -1)$ | | |
| 7 | $(2, -1, 1)$ | | |
| 8 | $(1, 0, 1)$ | | |
| 9 | $(5, 2, 1)$ | | |
| 10 | $(0, 2, 1)$ | | |

### S1.4. 3D panel (b): $B_{13,7}$

TABLE S5: Unnormalized vectors and all contexts of $B_{13,7}$.

| $i$ | $r_i$ | $j$ | Context |
|---|---|---|---|
| 1 | $(0,1,1)$ | 1 | $\{1,2,3\}$ |
| 2 | $(1,0,0)$ | 2 | $\{7,8,9\}$ |
| 3 | $(0,1,-1)$ | 3 | $\{2,8,5\}$ |
| 5 | $(0,0,1)$ | 4 | $\{3,11,14\}$ |
| 7 | $(1,0,1)$ | 5 | $\{7,11,15\}$ |
| 8 | $(0,1,0)$ | 6 | $\{9,10,23\}$ |
| 9 | $(1,0,-1)$ | 7 | $\{1,10,19\}$ |
| 10 | $(1,-1,1)$ | | |
| 11 | $(1,-1,-1)$ | | |
| 14 | $(2,1,1)$ | | |
| 15 | $(1,2,-1)$ | | |
| 19 | $(2,1,-1)$ | | |
| 23 | $(1,2,1)$ | | |

### S1.5. 3D panel (c): $D_{16,9}$

TABLE S6: Unnormalized vectors and all contexts of $D_{16,9}$.

| $i$ | $r_i$ | $j$ | Context |
|---|---|---|---|
| 1 | $(0,1,2)$ | 1 | $\{1,2,3\}$ |
| 2 | $(1,0,0)$ | 2 | $\{7,8,9\}$ |
| 3 | $(0,2,-1)$ | 3 | $\{4,5,6\}$ |
| 4 | $(1,2,0)$ | 4 | $\{2,8,5\}$ |
| 5 | $(0,0,1)$ | 5 | $\{3,11,14\}$ |
| 6 | $(2,-1,0)$ | 6 | $\{7,11,15\}$ |
| 7 | $(1,0,1)$ | 7 | $\{4,11,24\}$ |
| 8 | $(0,1,0)$ | 8 | $\{9,10,23\}$ |
| 9 | $(1,0,-1)$ | 9 | $\{4,10,18\}$ |
| 10 | $(2,-1,2)$ | | |
| 11 | $(2,-1,-2)$ | | |
| 14 | $(5,2,4)$ | | |
| 15 | $(1,4,-1)$ | | |
| 18 | $(4,-2,-5)$ | | |
| 23 | $(1,4,1)$ | | |
| 24 | $(4,-2,5)$ | | |

### S1.6. 3D panel (d): $I_{16,9}$

TABLE S7: Unnormalized vectors and all contexts of $I_{16,9}$.

| $i$ | $r_i$ | $j$ | Context |
|---|---|---|---|
| 1 | $(0,1,1)$ | 1 | $\{1,2,3\}$ |
| 2 | $(1,0,0)$ | 2 | $\{7,8,9\}$ |
| 3 | $(0,1,-1)$ | 3 | $\{2,8,5\}$ |
| 4 | $(4,5,3)$ | 4 | $\{3,11,14\}$ |
| 5 | $(0,0,1)$ | 5 | $\{7,11,15\}$ |
| 7 | $(1,0,2)$ | 6 | $\{4,11,24\}$ |
| 8 | $(0,1,0)$ | 7 | $\{9,10,23\}$ |
| 9 | $(2,0,-1)$ | 8 | $\{1,10,19\}$ |
| 10 | $(1,-2,2)$ | 9 | $\{4,10,18\}$ |
| 11 | $(2,-1,-1)$ | | |
| 14 | $(1,1,1)$ | | |
| 15 | $(2,5,-1)$ | | |
| 18 | $(16,-5,-13)$ | | |
| 19 | $(4,1,-1)$ | | |
| 23 | $(2,5,4)$ | | |
| 24 | $(1,-5,7)$ | | |

### S1.7. 3D panel (e): $T_{19,11}$

TABLE S8: Unnormalized vectors and all contexts of $T_{19,11}$.

| $i$ | $r_i$ | $j$ | Context |
|---|---|---|---|
| 1 | $(0,1,2)$ | 1 | $\{1,2,3\}$ |
| 2 | $(1,0,0)$ | 2 | $\{7,8,9\}$ |
| 3 | $(0,2,-1)$ | 3 | $\{4,5,6\}$ |
| 4 | $(1,2,0)$ | 4 | $\{2,8,5\}$ |
| 5 | $(0,0,1)$ | 5 | $\{3,11,14\}$ |
| 6 | $(2,-1,0)$ | 6 | $\{7,11,15\}$ |
| 7 | $(1,0,1)$ | 7 | $\{4,11,24\}$ |
| 8 | $(0,1,0)$ | 8 | $\{9,10,23\}$ |
| 9 | $(1,0,-1)$ | 9 | $\{4,10,18\}$ |
| 10 | $(2,-1,2)$ | 10 | $\{14,26,27\}$ |
| 11 | $(2,-1,-2)$ | 11 | $\{18,26,28\}$ |
| 14 | $(5,2,4)$ | | |
| 15 | $(1,4,-1)$ | | |
| 18 | $(4,-2,-5)$ | | |
| 23 | $(1,4,1)$ | | |
| 24 | $(4,-2,5)$ | | |
| 26 | $(2,-41,18)$ | | |
| 27 | $(200,-82,-209)$ | | |
| 28 | $(241,82,160)$ | | |

### S1.8. 4D panel (a): $W^{(4)}_{11,5}$

TABLE S9: Unnormalized vectors and all contexts of $W^{(4)}_{11,5}$.

| $i$ | $r_i$ | $j$ | Context |
|---|---|---|---|
| 0 | $(0,0,0,1)$ | 1 | $\{0,1,6,2\}$ |
| 1 | $(1,0,0,0)$ | 2 | $\{0,2,7,3\}$ |
| 2 | $(0,1,1,0)$ | 3 | $\{0,3,8,4\}$ |
| 3 | $(1,1,-1,0)$ | 4 | $\{0,4,9,5\}$ |
| 4 | $(1,-2,-1,0)$ | 5 | $\{0,5,10,1\}$ |
| 5 | $(0,1,-2,0)$ | | |
| 6 | $(0,1,-1,0)$ | | |
| 7 | $(2,-1,1,0)$ | | |
| 8 | $(1,0,1,0)$ | | |
| 9 | $(5,2,1,0)$ | | |
| 10 | $(0,2,1,0)$ | | |

### S1.9. 4D panel (b): $W^{(4)}_{16,6}$

TABLE S10: Unnormalized vectors and all contexts of $W^{(4)}_{16,6}$.

| $i$ | $r_i$ | $j$ | Context |
|---|---|---|---|
| 0 | $(0,0,0,1)$ | 1 | $\{2,0,6,1\}$ |
| 1 | $(1,0,0,0)$ | 2 | $\{2,7,13,3\}$ |
| 2 | $(0,1,1,0)$ | 3 | $\{3,0,8,4\}$ |
| 3 | $(1,1,-1,0)$ | 4 | $\{4,9,14,5\}$ |
| 4 | $(1,-2,-1,0)$ | 5 | $\{5,10,15,1\}$ |
| 5 | $(0,1,-2,0)$ | 6 | $\{0,11,12,5\}$ |
| 6 | $(0,1,-1,0)$ | | |
| 7 | $(2,-1,1,1)$ | | |
| 8 | $(1,0,1,0)$ | | |
| 9 | $(5,2,1,1)$ | | |
| 10 | $(0,4,2,1)$ | | |
| 11 | $(1,2,1,0)$ | | |
| 12 | $(5,-2,-1,0)$ | | |
| 13 | $(2,-1,1,-6)$ | | |
| 14 | $(5,2,1,-30)$ | | |
| 15 | $(0,2,1,-10)$ | | |

### S1.10. 4D panel (c): $B^{(4)}_{14,7}$

TABLE S11: Unnormalized vectors and all contexts of $B^{(4)}_{14,7}$.

| $i$ | $r_i$ | $j$ | Context |
|---|---|---|---|
| 0 | $(0,0,0,1)$ | 1 | $\{0,1,2,3\}$ |
| 1 | $(0,1,1,0)$ | 2 | $\{0,7,8,9\}$ |
| 2 | $(1,0,0,0)$ | 3 | $\{0,2,8,5\}$ |
| 3 | $(0,1,-1,0)$ | 4 | $\{0,3,11,14\}$ |
| 5 | $(0,0,1,0)$ | 5 | $\{0,7,11,15\}$ |
| 7 | $(1,0,1,0)$ | 6 | $\{0,1,10,19\}$ |
| 8 | $(0,1,0,0)$ | 7 | $\{0,9,10,23\}$ |
| 9 | $(1,0,-1,0)$ | | |
| 10 | $(1,-1,1,0)$ | | |
| 11 | $(1,-1,-1,0)$ | | |
| 14 | $(2,1,1,0)$ | | |
| 15 | $(1,2,-1,0)$ | | |
| 19 | $(2,1,-1,0)$ | | |
| 23 | $(1,2,1,0)$ | | |

### S1.11. 4D panel (d): $L^{(4)}_{18,7}$

TABLE S12: Unnormalized vectors and all contexts of $L^{(4)}_{18,7}$.

| $i$ | $r_i$ | $j$ | Context |
|---|---|---|---|
| 1 | $(1,0,0,0)$ | 1 | $\{1,2,4,3\}$ |
| 2 | $(0,1,0,0)$ | 2 | $\{5,8,7,6\}$ |
| 3 | $(0,0,1,0)$ | 3 | $\{4,9,10,8\}$ |
| 4 | $(0,0,0,1)$ | 4 | $\{11,1,12,5\}$ |
| 5 | $(0,1,-1,-1)$ | 5 | $\{11,14,13,2\}$ |
| 6 | $(4,-5,-11,6)$ | 6 | $\{15,3,16,7\}$ |
| 7 | $(1,-4,0,-4)$ | 7 | $\{15,17,18,6\}$ |
| 8 | $(4,1,1,0)$ | | |
| 9 | $(1,-5,1,0)$ | | |
| 10 | $(2,-1,-7,0)$ | | |
| 11 | $(0,0,1,-1)$ | | |
| 12 | $(0,2,1,1)$ | | |
| 13 | $(1,0,1,1)$ | | |
| 14 | $(2,0,-1,-1)$ | | |
| 15 | $(4,2,0,-1)$ | | |
| 16 | $(4,-5,0,6)$ | | |
| 17 | $(5,-9,7,2)$ | | |
| 18 | $(2,113,77,234)$ | | |

### S1.12. 4D panel (e): $D^{(4)}_{17,9}$

TABLE S13: Unnormalized vectors and all contexts of $D^{(4)}_{17,9}$.

| $i$ | $r_i$ | $j$ | Context |
|---|---|---|---|
| 0 | $(0,0,0,1)$ | 1 | $\{0,1,2,3\}$ |
| 1 | $(0,1,2,0)$ | 2 | $\{0,7,8,9\}$ |
| 2 | $(1,0,0,0)$ | 3 | $\{0,4,5,6\}$ |
| 3 | $(0,2,-1,0)$ | 4 | $\{0,2,8,5\}$ |
| 4 | $(1,2,0,0)$ | 5 | $\{0,3,11,14\}$ |
| 5 | $(0,0,1,0)$ | 6 | $\{0,7,11,15\}$ |
| 6 | $(2,-1,0,0)$ | 7 | $\{0,9,10,23\}$ |
| 7 | $(1,0,1,0)$ | 8 | $\{0,4,11,24\}$ |
| 8 | $(0,1,0,0)$ | 9 | $\{0,4,10,18\}$ |
| 9 | $(1,0,-1,0)$ | | |
| 10 | $(2,-1,2,0)$ | | |
| 11 | $(2,-1,-2,0)$ | | |
| 14 | $(5,2,4,0)$ | | |
| 15 | $(1,4,-1,0)$ | | |
| 18 | $(4,-2,-5,0)$ | | |
| 23 | $(1,4,1,0)$ | | |
| 24 | $(4,-2,5,0)$ | | |

### S1.13. 4D panel (f): $L^{(4)}_{22,9}$

TABLE S14: Unnormalized vectors and all contexts of $L^{(4)}_{22,9}$.

| $i$ | $r_i$ | $j$ | Context |
|---|---|---|---|
| 1 | $(1,0,0,0)$ | 1 | $\{1,2,4,3\}$ |
| 2 | $(0,1,0,0)$ | 2 | $\{5,8,7,6\}$ |
| 3 | $(0,0,1,0)$ | 3 | $\{4,9,10,8\}$ |
| 4 | $(0,0,0,1)$ | 4 | $\{11,1,12,5\}$ |
| 5 | $(0,1,-1,-1)$ | 5 | $\{11,14,13,2\}$ |
| 6 | $(4,-5,-11,6)$ | 6 | $\{15,3,16,7\}$ |
| 7 | $(1,-4,0,-4)$ | 7 | $\{15,17,18,6\}$ |
| 8 | $(4,1,1,0)$ | 8 | $\{11,21,22,19\}$ |
| 9 | $(1,-5,1,0)$ | 9 | $\{13,20,15,19\}$ |
| 10 | $(2,-1,-7,0)$ | | |
| 11 | $(0,0,1,-1)$ | | |
| 12 | $(0,2,1,1)$ | | |
| 13 | $(1,0,4,4)$ | | |
| 14 | $(8,0,-1,-1)$ | | |
| 15 | $(4,2,0,-1)$ | | |
| 16 | $(4,-5,0,6)$ | | |
| 17 | $(5,-9,7,2)$ | | |
| 18 | $(2,113,77,234)$ | | |
| 19 | $(16,-33,-2,-2)$ | | |
| 20 | $(4,2,-21,20)$ | | |
| 21 | $(22,12,-11,-11)$ | | |
| 22 | $(129,44,153,153)$ | | |

### S1.14. Hesse $H_{37}$: vectors, contexts, and non-collinearity

This appendix uses exactly the vertex numbers and context numbers of Fig. 4. All indices $a, k$ in the family notation are in $\mathbb{Z}_3$. The vectors are unnormalized and the inner product is Hermitian. Table S15 lists every vector, squared norm, color, and incident context. Table S16 lists all 24 contexts with their actual vectors. Together they specify the faithful orthogonal representation completely.

### *S1.14.1. All 37 vectors, their colors, and incident contexts*

TABLE S15: Complete enumeration of the 37 vectors of $H_{37}$. The color $c(i) \in \{0,1,2,3\}$ is the explicit four-coloring in the main text. Context indices refer to Table S16.

| $i$ | Ray | $v_i$ | $\|v_i\|^2$ | $c(i)$ | Incident contexts |
|---|---|---|---|---|---|
| 1 | $z$ | $(1,0,0)$ | 1 | 0 | $\{4,5,6\}$ |
| 2 | $M_{00}$ | $(1,1,1)$ | 3 | 0 | $\{1,7,16\}$ |
| 3 | $M_{01}$ | $(1,\zeta,\zeta^2)$ | 3 | 1 | $\{1,8,17\}$ |
| 4 | $M_{02}$ | $(1,\zeta^2,\zeta)$ | 3 | 2 | $\{1,9,18\}$ |
| 5 | $M_{10}$ | $(1,\zeta,\zeta)$ | 3 | 0 | $\{2,10,21\}$ |
| 6 | $M_{11}$ | $(1,\zeta^2,1)$ | 3 | 1 | $\{2,11,19\}$ |
| 7 | $M_{12}$ | $(1,1,\zeta^2)$ | 3 | 2 | $\{2,12,20\}$ |
| 8 | $M_{20}$ | $(1,\zeta^2,\zeta^2)$ | 3 | 0 | $\{3,13,23\}$ |
| 9 | $M_{21}$ | $(1,1,\zeta)$ | 3 | 1 | $\{3,14,24\}$ |
| 10 | $M_{22}$ | $(1,\zeta,1)$ | 3 | 2 | $\{3,15,22\}$ |
| 11 | $P_0$ | $(0,1,-1)$ | 2 | 3 | $\{4,7,10,13\}$ |
| 12 | $P_1$ | $(0,1,-\zeta)$ | 2 | 3 | $\{5,8,11,14\}$ |
| 13 | $P_2$ | $(0,1,-\zeta^2)$ | 2 | 3 | $\{6,9,12,15\}$ |
| 14 | $Q_0$ | $(-1,0,1)$ | 2 | 3 | $\{16,19,22\}$ |
| 15 | $Q_1$ | $(-\zeta,0,1)$ | 2 | 3 | $\{17,20,23\}$ |
| 16 | $Q_2$ | $(-\zeta^2,0,1)$ | 2 | 3 | $\{18,21,24\}$ |
| 17 | $A_0$ | $(0,1,1)$ | 2 | 1 | $\{4\}$ |
| 18 | $A_1$ | $(0,1,\zeta)$ | 2 | 1 | $\{5\}$ |
| 19 | $A_2$ | $(0,1,\zeta^2)$ | 2 | 1 | $\{6\}$ |
| 20 | $B_{00}$ | $(-2,1,1)$ | 6 | 1 | $\{7\}$ |
| 21 | $B_{01}$ | $(-2,\zeta,\zeta^2)$ | 6 | 2 | $\{8\}$ |
| 22 | $B_{02}$ | $(-2,\zeta^2,\zeta)$ | 6 | 0 | $\{9\}$ |
| 23 | $B_{10}$ | $(-2,\zeta,\zeta)$ | 6 | 1 | $\{10\}$ |
| 24 | $B_{11}$ | $(-2,\zeta^2,1)$ | 6 | 2 | $\{11\}$ |
| 25 | $B_{12}$ | $(-2,1,\zeta^2)$ | 6 | 0 | $\{12\}$ |
| 26 | $B_{20}$ | $(-2,\zeta^2,\zeta^2)$ | 6 | 1 | $\{13\}$ |
| 27 | $B_{21}$ | $(-2,1,\zeta)$ | 6 | 2 | $\{14\}$ |
| 28 | $B_{22}$ | $(-2,\zeta,1)$ | 6 | 0 | $\{15\}$ |
| 29 | $C_{00}$ | $(1,-2,1)$ | 6 | 1 | $\{16\}$ |
| 30 | $C_{01}$ | $(1,-2\zeta,\zeta^2)$ | 6 | 2 | $\{17\}$ |
| 31 | $C_{02}$ | $(1,-2\zeta^2,\zeta)$ | 6 | 0 | $\{18\}$ |
| 32 | $C_{10}$ | $(1,-2\zeta^2,1)$ | 6 | 2 | $\{19\}$ |
| 33 | $C_{11}$ | $(1,-2,\zeta^2)$ | 6 | 0 | $\{20\}$ |
| 34 | $C_{12}$ | $(1,-2\zeta,\zeta)$ | 6 | 1 | $\{21\}$ |
| 35 | $C_{20}$ | $(1,-2\zeta,1)$ | 6 | 0 | $\{22\}$ |
| 36 | $C_{21}$ | $(1,-2\zeta^2,\zeta^2)$ | 6 | 1 | $\{23\}$ |
| 37 | $C_{22}$ | $(1,-2,\zeta)$ | 6 | 2 | $\{24\}$ |

### *S1.14.2. All 24 contexts and their orthogonal vectors*

TABLE S16: All 24 orthogonal triads, numbered in the same order as the context colors in Fig. 4. Vectors are given in increasing order of their vertex numbers. Each triad contributes three orthogonality edges; the 72 resulting pairs are all the orthogonal pairs.

| Context | Family | Vertices | Unnormalized vectors |
|---|---|---|---|
| $e_1$ | $\mathcal{M}_0$ | $\{2,3,4\}$ | $\{(1,1,1),\ (1,\zeta,\zeta^2),\ (1,\zeta^2,\zeta)\}$ |
| $e_2$ | $\mathcal{M}_1$ | $\{5,6,7\}$ | $\{(1,\zeta,\zeta),\ (1,\zeta^2,1),\ (1,1,\zeta^2)\}$ |
| $e_3$ | $\mathcal{M}_2$ | $\{8,9,10\}$ | $\{(1,\zeta^2,\zeta^2),\ (1,1,\zeta),\ (1,\zeta,1)\}$ |
| $e_4$ | $\mathcal{A}_0$ | $\{1,11,17\}$ | $\{(1,0,0),\ (0,1,-1),\ (0,1,1)\}$ |
| $e_5$ | $\mathcal{A}_1$ | $\{1,12,18\}$ | $\{(1,0,0),\ (0,1,-\zeta),\ (0,1,\zeta)\}$ |
| $e_6$ | $\mathcal{A}_2$ | $\{1,13,19\}$ | $\{(1,0,0),\ (0,1,-\zeta^2),\ (0,1,\zeta^2)\}$ |
| $e_7$ | $\mathcal{B}_{00}$ | $\{2,11,20\}$ | $\{(1,1,1),\ (0,1,-1),\ (-2,1,1)\}$ |
| $e_8$ | $\mathcal{B}_{01}$ | $\{3,12,21\}$ | $\{(1,\zeta,\zeta^2),\ (0,1,-\zeta),\ (-2,\zeta,\zeta^2)\}$ |
| $e_9$ | $\mathcal{B}_{02}$ | $\{4,13,22\}$ | $\{(1,\zeta^2,\zeta),\ (0,1,-\zeta^2),\ (-2,\zeta^2,\zeta)\}$ |
| $e_{10}$ | $\mathcal{B}_{10}$ | $\{5,11,23\}$ | $\{(1,\zeta,\zeta),\ (0,1,-1),\ (-2,\zeta,\zeta)\}$ |
| $e_{11}$ | $\mathcal{B}_{11}$ | $\{6,12,24\}$ | $\{(1,\zeta^2,1),\ (0,1,-\zeta),\ (-2,\zeta^2,1)\}$ |
| $e_{12}$ | $\mathcal{B}_{12}$ | $\{7,13,25\}$ | $\{(1,1,\zeta^2),\ (0,1,-\zeta^2),\ (-2,1,\zeta^2)\}$ |
| $e_{13}$ | $\mathcal{B}_{20}$ | $\{8,11,26\}$ | $\{(1,\zeta^2,\zeta^2),\ (0,1,-1),\ (-2,\zeta^2,\zeta^2)\}$ |
| $e_{14}$ | $\mathcal{B}_{21}$ | $\{9,12,27\}$ | $\{(1,1,\zeta),\ (0,1,-\zeta),\ (-2,1,\zeta)\}$ |
| $e_{15}$ | $\mathcal{B}_{22}$ | $\{10,13,28\}$ | $\{(1,\zeta,1),\ (0,1,-\zeta^2),\ (-2,\zeta,1)\}$ |
| $e_{16}$ | $\mathcal{C}_{00}$ | $\{2,14,29\}$ | $\{(1,1,1),\ (-1,0,1),\ (1,-2,1)\}$ |

*Continued on the next page.*

TABLE S16: All 24 orthogonal triads, numbered in the same order as the context colors in Fig. 4. Vectors are given in increasing order of their vertex numbers. Each triad contributes three orthogonality edges; the 72 resulting pairs are all the orthogonal pairs. (continued).

| Context | Family | Vertices | Unnormalized vectors |
|---|---|---|---|
| $e_{17}$ | $\mathcal{C}_{01}$ | $\{3,15,30\}$ | $\{(1,\zeta,\zeta^2),\ (-\zeta,0,1),\ (1,-2\zeta,\zeta^2)\}$ |
| $e_{18}$ | $\mathcal{C}_{02}$ | $\{4,16,31\}$ | $\{(1,\zeta^2,\zeta),\ (-\zeta^2,0,1),\ (1,-2\zeta^2,\zeta)\}$ |
| $e_{19}$ | $\mathcal{C}_{10}$ | $\{6,14,32\}$ | $\{(1,\zeta^2,1),\ (-1,0,1),\ (1,-2\zeta^2,1)\}$ |
| $e_{20}$ | $\mathcal{C}_{11}$ | $\{7,15,33\}$ | $\{(1,1,\zeta^2),\ (-\zeta,0,1),\ (1,-2,\zeta^2)\}$ |
| $e_{21}$ | $\mathcal{C}_{12}$ | $\{5,16,34\}$ | $\{(1,\zeta,\zeta),\ (-\zeta^2,0,1),\ (1,-2\zeta,\zeta)\}$ |
| $e_{22}$ | $\mathcal{C}_{20}$ | $\{10,14,35\}$ | $\{(1,\zeta,1),\ (-1,0,1),\ (1,-2\zeta,1)\}$ |
| $e_{23}$ | $\mathcal{C}_{21}$ | $\{8,15,36\}$ | $\{(1,\zeta^2,\zeta^2),\ (-\zeta,0,1),\ (1,-2\zeta^2,\zeta^2)\}$ |
| $e_{24}$ | $\mathcal{C}_{22}$ | $\{9,16,37\}$ | $\{(1,1,\zeta),\ (-\zeta^2,0,1),\ (1,-2,\zeta)\}$ |

#### *S1.14.3. Norm shells and a non-collinearity certificate*

The four squared-norm shells have sizes

$$\#\{i : \|v_i\|^2 = 1\} = 1, \quad \#\{i : \|v_i\|^2 = 2\} = 9, \quad \#\{i : \|v_i\|^2 = 3\} = 9, \quad \#\{i : \|v_i\|^2 = 6\} = 18. \tag{S1}$$

Exact evaluation over the $\binom{37}{2} = 666$ unordered pairs gives

$$\max_{i<j} \frac{|\langle v_i, v_j\rangle|^2}{\|v_i\|^2\|v_j\|^2} = \frac{3}{4} < 1. \tag{S2}$$

For example, the maximum is attained by $v_{11} = P_0 = (0,1,-1)$ and $v_{29} = C_{00} = (1,-2,1)$:

$$\frac{|\langle v_{11}, v_{29}\rangle|^2}{\|v_{11}\|^2\|v_{29}\|^2} = \frac{(-3)^2}{2\cdot 6} = \frac{3}{4}. \tag{S3}$$

Thus no two printed vectors represent the same ray. The maximum and all 72 zero inner products are checked exactly by the accompanying verifier; the family-wise argument in the main text explains why there are no further orthogonalities.

### S1.15. Four-dimensional $H_{\mathrm{YO}}^{(4)}$ and $H_{65}$: vectors and contexts

For direct human and computational verification, this appendix enumerates all unnormalized integer vectors in $\mathbb{Z}^4$ and all four-element contexts for the two four-dimensional examples $H_{65}$ and $H_{\mathrm{YO}}^{(4)}$:

1. The *canonical linear hypergraph* $H_{65} = (V_{65}, E_{29})$ (Theorem 22, Appendix S1.15.1), which has exactly 65 **pairwise non-collinear integer vectors** (17 core vectors $v_1,\ldots,v_{17}$ and 48 degree-1 completion vectors $v_{18},\ldots,v_{65}$), 29 **four-element contexts** (5 core contexts $C_1,\ldots,C_5$ and 24 bridge-completion contexts $E_1,\ldots,E_{24}$), and 174 **orthogonality edges** ($29\times\binom{4}{2} = 174$).

2. The *non-linear warm-up apex suspension* $H_{\mathrm{YO}}^{(4)} = (V_{26}, E_{16}^{(4)})$ (Proposition 20, Appendix S1.15.2), which has exactly 26 **pairwise non-collinear integer vectors** (13 Yu–Oh rays $v_1,\ldots,v_{13}$, 12 completion rays $v_{14},\ldots,v_{25}$ in $\mathbb{R}^3\times\{0\}$, and 1 apex ray $v_{26}$), 16 **four-element contexts**, and 73 **orthogonality edges** ($16\times\binom{3}{2} = 48$ edges in $\mathbb{R}^3\times\{0\}$ plus 25 apex edges).

*Canonical primitive normal form and human-verifiable non-collinearity.* Every vector $v_i = (x_{i,1}, x_{i,2}, x_{i,3}, x_{i,4})^T \in \mathbb{Z}^4$ listed below in both $H_{65}$ and $H_{\mathrm{YO}}^{(4)}$ is written in *canonical primitive projective form*:

$$\begin{gathered}\gcd\big(|x_{i,1}|, |x_{i,2}|, |x_{i,3}|, |x_{i,4}|\big) = 1,\\ \text{first nonzero entry of } v_i > 0.\end{gathered} \tag{S4}$$

Because every projective ray $[v] \in \mathbb{P}(\mathbb{Q}^4)$ possesses a *unique* representative in $\mathbb{Z}^4$ satisfying Eq. (S4), two vectors $v_i, v_j$ in canonical primitive form are collinear (proportional over $\mathbb{R}$ or $\mathbb{C}$) *if and only if they are identical* 4*-tuples* ($v_i = v_j$) and, in particular, have the same squared Euclidean norm $\|v_i\|^2 = \|v_j\|^2$.

### *S1.15.1. The canonical linear 65-vertex, 29-context hypergraph $H_{65}$*

Table S17 enumerates all 65 unnormalized integer vectors $v_1, \ldots, v_{65} \in \mathbb{Z}^4$ of $H_{65}$, together with their squared norms $\|v_i\|^2$, their colors $c(i) \in \{0, 1, 2, 3, 4\}$ in the strong 5-coloring extending Eq. (S28), and the contexts in $E_{29}$ to which each vertex belongs. Table S18 enumerates all 29 contexts $C_1, \ldots, C_5, E_1, \ldots, E_{24} \in E_{29}$, displaying side by side both the vertex indices $\{i, j, k, \ell\}$ and the four mutually orthogonal integer vectors $\{v_i, v_j, v_k, v_\ell\} \subset \mathbb{Z}^4$.

To verify by hand that all 65 vectors in Table S17 are pairwise non-collinear across all $\binom{65}{2} = 2{,}080$ pairs, observe that all 65 tuples satisfy Eq. (S4) and partition by squared norm $\|v_i\|^2$ into seven small, disjoint norm shells ($4 + 6 + 7 + 12 + 12 + 12 + 12 = 65$):

- $\|v_i\|^2 = 1$ (4 vectors): indices $\{1, 2, 3, 4\}$, the 4 standard coordinate basis vectors;
- $\|v_i\|^2 = 2$ (6 vectors): indices $\{12, \ldots, 17\}$, one vector for each of the $\binom{4}{2} = 6$ coordinate-pair supports;
- $\|v_i\|^2 = 4$ (7 vectors): indices $\{5, \ldots, 11\}$, seven distinct $\pm 1$ vectors with $x_{i,1} = +1$;
- $\|v_i\|^2 = 11$ (12 vectors): indices $\{18, 20, 22, 25, 26, 28, 30, 33, 34, 36, 39, 41\}$, each having a single zero coordinate (three each with $x_1 = 0$, $x_2 = 0$, $x_3 = 0$, $x_4 = 0$) and distinct sign/entry patterns of $\{0, \pm 1, \pm 1, \pm 3\}$;
- $\|v_i\|^2 = 18$ (12 vectors): indices $\{43, 45, 46, 49, 51, 53, 54, 56, 59, 61, 62, 65\}$, twelve distinct permutations and sign patterns of $\{\pm 1, \pm 2, \pm 2, \pm 3\}$;
- $\|v_i\|^2 = 22$ (12 vectors): indices $\{19, 21, 23, 24, 27, 29, 31, 32, 35, 37, 38, 40\}$, each having a single zero coordinate (three each with $x_1 = 0$, $x_2 = 0$, $x_3 = 0$, $x_4 = 0$) and distinct patterns of $\{0, \pm 2, \pm 3, \pm 3\}$;
- $\|v_i\|^2 = 36$ (12 vectors): indices $\{42, 44, 47, 48, 50, 52, 55, 57, 58, 60, 63, 64\}$, twelve distinct permutations and sign patterns of $\{\pm 1, \pm 1, \pm 3, \pm 5\}$.

Because no two 4-tuples within any of the seven norm shells coincide, no two of the 65 vectors are collinear; quantitatively, the maximum squared cosine over all $\binom{65}{2} = 2{,}080$ pairs is $\max_{1 \le i < j \le 65} (v_i \cdot v_j)^2 / (\|v_i\|^2 \|v_j\|^2) = 19^2/(11 \times 36) = 361/396 \approx 0.9116 < 1$, attained by $(v_{41}, v_{47})$, whose squared norms are 11 and 36 and whose inner product is 19.

TABLE S17: Complete enumeration of the 65 unnormalized integer vectors $v_i \in \mathbb{Z}^4$ of $H_{65}$, their squared norms $\|v_i\|^2$, their colors $c(i) \in \{0, 1, 2, 3, 4\}$, and the contexts in $E_{29}$ containing vertex $i$.

| $i$ | $v_i \in \mathbb{Z}^4$ | $\|v_i\|^2$ | $c(i)$ | Contexts containing $i$ | $i$ | $v_i \in \mathbb{Z}^4$ | $\|v_i\|^2$ | $c(i)$ | Contexts containing $i$ |
|---|---|---|---|---|---|---|---|---|---|
| 1 | $(1,0,0,0)$ | 1 | 0 | $C_1, E_3, E_7, E_{11}$ | 34 | $(1,-1,0,3)$ | 11 | 1 | $E_9 = \{3,16,34,35\}$ |
| 2 | $(0,1,0,0)$ | 1 | 1 | $C_1, E_1, E_5, E_{12}$ | 35 | $(3,-3,0,-2)$ | 22 | 3 | $E_9 = \{3,16,34,35\}$ |
| 3 | $(0,0,1,0)$ | 1 | 2 | $C_1, E_2, E_8, E_9$ | 36 | $(1,-1,3,0)$ | 11 | 1 | $E_{10} = \{4,16,36,37\}$ |
| 4 | $(0,0,0,1)$ | 1 | 3 | $C_1, E_4, E_6, E_{10}$ | 37 | $(3,-3,-2,0)$ | 22 | 2 | $E_{10} = \{4,16,36,37\}$ |
| 5 | $(1,1,1,1)$ | 4 | 0 | $C_2, E_{13}, E_{19}, E_{23}$ | 38 | $(0,2,3,3)$ | 22 | 1 | $E_{11} = \{1,17,38,39\}$ |
| 6 | $(1,1,-1,-1)$ | 4 | 1 | $C_2, E_{15}, E_{17}, E_{24}$ | 39 | $(0,3,-1,-1)$ | 11 | 3 | $E_{11} = \{1,17,38,39\}$ |
| 7 | $(1,-1,1,-1)$ | 4 | 2 | $C_2, E_{16}, E_{20}, E_{21}$ | 40 | $(2,0,-3,-3)$ | 22 | 0 | $E_{12} = \{2,17,40,41\}$ |
| 8 | $(1,-1,-1,1)$ | 4 | 4 | $C_2, E_{14}, E_{18}, E_{22}$ | 41 | $(3,0,1,1)$ | 11 | 3 | $E_{12} = \{2,17,40,41\}$ |
| 9 | $(1,1,-1,1)$ | 4 | 0 | $C_3, C_4$ | 42 | $(1,-5,3,1)$ | 36 | 1 | $E_{13} = \{5,12,42,43\}$ |
| 10 | $(1,-1,1,1)$ | 4 | 1 | $C_3, C_5$ | 43 | $(2,-1,-3,2)$ | 18 | 2 | $E_{13} = \{5,12,42,43\}$ |
| 11 | $(1,-1,-1,-1)$ | 4 | 3 | $C_4, C_5$ | 44 | $(1,-3,5,1)$ | 36 | 0 | $E_{14} = \{8,12,44,45\}$ |
| 12 | $(1,0,0,-1)$ | 2 | 3 | $C_3, E_1, E_2, E_{13}, E_{14}$ | 45 | $(2,3,1,2)$ | 18 | 1 | $E_{14} = \{8,12,44,45\}$ |
| 13 | $(0,1,1,0)$ | 2 | 4 | $C_3, E_3, E_4, E_{15}, E_{16}$ | 46 | $(1,-2,2,-3)$ | 18 | 0 | $E_{15} = \{6,13,46,47\}$ |
| 14 | $(1,0,1,0)$ | 2 | 2 | $C_4, E_5, E_6, E_{17}, E_{18}$ | 47 | $(5,-1,1,3)$ | 36 | 2 | $E_{15} = \{6,13,46,47\}$ |
| 15 | $(0,1,0,-1)$ | 2 | 1 | $C_4, E_7, E_8, E_{19}, E_{20}$ | 48 | $(3,-1,1,5)$ | 36 | 0 | $E_{16} = \{7,13,48,49\}$ |
| 16 | $(1,1,0,0)$ | 2 | 0 | $C_5, E_9, E_{10}, E_{21}, E_{22}$ | 49 | $(3,2,-2,-1)$ | 18 | 1 | $E_{16} = \{7,13,48,49\}$ |
| 17 | $(0,0,1,-1)$ | 2 | 2 | $C_5, E_{11}, E_{12}, E_{23}, E_{24}$ | 50 | $(1,3,-1,5)$ | 36 | 0 | $E_{17} = \{6,14,50,51\}$ |
| 18 | $(1,0,-3,1)$ | 11 | 0 | $E_1 = \{2,12,18,19\}$ | 51 | $(2,-3,-2,1)$ | 18 | 3 | $E_{17} = \{6,14,50,51\}$ |
| 19 | $(3,0,2,3)$ | 22 | 2 | $E_1 = \{2,12,18,19\}$ | 52 | $(1,5,-1,3)$ | 36 | 0 | $E_{18} = \{8,14,52,53\}$ |
| 20 | $(1,3,0,1)$ | 11 | 0 | $E_2 = \{3,12,20,21\}$ | 53 | $(2,1,-2,-3)$ | 18 | 1 | $E_{18} = \{8,14,52,53\}$ |
| 21 | $(3,-2,0,3)$ | 22 | 1 | $E_2 = \{3,12,20,21\}$ | 54 | $(3,-2,1,-2)$ | 18 | 2 | $E_{19} = \{5,15,54,55\}$ |
| 22 | $(0,1,-1,3)$ | 11 | 1 | $E_3 = \{1,13,22,23\}$ | 55 | $(3,1,-5,1)$ | 36 | 3 | $E_{19} = \{5,15,54,55\}$ |
| 23 | $(0,3,-3,-2)$ | 22 | 2 | $E_3 = \{1,13,22,23\}$ | 56 | $(1,2,3,2)$ | 18 | 0 | $E_{20} = \{7,15,56,57\}$ |
| 24 | $(2,-3,3,0)$ | 22 | 0 | $E_4 = \{4,13,24,25\}$ | 57 | $(5,1,-3,1)$ | 36 | 3 | $E_{20} = \{7,15,56,57\}$ |

*Continued on the next page.*

TABLE S17: All 65 vectors of $H_{65}$ (continued).

| $i$ | $v_i \in \mathbb{Z}^4$ | $\|v_i\|^2$ | $c(i)$ | Contexts containing $i$ | $i$ | $v_i \in \mathbb{Z}^4$ | $\|v_i\|^2$ | $c(i)$ | Contexts containing $i$ |
|---|---|---|---|---|---|---|---|---|---|
| 25 | $(3,1,-1,0)$ | 11 | 1 | $E_4 = \{4,13,24,25\}$ | 58 | $(1,-1,-5,-3)$ | 36 | 1 | $E_{21} = \{7,16,58,59\}$ |
| 26 | $(1,0,-1,-3)$ | 11 | 0 | $E_5 = \{2,14,26,27\}$ | 59 | $(2,-2,-1,3)$ | 18 | 3 | $E_{21} = \{7,16,58,59\}$ |
| 27 | $(3,0,-3,2)$ | 22 | 3 | $E_5 = \{2,14,26,27\}$ | 60 | $(1,-1,-3,-5)$ | 36 | 1 | $E_{22} = \{8,16,60,61\}$ |
| 28 | $(1,-3,-1,0)$ | 11 | 0 | $E_6 = \{4,14,28,29\}$ | 61 | $(2,-2,3,-1)$ | 18 | 2 | $E_{22} = \{8,16,60,61\}$ |
| 29 | $(3,2,-3,0)$ | 22 | 1 | $E_6 = \{4,14,28,29\}$ | 62 | $(1,3,-2,-2)$ | 18 | 1 | $E_{23} = \{5,17,62,63\}$ |
| 30 | $(0,1,3,1)$ | 11 | 2 | $E_7 = \{1,15,30,31\}$ | 63 | $(5,-3,-1,-1)$ | 36 | 3 | $E_{23} = \{5,17,62,63\}$ |
| 31 | $(0,3,-2,3)$ | 22 | 3 | $E_7 = \{1,15,30,31\}$ | 64 | $(3,-5,-1,-1)$ | 36 | 0 | $E_{24} = \{6,17,64,65\}$ |
| 32 | $(2,3,0,3)$ | 22 | 0 | $E_8 = \{3,15,32,33\}$ | 65 | $(3,1,2,2)$ | 18 | 3 | $E_{24} = \{6,17,64,65\}$ |
| 33 | $(3,-1,0,-1)$ | 11 | 3 | $E_8 = \{3,15,32,33\}$ | | | | | |

TABLE S18: Complete enumeration of the 29 four-element contexts of $H_{65} = (V_{65}, E_{29})$, listing for each context both its vertex indices $\{i,j,k,\ell\}$ and the four mutually orthogonal unnormalized integer vectors $\{v_i, v_j, v_k, v_\ell\} \subset \mathbb{Z}^4$.

| Context | Vertex set | $v_i$ | $v_j$ | $v_k$ | $v_\ell$ |
|---|---|---|---|---|---|
| *Five core 4-contexts $C_1, \ldots, C_5$ on $V_{\text{core}} = \{1, \ldots, 17\}$:* | | | | | |
| $C_1$ | $\{1,2,3,4\}$ | $v_1 = (1,0,0,0)$ | $v_2 = (0,1,0,0)$ | $v_3 = (0,0,1,0)$ | $v_4 = (0,0,0,1)$ |
| $C_2$ | $\{5,6,7,8\}$ | $v_5 = (1,1,1,1)$ | $v_6 = (1,1,-1,-1)$ | $v_7 = (1,-1,1,-1)$ | $v_8 = (1,-1,-1,1)$ |
| $C_3$ | $\{9,10,12,13\}$ | $v_9 = (1,1,-1,1)$ | $v_{10} = (1,-1,1,1)$ | $v_{12} = (1,0,0,-1)$ | $v_{13} = (0,1,1,0)$ |
| $C_4$ | $\{9,11,14,15\}$ | $v_9 = (1,1,-1,1)$ | $v_{11} = (1,-1,-1,-1)$ | $v_{14} = (1,0,1,0)$ | $v_{15} = (0,1,0,-1)$ |
| $C_5$ | $\{10,11,16,17\}$ | $v_{10} = (1,-1,1,1)$ | $v_{11} = (1,-1,-1,-1)$ | $v_{16} = (1,1,0,0)$ | $v_{17} = (0,0,1,-1)$ |
| *Twelve coordinate-to-bridge completion contexts $E_1, \ldots, E_{12}$ (between $C_1$ and $B_1 \cup B_2 \cup B_3$):* | | | | | |
| $E_1$ | $\{2,12,18,19\}$ | $v_2 = (0,1,0,0)$ | $v_{12} = (1,0,0,-1)$ | $v_{18} = (1,0,-3,1)$ | $v_{19} = (3,0,2,3)$ |
| $E_2$ | $\{3,12,20,21\}$ | $v_3 = (0,0,1,0)$ | $v_{12} = (1,0,0,-1)$ | $v_{20} = (1,3,0,1)$ | $v_{21} = (3,-2,0,3)$ |
| $E_3$ | $\{1,13,22,23\}$ | $v_1 = (1,0,0,0)$ | $v_{13} = (0,1,1,0)$ | $v_{22} = (0,1,-1,3)$ | $v_{23} = (0,3,-3,-2)$ |
| $E_4$ | $\{4,13,24,25\}$ | $v_4 = (0,0,0,1)$ | $v_{13} = (0,1,1,0)$ | $v_{24} = (2,-3,3,0)$ | $v_{25} = (3,1,-1,0)$ |
| $E_5$ | $\{2,14,26,27\}$ | $v_2 = (0,1,0,0)$ | $v_{14} = (1,0,1,0)$ | $v_{26} = (1,0,-1,-3)$ | $v_{27} = (3,0,-3,2)$ |
| $E_6$ | $\{4,14,28,29\}$ | $v_4 = (0,0,0,1)$ | $v_{14} = (1,0,1,0)$ | $v_{28} = (1,-3,-1,0)$ | $v_{29} = (3,2,-3,0)$ |
| $E_7$ | $\{1,15,30,31\}$ | $v_1 = (1,0,0,0)$ | $v_{15} = (0,1,0,-1)$ | $v_{30} = (0,1,3,1)$ | $v_{31} = (0,3,-2,3)$ |
| $E_8$ | $\{3,15,32,33\}$ | $v_3 = (0,0,1,0)$ | $v_{15} = (0,1,0,-1)$ | $v_{32} = (2,3,0,3)$ | $v_{33} = (3,-1,0,-1)$ |
| $E_9$ | $\{3,16,34,35\}$ | $v_3 = (0,0,1,0)$ | $v_{16} = (1,1,0,0)$ | $v_{34} = (1,-1,0,3)$ | $v_{35} = (3,-3,0,-2)$ |
| $E_{10}$ | $\{4,16,36,37\}$ | $v_4 = (0,0,0,1)$ | $v_{16} = (1,1,0,0)$ | $v_{36} = (1,-1,3,0)$ | $v_{37} = (3,-3,-2,0)$ |
| $E_{11}$ | $\{1,17,38,39\}$ | $v_1 = (1,0,0,0)$ | $v_{17} = (0,0,1,-1)$ | $v_{38} = (0,2,3,3)$ | $v_{39} = (0,3,-1,-1)$ |
| $E_{12}$ | $\{2,17,40,41\}$ | $v_2 = (0,1,0,0)$ | $v_{17} = (0,0,1,-1)$ | $v_{40} = (2,0,-3,-3)$ | $v_{41} = (3,0,1,1)$ |
| *Twelve Hadamard-to-bridge completion contexts $E_{13}, \ldots, E_{24}$ (between $C_2$ and $B_1 \cup B_2 \cup B_3$):* | | | | | |
| $E_{13}$ | $\{5,12,42,43\}$ | $v_5 = (1,1,1,1)$ | $v_{12} = (1,0,0,-1)$ | $v_{42} = (1,-5,3,1)$ | $v_{43} = (2,-1,-3,2)$ |
| $E_{14}$ | $\{8,12,44,45\}$ | $v_8 = (1,-1,-1,1)$ | $v_{12} = (1,0,0,-1)$ | $v_{44} = (1,-3,5,1)$ | $v_{45} = (2,3,1,2)$ |
| $E_{15}$ | $\{6,13,46,47\}$ | $v_6 = (1,1,-1,-1)$ | $v_{13} = (0,1,1,0)$ | $v_{46} = (1,-2,2,-3)$ | $v_{47} = (5,-1,1,3)$ |
| $E_{16}$ | $\{7,13,48,49\}$ | $v_7 = (1,-1,1,-1)$ | $v_{13} = (0,1,1,0)$ | $v_{48} = (3,-1,1,5)$ | $v_{49} = (3,2,-2,-1)$ |
| $E_{17}$ | $\{6,14,50,51\}$ | $v_6 = (1,1,-1,-1)$ | $v_{14} = (1,0,1,0)$ | $v_{50} = (1,3,-1,5)$ | $v_{51} = (2,-3,-2,1)$ |
| $E_{18}$ | $\{8,14,52,53\}$ | $v_8 = (1,-1,-1,1)$ | $v_{14} = (1,0,1,0)$ | $v_{52} = (1,5,-1,3)$ | $v_{53} = (2,1,-2,-3)$ |
| $E_{19}$ | $\{5,15,54,55\}$ | $v_5 = (1,1,1,1)$ | $v_{15} = (0,1,0,-1)$ | $v_{54} = (3,-2,1,-2)$ | $v_{55} = (3,1,-5,1)$ |
| $E_{20}$ | $\{7,15,56,57\}$ | $v_7 = (1,-1,1,-1)$ | $v_{15} = (0,1,0,-1)$ | $v_{56} = (1,2,3,2)$ | $v_{57} = (5,1,-3,1)$ |
| $E_{21}$ | $\{7,16,58,59\}$ | $v_7 = (1,-1,1,-1)$ | $v_{16} = (1,1,0,0)$ | $v_{58} = (1,-1,-5,-3)$ | $v_{59} = (2,-2,-1,3)$ |
| $E_{22}$ | $\{8,16,60,61\}$ | $v_8 = (1,-1,-1,1)$ | $v_{16} = (1,1,0,0)$ | $v_{60} = (1,-1,-3,-5)$ | $v_{61} = (2,-2,3,-1)$ |
| $E_{23}$ | $\{5,17,62,63\}$ | $v_5 = (1,1,1,1)$ | $v_{17} = (0,0,1,-1)$ | $v_{62} = (1,3,-2,-2)$ | $v_{63} = (5,-3,-1,-1)$ |
| $E_{24}$ | $\{6,17,64,65\}$ | $v_6 = (1,1,-1,-1)$ | $v_{17} = (0,0,1,-1)$ | $v_{64} = (3,-5,-1,-1)$ | $v_{65} = (3,1,2,2)$ |

### *S1.15.2. The non-linear warm-up 26-vertex, 16-context apex suspension $H_{\mathrm{YO}}^{(4)}$*

For completeness, the 26 unnormalized integer vectors $v_1, \dots, v_{26} \in \mathbb{Z}^4$ of the non-linear warm-up hypergraph $H_{\mathrm{YO}}^{(4)} = (V_{26}, E_{16}^{(4)})$ in Proposition 20 (written in canonical primitive form satisfying Eq. (S4)) are:

$$\begin{aligned}
v_1 &= (0,1,1,0), & v_2 &= (1,0,0,0),\\
v_3 &= (0,1,-1,0), & v_4 &= (1,1,0,0),\\
v_5 &= (0,0,1,0), & v_6 &= (1,-1,0,0),\\
v_7 &= (1,0,1,0), & v_8 &= (0,1,0,0),\\
v_9 &= (1,0,-1,0), & v_{10} &= (1,-1,1,0),\\
v_{11} &= (1,-1,-1,0), & v_{12} &= (1,1,1,0),\\
v_{13} &= (1,1,-1,0), & v_{14} &= (2,1,1,0),\\
v_{15} &= (1,2,-1,0), & v_{16} &= (1,-2,1,0),\\
v_{17} &= (1,1,-2,0), & v_{18} &= (1,-1,-2,0),\\
v_{19} &= (2,1,-1,0), & v_{20} &= (2,-1,1,0),\\
v_{21} &= (1,-2,-1,0), & v_{22} &= (1,1,2,0),\\
v_{23} &= (1,2,1,0), & v_{24} &= (1,-1,2,0),\\
v_{25} &= (2,-1,-1,0), & v_{26} &= (0,0,0,1).
\end{aligned} \tag{S5}$$

These 26 canonical primitive vectors partition by squared norm $\|v_i\|^2$ into four disjoint norm shells ($4+6+4+12 = 26$):

- $\|v_i\|^2 = 1$ (4 vectors): $\{v_2, v_5, v_8, v_{26}\}$, the four coordinate basis vectors of $\mathbb{R}^4$;
- $\|v_i\|^2 = 2$ (6 vectors): $\{v_1, v_3, v_4, v_6, v_7, v_9\}$, the six binary rays in $\mathbb{R}^3 \times \{0\}$;
- $\|v_i\|^2 = 3$ (4 vectors): $\{v_{10}, v_{11}, v_{12}, v_{13}\}$, the four triad rays $(1, \pm 1, \pm 1, 0)^T$;
- $\|v_i\|^2 = 6$ (12 vectors): $\{v_{14}, \dots, v_{25}\}$, all $3 \times 4 = 12$ canonical primitive permutations and sign patterns of $(\pm 2, \pm 1, \pm 1, 0)^T$.

Since all 26 canonical 4-tuples are distinct, all $\binom{26}{2} = 325$ pairs of vectors are strictly non-collinear, with maximum squared cosine $\max_{1 \le i < j \le 26} (v_i \cdot v_j)^2 / (\|v_i\|^2 \|v_j\|^2) = 4^2/(3 \times 6) = 8/9 \approx 0.8889 < 1$. Table S19 lists all 16 four-element contexts of $E_{16}^{(4)}$ side by side with their four mutually orthogonal integer vectors.

TABLE S19: Complete enumeration of the 16 four-element contexts of the non-linear warm-up apex suspension $H_{\mathrm{YO}}^{(4)} = (V_{26}, E_{16}^{(4)})$, listing for each context both its vertex indices $\{i, j, k, 26\}$ and the four mutually orthogonal unnormalized integer vectors $\{v_i, v_j, v_k, v_{26}\} \subset \mathbb{Z}^4$.

| Context | Vertex set | $v_i$ | $v_j$ | $v_k$ | $v_{26}$ |
|---|---|---|---|---|---|
| $e_1^{(4)}$ | $\{1,2,3,26\}$ | $v_1 = (0,1,1,0)$ | $v_2 = (1,0,0,0)$ | $v_3 = (0,1,-1,0)$ | $v_{26} = (0,0,0,1)$ |
| $e_2^{(4)}$ | $\{7,8,9,26\}$ | $v_7 = (1,0,1,0)$ | $v_8 = (0,1,0,0)$ | $v_9 = (1,0,-1,0)$ | $v_{26} = (0,0,0,1)$ |
| $e_3^{(4)}$ | $\{4,5,6,26\}$ | $v_4 = (1,1,0,0)$ | $v_5 = (0,0,1,0)$ | $v_6 = (1,-1,0,0)$ | $v_{26} = (0,0,0,1)$ |
| $e_4^{(4)}$ | $\{2,5,8,26\}$ | $v_2 = (1,0,0,0)$ | $v_5 = (0,0,1,0)$ | $v_8 = (0,1,0,0)$ | $v_{26} = (0,0,0,1)$ |
| $e_5^{(4)}$ | $\{3,12,25,26\}$ | $v_3 = (0,1,-1,0)$ | $v_{12} = (1,1,1,0)$ | $v_{25} = (2,-1,-1,0)$ | $v_{26} = (0,0,0,1)$ |
| $e_6^{(4)}$ | $\{9,12,16,26\}$ | $v_9 = (1,0,-1,0)$ | $v_{12} = (1,1,1,0)$ | $v_{16} = (1,-2,1,0)$ | $v_{26} = (0,0,0,1)$ |
| $e_7^{(4)}$ | $\{6,12,17,26\}$ | $v_6 = (1,-1,0,0)$ | $v_{12} = (1,1,1,0)$ | $v_{17} = (1,1,-2,0)$ | $v_{26} = (0,0,0,1)$ |
| $e_8^{(4)}$ | $\{3,11,14,26\}$ | $v_3 = (0,1,-1,0)$ | $v_{11} = (1,-1,-1,0)$ | $v_{14} = (2,1,1,0)$ | $v_{26} = (0,0,0,1)$ |
| $e_9^{(4)}$ | $\{7,11,15,26\}$ | $v_7 = (1,0,1,0)$ | $v_{11} = (1,-1,-1,0)$ | $v_{15} = (1,2,-1,0)$ | $v_{26} = (0,0,0,1)$ |
| $e_{10}^{(4)}$ | $\{4,11,24,26\}$ | $v_4 = (1,1,0,0)$ | $v_{11} = (1,-1,-1,0)$ | $v_{24} = (1,-1,2,0)$ | $v_{26} = (0,0,0,1)$ |
| $e_{11}^{(4)}$ | $\{9,10,23,26\}$ | $v_9 = (1,0,-1,0)$ | $v_{10} = (1,-1,1,0)$ | $v_{23} = (1,2,1,0)$ | $v_{26} = (0,0,0,1)$ |
| $e_{12}^{(4)}$ | $\{1,10,19,26\}$ | $v_1 = (0,1,1,0)$ | $v_{10} = (1,-1,1,0)$ | $v_{19} = (2,1,-1,0)$ | $v_{26} = (0,0,0,1)$ |
| $e_{13}^{(4)}$ | $\{4,10,18,26\}$ | $v_4 = (1,1,0,0)$ | $v_{10} = (1,-1,1,0)$ | $v_{18} = (1,-1,-2,0)$ | $v_{26} = (0,0,0,1)$ |
| $e_{14}^{(4)}$ | $\{1,13,20,26\}$ | $v_1 = (0,1,1,0)$ | $v_{13} = (1,1,-1,0)$ | $v_{20} = (2,-1,1,0)$ | $v_{26} = (0,0,0,1)$ |
| $e_{15}^{(4)}$ | $\{7,13,21,26\}$ | $v_7 = (1,0,1,0)$ | $v_{13} = (1,1,-1,0)$ | $v_{21} = (1,-2,-1,0)$ | $v_{26} = (0,0,0,1)$ |
| $e_{16}^{(4)}$ | $\{6,13,22,26\}$ | $v_6 = (1,-1,0,0)$ | $v_{13} = (1,1,-1,0)$ | $v_{22} = (1,1,2,0)$ | $v_{26} = (0,0,0,1)$ |

## S1.16. Explicit parent configurations and inclusion maps

### *S1.16.1. The 49-ray real Yu–Oh parent and published Bub labels*

Table S20 lists all rays, including the original 25. The final column gives the exact labels in the published full 49-ray encoding[1] (p. 14, Sec. 5(xi)) of Bub's set[2]. Under this map the 36 bases in Table S21 equal all 36 published contexts; their 108 pairs equal all geometric orthogonalities. In this labeling 26–33 are Eq. (S31), and 34–49 complete the sixteen remaining orthogonal pairs. The first 25 vectors are those of Table S1.

TABLE S20: All unnormalized vectors of the real 49-ray parent, with the verified published Bub labels.

| $i$ | $v_i$ | Published label |
|---|---|---|
| 1 | $(0,1,1)$ | `G` |
| 2 | $(1,0,0)$ | `1` |
| 3 | $(0,1,-1)$ | `P` |
| 4 | $(1,1,0)$ | `H` |
| 5 | $(0,0,1)$ | `3` |
| 6 | $(1,-1,0)$ | `U` |
| 7 | $(1,0,1)$ | `Q` |
| 8 | $(0,1,0)$ | `2` |
| 9 | $(1,0,-1)$ | `e` |
| 10 | $(1,-1,1)$ | `D` |
| 11 | $(1,-1,-1)$ | `h` |
| 12 | $(1,1,1)$ | `S` |
| 13 | $(1,1,-1)$ | `a` |
| 14 | $(2,1,1)$ | `g` |
| 15 | $(1,2,-1)$ | `n` |
| 16 | $(1,-2,1)$ | `d` |
| 17 | $(1,1,-2)$ | `T` |
| 18 | $(1,-1,-2)$ | `F` |
| 19 | $(2,1,-1)$ | `E` |
| 20 | $(2,-1,1)$ | `I` |
| 21 | $(1,-2,-1)$ | `k` |
| 22 | $(1,1,2)$ | `b` |
| 23 | $(1,2,1)$ | `A` |
| 24 | $(1,-1,2)$ | `N` |
| 25 | $(2,-1,-1)$ | `R` |
| 26 | $(0,1,2)$ | `i` |
| 27 | $(0,2,-1)$ | `O` |
| 28 | $(0,1,-2)$ | `6` |
| 29 | $(0,2,1)$ | `7` |
| 30 | $(1,2,0)$ | `5` |
| 31 | $(2,-1,0)$ | `4` |
| 32 | $(1,-2,0)$ | `V` |
| 33 | $(2,1,0)$ | `j` |
| 34 | $(2,1,-5)$ | `Z` |
| 35 | $(5,-2,1)$ | `m` |
| 36 | $(1,2,5)$ | `W` |
| 37 | $(5,2,-1)$ | `c` |
| 38 | $(1,-2,-5)$ | `f` |
| 39 | $(5,-1,2)$ | `L` |
| 40 | $(5,1,2)$ | `9` |
| 41 | $(2,1,5)$ | `8` |
| 42 | $(2,-1,-5)$ | `J` |
| 43 | $(5,2,1)$ | `X` |
| 44 | $(1,-2,5)$ | `l` |
| 45 | $(5,-1,-2)$ | `Y` |
| 46 | $(5,-2,-1)$ | `B` |
| 47 | $(1,2,-5)$ | `C` |
| 48 | $(5,1,-2)$ | `M` |
| 49 | $(2,-1,5)$ | `K` |

TABLE S21: All 36 complete bases of the real parent.

| $e$ | Context | $e$ | Context | $e$ | Context |
|---|---|---|---|---|---|
| 1 | $\{1,2,3\}$ | 2 | $\{1,10,19\}$ | 3 | $\{1,13,20\}$ |
| 4 | $\{2,5,8\}$ | 5 | $\{2,26,27\}$ | 6 | $\{2,28,29\}$ |
| 7 | $\{3,11,14\}$ | 8 | $\{3,12,25\}$ | 9 | $\{4,5,6\}$ |
| 10 | $\{4,10,18\}$ | 11 | $\{4,11,24\}$ | 12 | $\{5,30,31\}$ |
| 13 | $\{5,32,33\}$ | 14 | $\{6,12,17\}$ | 15 | $\{6,13,22\}$ |
| 16 | $\{7,8,9\}$ | 17 | $\{7,11,15\}$ | 18 | $\{7,13,21\}$ |
| 19 | $\{9,10,23\}$ | 20 | $\{9,12,16\}$ | 21 | $\{14,32,34\}$ |
| 22 | $\{15,26,35\}$ | 23 | $\{15,31,36\}$ | 24 | $\{16,26,37\}$ |
| 25 | $\{16,33,38\}$ | 26 | $\{17,29,39\}$ | 27 | $\{18,27,40\}$ |
| 28 | $\{19,32,41\}$ | 29 | $\{20,30,42\}$ | 30 | $\{21,28,43\}$ |
| 31 | $\{21,33,44\}$ | 32 | $\{22,27,45\}$ | 33 | $\{23,28,46\}$ |
| 34 | $\{23,31,47\}$ | 35 | $\{24,29,48\}$ | 36 | $\{25,30,49\}$ |

### *S1.16.2. The 87-ray Hesse parent*

Use $\zeta = \exp(2\pi i/3)$ and $\zeta^2 + \zeta + 1 = 0$. The first 37 rays retain exactly the Hesse numbering; projective representatives have their first nonzero coordinate one. The inventory labels use the generation convention of Sec. VII B. Tables S22 and S23 list all 87 rays and all 64 complete bases. The six additional orthogonal pairs outside the 64 complete bases are listed separately: this larger parent's full orthogonality graph need not equal its context two-section. The retained Hesse restriction has all its original orthogonalities covered by contexts.

TABLE S22: All 87 unnormalized rays of the first Hesse/lineage union.

| $i$ | $v_i$ | Inventory label |
|---|---|---|
| 1 | $(1,0,0)$ | $a_{11}$ |
| 2 | $(1,1,1)$ | $u_1$ |
| 3 | $(1,\zeta,-1-\zeta)$ | $u_2$ |
| 4 | $(1,-1-\zeta,\zeta)$ | $u_3$ |
| 5 | $(1,\zeta,\zeta)$ | $u_4$ |
| 6 | $(1,-1-\zeta,1)$ | $u_5$ |
| 7 | $(1,1,-1-\zeta)$ | $u_6$ |
| 8 | $(1,-1-\zeta,-1-\zeta)$ | $u_7$ |
| 9 | $(1,1,\zeta)$ | $u_9$ |
| 10 | $(1,\zeta,1)$ | $u_8$ |
| 11 | $(0,1,-1)$ | $c_{11}$ |
| 12 | $(0,1,-\zeta)$ | $c_{12}$ |
| 13 | $(0,1,1+\zeta)$ | $c_{13}$ |
| 14 | $(1,0,-1)$ | $c_{21}$ |
| 15 | $(1,0,1+\zeta)$ | $c_{22}$ |
| 16 | $(1,0,-\zeta)$ | $c_{23}$ |
| 17 | $(0,1,1)$ | $b_{11}$ |
| 18 | $(0,1,\zeta)$ | $b_{12}$ |
| 19 | $(0,1,-1-\zeta)$ | $b_{13}$ |
| 20 | $(1,\frac{-1}{2},\frac{-1}{2})$ | $d_{11}$ |
| 21 | $(1,-\frac{1}{2}\zeta,\frac{1}{2}+\frac{1}{2}\zeta)$ | $d_{12}$ |
| 22 | $(1,\frac{1}{2}+\frac{1}{2}\zeta,-\frac{1}{2}\zeta)$ | $d_{13}$ |
| 23 | $(1,-\frac{1}{2}\zeta,-\frac{1}{2}\zeta)$ | $d_{14}$ |
| 24 | $(1,\frac{1}{2}+\frac{1}{2}\zeta,\frac{-1}{2})$ | $d_{15}$ |
| 25 | $(1,\frac{-1}{2},\frac{1}{2}+\frac{1}{2}\zeta)$ | $d_{16}$ |
| 26 | $(1,\frac{1}{2}+\frac{1}{2}\zeta,\frac{1}{2}+\frac{1}{2}\zeta)$ | $d_{17}$ |
| 27 | $(1,\frac{-1}{2},-\frac{1}{2}\zeta)$ | $d_{19}$ |
| 28 | $(1,-\frac{1}{2}\zeta,\frac{-1}{2})$ | $d_{18}$ |
| 29 | $(1,-2,1)$ | $d_{21}$ |
| 30 | $(1,-2\zeta,-1-\zeta)$ | $d_{22}$ |
| 31 | $(1,2+2\zeta,\zeta)$ | $d_{23}$ |
| 32 | $(1,2+2\zeta,1)$ | $d_{25}$ |
| 33 | $(1,-2,-1-\zeta)$ | $d_{26}$ |

| $i$ | $v_i$ | Inventory label |
|---|---|---|
| 34 | $(1, -2\zeta, \zeta)$ | $d_{24}$ |
| 35 | $(1, -2\zeta, 1)$ | $d_{28}$ |
| 36 | $(1, 2+2\zeta, -1-\zeta)$ | $d_{27}$ |
| 37 | $(1, -2, \zeta)$ | $d_{29}$ |
| 38 | $(0, 1, 0)$ | $a_{21}$ |
| 39 | $(0, 0, 1)$ | $a_{31}$ |
| 40 | $(1, \frac{-1}{2}, \frac{1}{2})$ | $b_{1121}$ |
| 41 | $(1, -\frac{1}{2}\zeta, \frac{-1}{2} - \frac{1}{2}\zeta)$ | $b_{1122}$ |
| 42 | $(1, \frac{1}{2} + \frac{1}{2}\zeta, \frac{1}{2}\zeta)$ | $b_{1123}$ |
| 43 | $(1, \frac{1}{2}, \frac{-1}{2})$ | $b_{1131}$ |
| 44 | $(1, \frac{1}{2}\zeta, \frac{1}{2} + \frac{1}{2}\zeta)$ | $b_{1132}$ |
| 45 | $(1, \frac{-1}{2} - \frac{1}{2}\zeta, -\frac{1}{2}\zeta)$ | $b_{1133}$ |
| 46 | $(1, 1, -1)$ | $b_{121}$ |
| 47 | $(1, \zeta, 1+\zeta)$ | $b_{122}$ |
| 48 | $(1, -1-\zeta, -\zeta)$ | $b_{123}$ |
| 49 | $(1, -1, 1)$ | $b_{131}$ |
| 50 | $(1, -\zeta, -1-\zeta)$ | $b_{132}$ |
| 51 | $(1, 1+\zeta, \zeta)$ | $b_{133}$ |
| 52 | $(1, 0, 1)$ | $b_{21}$ |
| 53 | $(1, -2, -1)$ | $b_{2121}$ |
| 54 | $(1, -2\zeta, 1+\zeta)$ | $b_{2122}$ |
| 55 | $(1, 2+2\zeta, -\zeta)$ | $b_{2123}$ |
| 56 | $(1, 0, -1-\zeta)$ | $b_{22}$ |
| 57 | $(1, 2, -1)$ | $b_{2231}$ |
| 58 | $(1, 2\zeta, 1+\zeta)$ | $b_{2232}$ |
| 59 | $(1, -2-2\zeta, -\zeta)$ | $b_{2233}$ |
| 60 | $(1, 0, \zeta)$ | $b_{23}$ |
| 61 | $(1, -1, -1)$ | $b_{231}$ |
| 62 | $(1, -\zeta, 1+\zeta)$ | $b_{232}$ |
| 63 | $(1, 1+\zeta, -\zeta)$ | $b_{233}$ |
| 64 | $(1, 1, 0)$ | $b_{31}$ |
| 65 | $(1, -1, -2)$ | $b_{3131}$ |
| 66 | $(1, -\zeta, 2+2\zeta)$ | $b_{3132}$ |
| 67 | $(1, 1+\zeta, -2\zeta)$ | $b_{3133}$ |
| 68 | $(1, \zeta, 0)$ | $b_{32}$ |
| 69 | $(1, -1, 2)$ | $b_{3231}$ |
| 70 | $(1, -\zeta, -2-2\zeta)$ | $b_{3232}$ |
| 71 | $(1, 1+\zeta, 2\zeta)$ | $b_{3233}$ |
| 72 | $(1, -1-\zeta, 0)$ | $b_{33}$ |
| 73 | $(1, -1, 0)$ | $c_{31}$ |
| 74 | $(1, -\zeta, 0)$ | $c_{32}$ |
| 75 | $(1, 1+\zeta, 0)$ | $c_{33}$ |
| 76 | $(1, 1, -2)$ | $d_{31}$ |
| 77 | $(1, \zeta, 2+2\zeta)$ | $d_{32}$ |
| 78 | $(1, -1-\zeta, -2\zeta)$ | $d_{33}$ |
| 79 | $(1, 2, 1)$ | $e_{11}$ |
| 80 | $(1, 2\zeta, -1-\zeta)$ | $e_{12}$ |
| 81 | $(1, -2-2\zeta, \zeta)$ | $e_{13}$ |
| 82 | $(1, 1, 2)$ | $e_{21}$ |
| 83 | $(1, \zeta, -2-2\zeta)$ | $e_{22}$ |
| 84 | $(1, -1-\zeta, 2\zeta)$ | $e_{23}$ |
| 85 | $(1, \frac{1}{2}, \frac{1}{2})$ | $e_{31}$ |
| 86 | $(1, \frac{1}{2}\zeta, \frac{-1}{2} - \frac{1}{2}\zeta)$ | $e_{32}$ |
| 87 | $(1, \frac{-1}{2} - \frac{1}{2}\zeta, \frac{1}{2}\zeta)$ | $e_{33}$ |

TABLE S23: All 64 complete bases of the first 87-ray Hesse parent.

| $e$ | Context | $e$ | Context | $e$ | Context |
|---|---|---|---|---|---|
| 1 | $\{1,11,17\}$ | 2 | $\{1,12,18\}$ | 3 | $\{1,13,19\}$ |
| 4 | $\{1,38,39\}$ | 5 | $\{2,3,4\}$ | 6 | $\{2,11,20\}$ |
| 7 | $\{2,14,29\}$ | 8 | $\{2,73,76\}$ | 9 | $\{3,12,21\}$ |
| 10 | $\{3,15,30\}$ | 11 | $\{3,74,77\}$ | 12 | $\{4,13,22\}$ |
| 13 | $\{4,16,31\}$ | 14 | $\{4,75,78\}$ | 15 | $\{5,6,7\}$ |
| 16 | $\{5,11,23\}$ | 17 | $\{5,16,34\}$ | 18 | $\{6,12,24\}$ |
| 19 | $\{6,14,32\}$ | 20 | $\{7,13,25\}$ | 21 | $\{7,15,33\}$ |
| 22 | $\{8,9,10\}$ | 23 | $\{8,11,26\}$ | 24 | $\{8,15,36\}$ |
| 25 | $\{9,12,27\}$ | 26 | $\{9,16,37\}$ | 27 | $\{10,13,28\}$ |
| 28 | $\{10,14,35\}$ | 29 | $\{11,61,85\}$ | 30 | $\{12,62,86\}$ |
| 31 | $\{13,63,87\}$ | 32 | $\{14,38,52\}$ | 33 | $\{14,49,79\}$ |
| 34 | $\{15,38,56\}$ | 35 | $\{15,50,80\}$ | 36 | $\{16,38,60\}$ |
| 37 | $\{16,51,81\}$ | 38 | $\{17,40,46\}$ | 39 | $\{17,43,49\}$ |
| 40 | $\{18,41,47\}$ | 41 | $\{18,44,50\}$ | 42 | $\{19,42,48\}$ |
| 43 | $\{19,45,51\}$ | 44 | $\{39,64,73\}$ | 45 | $\{39,68,74\}$ |
| 46 | $\{39,72,75\}$ | 47 | $\{46,47,48\}$ | 48 | $\{46,52,53\}$ |
| 49 | $\{46,73,82\}$ | 50 | $\{47,54,56\}$ | 51 | $\{47,74,83\}$ |
| 52 | $\{48,55,60\}$ | 53 | $\{48,75,84\}$ | 54 | $\{49,50,51\}$ |
| 55 | $\{49,64,65\}$ | 56 | $\{50,66,68\}$ | 57 | $\{51,67,72\}$ |
| 58 | $\{52,57,61\}$ | 59 | $\{56,58,62\}$ | 60 | $\{59,60,63\}$ |
| 61 | $\{61,62,63\}$ | 62 | $\{61,64,69\}$ | 63 | $\{62,68,70\}$ |
| 64 | $\{63,71,72\}$ | | | | |

Its six further orthogonal pairs are $\{5,74\}$, $\{6,75\}$, $\{7,73\}$, $\{8,75\}$, $\{9,73\}$, $\{10,74\}$.

### *S1.16.3. Peres24 and the complete four-dimensional extensions*

The coordinates of $H_{65}$ and $H_{\mathrm{YO}}^{(4)}$ are those already listed above. Table S24 explicitly enumerates the known Peres rays. In each union, retain the original local labels and append the missing rays in the order displayed in Table S25. Tables S26 and S27 list every available complete basis. Full orthogonality-pair lists accompany the coordinates in the data; the 40-ray union has additional pairs outside its complete contexts.

TABLE S24: All 24 primitive unnormalized Peres rays, in the certificate order.

| $i$ | $p_i$ |
|---|---|
| 1 | $(1,0,0,0)$ |
| 2 | $(0,1,0,0)$ |
| 3 | $(0,0,1,0)$ |
| 4 | $(0,0,0,1)$ |
| 5 | $(1,-1,0,0)$ |
| 6 | $(1,1,0,0)$ |
| 7 | $(1,0,-1,0)$ |
| 8 | $(1,0,1,0)$ |
| 9 | $(1,0,0,-1)$ |
| 10 | $(1,0,0,1)$ |
| 11 | $(0,1,-1,0)$ |
| 12 | $(0,1,1,0)$ |
| 13 | $(0,1,0,-1)$ |
| 14 | $(0,1,0,1)$ |
| 15 | $(0,0,1,-1)$ |
| 16 | $(0,0,1,1)$ |

| $i$ | $p_i$ |
|---|---|
| 17 | $(1,-1,-1,-1)$ |
| 18 | $(1,-1,-1,1)$ |
| 19 | $(1,-1,1,-1)$ |
| 20 | $(1,-1,1,1)$ |
| 21 | $(1,1,-1,-1)$ |
| 22 | $(1,1,-1,1)$ |
| 23 | $(1,1,1,-1)$ |
| 24 | $(1,1,1,1)$ |

TABLE S25: Every ray added to the original four-dimensional examples to obtain the stated parents.

| Original | New label | Added vector |
|---|---|---|
| $H_{\mathrm{YO}}^{(4)}$ | 27 | $(1,0,0,-1)$ |
| $H_{\mathrm{YO}}^{(4)}$ | 28 | $(1,0,0,1)$ |
| $H_{\mathrm{YO}}^{(4)}$ | 29 | $(0,1,0,-1)$ |
| $H_{\mathrm{YO}}^{(4)}$ | 30 | $(0,1,0,1)$ |
| $H_{\mathrm{YO}}^{(4)}$ | 31 | $(0,0,1,-1)$ |
| $H_{\mathrm{YO}}^{(4)}$ | 32 | $(0,0,1,1)$ |
| $H_{\mathrm{YO}}^{(4)}$ | 33 | $(1,-1,-1,-1)$ |
| $H_{\mathrm{YO}}^{(4)}$ | 34 | $(1,-1,-1,1)$ |
| $H_{\mathrm{YO}}^{(4)}$ | 35 | $(1,-1,1,-1)$ |
| $H_{\mathrm{YO}}^{(4)}$ | 36 | $(1,-1,1,1)$ |
| $H_{\mathrm{YO}}^{(4)}$ | 37 | $(1,1,-1,-1)$ |
| $H_{\mathrm{YO}}^{(4)}$ | 38 | $(1,1,-1,1)$ |
| $H_{\mathrm{YO}}^{(4)}$ | 39 | $(1,1,1,-1)$ |
| $H_{\mathrm{YO}}^{(4)}$ | 40 | $(1,1,1,1)$ |
| $H_{65}$ | 66 | $(1,-1,0,0)$ |
| $H_{65}$ | 67 | $(1,0,-1,0)$ |
| $H_{65}$ | 68 | $(1,0,0,1)$ |
| $H_{65}$ | 69 | $(0,1,-1,0)$ |
| $H_{65}$ | 70 | $(0,1,0,1)$ |
| $H_{65}$ | 71 | $(0,0,1,1)$ |
| $H_{65}$ | 72 | $(1,1,1,-1)$ |

TABLE S26: All 36 complete bases of the 40-ray parent.

| $e$ | Context | $e$ | Context |
|---|---|---|---|
| 1 | $\{1,2,3,26\}$ | 2 | $\{1,3,27,28\}$ |
| 3 | $\{1,10,19,26\}$ | 4 | $\{1,13,20,26\}$ |
| 5 | $\{1,27,36,38\}$ | 6 | $\{1,28,35,37\}$ |
| 7 | $\{2,5,8,26\}$ | 8 | $\{2,5,29,30\}$ |
| 9 | $\{2,8,31,32\}$ | 10 | $\{3,11,14,26\}$ |
| 11 | $\{3,12,25,26\}$ | 12 | $\{3,27,34,40\}$ |
| 13 | $\{3,28,33,39\}$ | 14 | $\{4,5,6,26\}$ |
| 15 | $\{4,6,31,32\}$ | 16 | $\{4,10,18,26\}$ |
| 17 | $\{4,11,24,26\}$ | 18 | $\{4,31,33,36\}$ |
| 19 | $\{4,32,34,35\}$ | 20 | $\{5,8,27,28\}$ |
| 21 | $\{6,12,17,26\}$ | 22 | $\{6,13,22,26\}$ |
| 23 | $\{6,31,37,40\}$ | 24 | $\{6,32,38,39\}$ |
| 25 | $\{7,8,9,26\}$ | 26 | $\{7,9,29,30\}$ |
| 27 | $\{7,11,15,26\}$ | 28 | $\{7,13,21,26\}$ |
| 29 | $\{7,29,33,38\}$ | 30 | $\{7,30,34,37\}$ |
| 31 | $\{9,10,23,26\}$ | 32 | $\{9,12,16,26\}$ |
| 33 | $\{9,29,35,40\}$ | 34 | $\{9,30,36,39\}$ |
| 35 | $\{33,36,38,39\}$ | 36 | $\{34,35,37,40\}$ |

TABLE S27: All 48 complete bases of the 72-ray parent.

| $e$ | Context | $e$ | Context |
|---|---|---|---|
| 1 | $\{1,2,3,4\}$ | 2 | $\{1,2,17,71\}$ |
| 3 | $\{1,3,15,70\}$ | 4 | $\{1,4,13,69\}$ |
| 5 | $\{1,13,22,23\}$ | 6 | $\{1,15,30,31\}$ |
| 7 | $\{1,17,38,39\}$ | 8 | $\{2,3,12,68\}$ |
| 9 | $\{2,4,14,67\}$ | 10 | $\{2,12,18,19\}$ |
| 11 | $\{2,14,26,27\}$ | 12 | $\{2,17,40,41\}$ |
| 13 | $\{3,4,16,66\}$ | 14 | $\{3,12,20,21\}$ |
| 15 | $\{3,15,32,33\}$ | 16 | $\{3,16,34,35\}$ |
| 17 | $\{4,13,24,25\}$ | 18 | $\{4,14,28,29\}$ |
| 19 | $\{4,16,36,37\}$ | 20 | $\{5,6,7,8\}$ |
| 21 | $\{5,6,17,66\}$ | 22 | $\{5,7,15,67\}$ |
| 23 | $\{5,8,12,69\}$ | 24 | $\{5,12,42,43\}$ |
| 25 | $\{5,15,54,55\}$ | 26 | $\{5,17,62,63\}$ |
| 27 | $\{6,7,13,68\}$ | 28 | $\{6,8,14,70\}$ |
| 29 | $\{6,13,46,47\}$ | 30 | $\{6,14,50,51\}$ |
| 31 | $\{6,17,64,65\}$ | 32 | $\{7,8,16,71\}$ |
| 33 | $\{7,13,48,49\}$ | 34 | $\{7,15,56,57\}$ |
| 35 | $\{7,16,58,59\}$ | 36 | $\{8,12,44,45\}$ |
| 37 | $\{8,14,52,53\}$ | 38 | $\{8,16,60,61\}$ |
| 39 | $\{9,10,11,72\}$ | 40 | $\{9,10,12,13\}$ |
| 41 | $\{9,11,14,15\}$ | 42 | $\{9,66,71,72\}$ |
| 43 | $\{10,11,16,17\}$ | 44 | $\{10,67,70,72\}$ |
| 45 | $\{11,68,69,72\}$ | 46 | $\{12,13,68,69\}$ |
| 47 | $\{14,15,67,70\}$ | 48 | $\{16,17,66,71\}$ |

## S1.17. Cabello completions and the two 68-ray parents

### *S1.17.1. The eighteen-ray input and faithful 36-ray completion*

All labels in this subsection are local. In $C_{18,9}$ and $C_{36,18}$, $18 = a$, $19 = b$, and $20 = c$; labels 4 and 5 are absent. They do not denote the completion rays of $H_{65}$. The core is the first eighteen rows of Table S28; labels 21–38 are fresh. The contexts of $C_{18,9}$ are the first nine rows of Table S29.

TABLE S28: All 36 unnormalized vectors of the faithful Cabello completion.

| $i$ | $v_i$ |
|---|---|
| 1 | $(1,0,0,0)$ |
| 2 | $(0,1,0,0)$ |
| 3 | $(0,0,1,0)$ |
| 6 | $(1,1,-1,-1)$ |
| 7 | $(1,-1,1,-1)$ |
| 8 | $(1,-1,-1,1)$ |
| 9 | $(1,1,-1,1)$ |
| 10 | $(1,-1,1,1)$ |
| 11 | $(1,-1,-1,-1)$ |
| 12 | $(1,0,0,-1)$ |
| 13 | $(0,1,1,0)$ |
| 14 | $(1,0,1,0)$ |
| 15 | $(0,1,0,-1)$ |
| 16 | $(1,1,0,0)$ |
| 17 | $(0,0,1,-1)$ |
| 18 | $(1,0,0,1)$ |
| 19 | $(0,1,0,1)$ |
| 20 | $(0,0,1,1)$ |
| 21 | $(0,1,-1,3)$ |
| 22 | $(0,3,-3,-2)$ |
| 23 | $(1,0,-1,3)$ |
| 24 | $(3,0,-3,-2)$ |
| 25 | $(1,-1,0,3)$ |
| 26 | $(3,-3,0,-2)$ |
| 27 | $(3,1,2,2)$ |
| 28 | $(3,-5,-1,-1)$ |
| 29 | $(3,2,1,2)$ |
| 30 | $(3,-1,-5,-1)$ |
| 31 | $(2,3,1,2)$ |
| 32 | $(1,-3,5,1)$ |
| 33 | $(4,2,3,-3)$ |
| 34 | $(5,-7,-1,1)$ |
| 35 | $(4,3,2,-3)$ |
| 36 | $(5,-1,-7,1)$ |
| 37 | $(3,4,2,-3)$ |
| 38 | $(1,-5,7,-1)$ |

TABLE S29: All eighteen full bases of the faithful Cabello completion.

| $e$ | Context |
|---|---|
| 1 | $\{9, 10, 12, 13\}$ |
| 2 | $\{9, 11, 14, 15\}$ |
| 3 | $\{10, 11, 16, 17\}$ |
| 4 | $\{2, 3, 12, 18\}$ |
| 5 | $\{1, 3, 15, 19\}$ |
| 6 | $\{1, 2, 17, 20\}$ |
| 7 | $\{6, 7, 13, 18\}$ |
| 8 | $\{6, 8, 14, 19\}$ |
| 9 | $\{7, 8, 16, 20\}$ |
| 10 | $\{1, 13, 21, 22\}$ |
| 11 | $\{2, 14, 23, 24\}$ |
| 12 | $\{3, 16, 25, 26\}$ |
| 13 | $\{6, 17, 27, 28\}$ |
| 14 | $\{7, 15, 29, 30\}$ |
| 15 | $\{8, 12, 31, 32\}$ |
| 16 | $\{9, 20, 33, 34\}$ |
| 17 | $\{10, 19, 35, 36\}$ |
| 18 | $\{11, 18, 37, 38\}$ |

The orthogonal matrix $Q = M/\sqrt{2}$, with

$$M = \begin{pmatrix} 0 & 1 & 0 & -1 \\ -1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \qquad M^T M = 2I_4, \tag{S6}$$

maps the published eighteen-ray set and all nine bases of Ref.[3] (Eqs. (1)–(9)) onto this realization. Table S30 gives the complete ray map; the input representatives are unnormalized and signs are immaterial.

TABLE S30: Exact ray map from the originally published Cabello coordinates.

| Published vector $u$ | Local label of $[Qu]$ |
|---|---|
| $(0, 0, 0, 1)$ | 12 |
| $(0, 0, 1, 0)$ | 13 |
| $(0, 0, 1, 1)$ | 11 |
| $(0, 1, -1, 0)$ | 8 |
| $(0, 1, 0, -1)$ | 1 |
| $(0, 1, 0, 0)$ | 18 |
| $(1, -1, -1, -1)$ | 19 |
| $(1, -1, -1, 1)$ | 16 |
| $(1, -1, 0, 0)$ | 9 |
| $(1, -1, 1, -1)$ | 17 |
| $(1, 0, -1, 0)$ | 2 |
| $(1, 0, 0, -1)$ | 7 |
| $(1, 0, 0, 1)$ | 6 |
| $(1, 0, 1, 0)$ | 3 |
| $(1, 1, -1, 1)$ | 15 |
| $(1, 1, 0, 0)$ | 10 |
| $(1, 1, 1, -1)$ | 14 |
| $(1, 1, 1, 1)$ | 20 |

### *S1.17.2. An injective 45-ray completion after deleting $c$*

Delete local ray 20 and contexts 6 and 9 from the first nine contexts above. The seven remaining bases are followed by fourteen fresh-pair contexts in Table S32. The next table prints all 45 vectors; 4, 5, and 20 are absent and 21–48 are fresh. Its only uncovered orthogonal pairs are $\{16, 21\}, \{16, 22\}, \{17, 35\}, \{17, 36\}$.

TABLE S31: All 45 unnormalized vectors of the vertex-deleted completion.

| $i$ | $v_i$ |
|---|---|
| 1 | $(1, 0, 0, 0)$ |
| 2 | $(0, 1, 0, 0)$ |
| 3 | $(0, 0, 1, 0)$ |
| 6 | $(1, 1, -1, -1)$ |
| 7 | $(1, -1, 1, -1)$ |
| 8 | $(1, -1, -1, 1)$ |
| 9 | $(1, 1, -1, 1)$ |
| 10 | $(1, -1, 1, 1)$ |
| 11 | $(1, -1, -1, -1)$ |
| 12 | $(1, 0, 0, -1)$ |
| 13 | $(0, 1, 1, 0)$ |
| 14 | $(1, 0, 1, 0)$ |
| 15 | $(0, 1, 0, -1)$ |
| 16 | $(1, 1, 0, 0)$ |
| 17 | $(0, 0, 1, -1)$ |
| 18 | $(1, 0, 0, 1)$ |
| 19 | $(0, 1, 0, 1)$ |
| 21 | $(0, 0, 1, 2)$ |
| 22 | $(0, 0, 2, -1)$ |
| 23 | $(0, 1, -1, 3)$ |
| 24 | $(0, 3, -3, -2)$ |
| 25 | $(0, 1, 2, 2)$ |
| 26 | $(0, 4, -1, -1)$ |
| 27 | $(1, 0, -1, 3)$ |
| 28 | $(3, 0, -3, -2)$ |
| 29 | $(1, 0, 2, 2)$ |
| 30 | $(4, 0, -1, -1)$ |
| 31 | $(1, -1, 0, 3)$ |
| 32 | $(3, -3, 0, -2)$ |
| 33 | $(5, 3, 4, 4)$ |
| 34 | $(7, -9, -1, -1)$ |
| 35 | $(1, 1, 3, 3)$ |
| 36 | $(3, 3, -1, -1)$ |
| 37 | $(5, 4, 3, 4)$ |
| 38 | $(7, -1, -9, -1)$ |
| 39 | $(2, -2, -1, 3)$ |
| 40 | $(1, -1, -5, -3)$ |
| 41 | $(4, 5, 3, 4)$ |
| 42 | $(1, -7, 9, 1)$ |
| 43 | $(4, -4, -1, -9)$ |
| 44 | $(5, -5, 13, 3)$ |
| 45 | $(6, 5, 4, -5)$ |
| 46 | $(9, -1, -11, 1)$ |
| 47 | $(3, 4, 2, -3)$ |
| 48 | $(1, -5, 7, -1)$ |

TABLE S32: All 21 full bases of the vertex-deleted completion.

| $e$ | Context |
|---|---|
| 1 | $\{9, 10, 12, 13\}$ |
| 2 | $\{9, 11, 14, 15\}$ |
| 3 | $\{10, 11, 16, 17\}$ |
| 4 | $\{2, 3, 12, 18\}$ |
| 5 | $\{1, 3, 15, 19\}$ |
| 6 | $\{6, 7, 13, 18\}$ |
| 7 | $\{6, 8, 14, 19\}$ |
| 8 | $\{1, 2, 21, 22\}$ |
| 9 | $\{1, 13, 23, 24\}$ |
| 10 | $\{1, 17, 25, 26\}$ |
| 11 | $\{2, 14, 27, 28\}$ |
| 12 | $\{2, 17, 29, 30\}$ |
| 13 | $\{3, 16, 31, 32\}$ |
| 14 | $\{6, 17, 33, 34\}$ |
| 15 | $\{7, 8, 35, 36\}$ |
| 16 | $\{7, 15, 37, 38\}$ |
| 17 | $\{7, 16, 39, 40\}$ |
| 18 | $\{8, 12, 41, 42\}$ |
| 19 | $\{8, 16, 43, 44\}$ |
| 20 | $\{10, 19, 45, 46\}$ |
| 21 | $\{11, 18, 47, 48\}$ |

### *S1.17.3. All vectors and bases of the two 68-ray parents*

In each parent retain all 65 vectors of Appendix S1.15.1, with their original labels, and add the three vectors in the corresponding rows of Table S33; Tables S34 and S35 list all 35 bases of each parent. Thus the six rows specify every added ray; the two parent inventories use distinct local meanings for labels 66–68. These labels also differ from the 72-ray parent numbering in Appendix S1.16.

TABLE S33: The three added rays in each local 68-ray parent inventory.

| Parent variant | $i$ | $v_i$ |
|---|---|---|
| 18–9 subsystem | 66 | $(1, 0, 0, 1)$ |
| 18–9 subsystem | 67 | $(0, 1, 0, 1)$ |
| 18–9 subsystem | 68 | $(0, 0, 1, 1)$ |
| 20–11 subsystem | 66 | $(0, 1, -1, 0)$ |
| 20–11 subsystem | 67 | $(1, 0, -1, 0)$ |
| 20–11 subsystem | 68 | $(1, -1, 0, 0)$ |

TABLE S34: All 35 full bases of 68-ray parent variant 1.

| $e$ | Context | $e$ | Context |
|---|---|---|---|
| 1 | $\{1,2,3,4\}$ | 2 | $\{1,2,17,68\}$ |
| 3 | $\{1,3,15,67\}$ | 4 | $\{1,13,22,23\}$ |
| 5 | $\{1,15,30,31\}$ | 6 | $\{1,17,38,39\}$ |
| 7 | $\{2,3,12,66\}$ | 8 | $\{2,12,18,19\}$ |
| 9 | $\{2,14,26,27\}$ | 10 | $\{2,17,40,41\}$ |
| 11 | $\{3,12,20,21\}$ | 12 | $\{3,15,32,33\}$ |
| 13 | $\{3,16,34,35\}$ | 14 | $\{4,13,24,25\}$ |
| 15 | $\{4,14,28,29\}$ | 16 | $\{4,16,36,37\}$ |
| 17 | $\{5,6,7,8\}$ | 18 | $\{5,12,42,43\}$ |
| 19 | $\{5,15,54,55\}$ | 20 | $\{5,17,62,63\}$ |
| 21 | $\{6,7,13,66\}$ | 22 | $\{6,8,14,67\}$ |
| 23 | $\{6,13,46,47\}$ | 24 | $\{6,14,50,51\}$ |
| 25 | $\{6,17,64,65\}$ | 26 | $\{7,8,16,68\}$ |
| 27 | $\{7,13,48,49\}$ | 28 | $\{7,15,56,57\}$ |
| 29 | $\{7,16,58,59\}$ | 30 | $\{8,12,44,45\}$ |
| 31 | $\{8,14,52,53\}$ | 32 | $\{8,16,60,61\}$ |
| 33 | $\{9,10,12,13\}$ | 34 | $\{9,11,14,15\}$ |
| 35 | $\{10,11,16,17\}$ | | |

TABLE S35: All 35 full bases of 68-ray parent variant 2.

| $e$ | Context | $e$ | Context |
|---|---|---|---|
| 1 | $\{1,2,3,4\}$ | 2 | $\{1,4,13,66\}$ |
| 3 | $\{1,13,22,23\}$ | 4 | $\{1,15,30,31\}$ |
| 5 | $\{1,17,38,39\}$ | 6 | $\{2,4,14,67\}$ |
| 7 | $\{2,12,18,19\}$ | 8 | $\{2,14,26,27\}$ |
| 9 | $\{2,17,40,41\}$ | 10 | $\{3,4,16,68\}$ |
| 11 | $\{3,12,20,21\}$ | 12 | $\{3,15,32,33\}$ |
| 13 | $\{3,16,34,35\}$ | 14 | $\{4,13,24,25\}$ |
| 15 | $\{4,14,28,29\}$ | 16 | $\{4,16,36,37\}$ |
| 17 | $\{5,6,7,8\}$ | 18 | $\{5,6,17,68\}$ |
| 19 | $\{5,7,15,67\}$ | 20 | $\{5,8,12,66\}$ |
| 21 | $\{5,12,42,43\}$ | 22 | $\{5,15,54,55\}$ |
| 23 | $\{5,17,62,63\}$ | 24 | $\{6,13,46,47\}$ |
| 25 | $\{6,14,50,51\}$ | 26 | $\{6,17,64,65\}$ |
| 27 | $\{7,13,48,49\}$ | 28 | $\{7,15,56,57\}$ |
| 29 | $\{7,16,58,59\}$ | 30 | $\{8,12,44,45\}$ |
| 31 | $\{8,14,52,53\}$ | 32 | $\{8,16,60,61\}$ |
| 33 | $\{9,10,12,13\}$ | 34 | $\{9,11,14,15\}$ |
| 35 | $\{10,11,16,17\}$ | | |

For variant 1 the nine-context parity subsystem consists of exactly $C_3, C_4, C_5, D_1, \ldots, D_6$ with $a = 66, b = 67, c = 68$. For variant 2 the twenty-ray subsystem consists of all rows whose labels lie in $\{1, \ldots, 17, 66, 67, 68\}$. The machine-readable inventories repeat every coordinate, basis, and inclusion map.

## S2. TWO-VALUED STATES, EXTENDABILITY CERTIFICATES, COLORINGS, AND SEPARATION PROFILES

### S2.1. State, coloring, odd-cycle certificate, and profile tables for chromatically complete configurations ($\chi(H) = d$)

This subsection provides the complete state and coloring enumerations and odd-cycle nonextendability certificates for $H_{6,3}$ (Table S36), $P_5$ (Table S37), $H_8$ (Tables S38 and S39), $H_{17,10}$ (Tables S40 and S41), $H_{21,13}$ (Tables S42, S43, and S44), the vertex separation profiles and Born probability decomposition on $H_{17,10}$ and $H_{21,13}$ (Tables S45–S47), the complete state and normalized coloring tables for all faithful 3- and 4-dimensional logical gadgets (Tables S48–S69 and S70), and the ten chromatically inseparable faithful restrictions of $H_{\mathrm{YO}}^{+}$ (Table S71).

TABLE S36: All two-valued states of $H_{6,3}$. The last column gives an odd cycle in the complement of each nonextendable support.

| $j$ | support of $s_j$ | extendable | odd cycle |
|---|---|---|---|
| 1 | $\{1,4\}$ | yes | – |
| 2 | $\{2,4,6\}$ | no | $(3,1,5)$ |
| 3 | $\{2,5\}$ | yes | – |
| 4 | $\{3,6\}$ | yes | – |

TABLE S37: All two-valued states of $P_5$. The last column gives an odd cycle in the complement of each nonextendable support.

| $j$ | support of $s_j$ | extendable | odd cycle |
|---|---|---|---|
| 1 | $\{1,3,9\}$ | yes | – |
| 2 | $\{1,4,7\}$ | yes | – |
| 3 | $\{1,7,8,9\}$ | yes | – |
| 4 | $\{2,4,10\}$ | yes | – |
| 5 | $\{2,5,8\}$ | yes | – |
| 6 | $\{2,8,9,10\}$ | yes | – |
| 7 | $\{3,5,6\}$ | yes | – |
| 8 | $\{3,6,9,10\}$ | yes | – |
| 9 | $\{4,6,7,10\}$ | yes | – |
| 10 | $\{5,6,7,8\}$ | yes | – |
| 11 | $\{6,7,8,9,10\}$ | no | $(3,2,1,5,4)$ |

TABLE S38: All two-valued states of $H_8$. The last column gives an odd cycle in the complement of each nonextendable support.

| $j$ | support of $s_j$ | extendable | odd cycle |
|---|---|---|---|
| 1 | $\{1,4,6,9\}$ | yes | – |
| 2 | $\{1,5,8,10,14\}$ | no | $(9,7,6,12,11)$ |
| 3 | $\{1,5,9,13,14\}$ | yes | – |
| 4 | $\{2,4,6,8,11\}$ | yes | – |
| 5 | $\{2,4,7,11,14\}$ | no | $(5,3,1,12,6)$ |
| 6 | $\{2,5,8,10,12\}$ | yes | – |
| 7 | $\{2,5,8,11,13,14\}$ | yes | – |
| 8 | $\{2,5,9,12,13\}$ | no | $(4,3,1,11,10)$ |
| 9 | $\{3,6,8,11,13\}$ | no | $(7,5,4,10,9)$ |
| 10 | $\{3,7,10,12\}$ | yes | – |
| 11 | $\{3,7,11,13,14\}$ | yes | – |

TABLE S39. All strong three-colorings of $H_8$ up to color permutation. The anchor is $(1,2,3) = (0,1,2)$.

| $i$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $c_1$ | 0 | 1 | 2 | 0 | 1 | 0 | 2 | 1 | 0 | 1 | 2 | 1 | 2 | 2 |
| $c_2$ | 0 | 1 | 2 | 0 | 1 | 0 | 2 | 1 | 0 | 2 | 1 | 2 | 1 | 1 |
| $c_3$ | 0 | 1 | 2 | 1 | 0 | 1 | 2 | 1 | 0 | 2 | 1 | 2 | 0 | 0 |

TABLE S40: All two-valued states of $H_{17,10}$. The last column gives an odd cycle in the complement of each nonextendable support.

| $j$ | support of $s_j$ | extendable | odd cycle |
|---|---|---|---|
| 1 | $\{1,4,8,14,15,23\}$ | yes | – |
| 2 | $\{1,5,7,14,18,23,24\}$ | no | $(4,11,3,2,8,9,10)$ |
| 3 | $\{1,5,9,11,18\}$ | yes | – |
| 4 | $\{1,5,9,14,15,18,24\}$ | no | $(11,3,2,8,7)$ |
| 5 | $\{1,6,8,11,18,23\}$ | yes | – |
| 6 | $\{1,6,8,14,15,18,23,24\}$ | no | $(11,3,2,5,4)$ |
| 7 | $\{2,4,7,14,19,23\}$ | yes | – |
| 8 | $\{2,4,9,14,15,19\}$ | yes | – |
| 9 | $\{2,6,7,10,14,24\}$ | yes | – |
| 10 | $\{2,6,7,14,18,19,23,24\}$ | no | $(11,3,1,10,4)$ |
| 11 | $\{2,6,9,11,18,19\}$ | yes | – |
| 12 | $\{2,6,9,14,15,18,19,24\}$ | no | $(11,3,1,10,4)$ |
| 13 | $\{3,4,8,15,19,23\}$ | yes | – |
| 14 | $\{3,5,7,10,24\}$ | yes | – |
| 15 | $\{3,5,7,18,19,23,24\}$ | no | $(8,2,1,10,9)$ |
| 16 | $\{3,5,9,15,18,19,24\}$ | no | $(7,8,2,1,10,4,11)$ |
| 17 | $\{3,6,8,10,15,24\}$ | yes | – |
| 18 | $\{3,6,8,15,18,19,23,24\}$ | no | $(5,2,1,10,4)$ |

TABLE S41. All strong three-colorings of $H_{17,10}$ up to color permutation. The anchor is $(2,8,5)=(0,1,2)$.

| $i$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 14 | 15 | 18 | 19 | 23 | 24 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $c_1$ | 1 | 0 | 2 | 0 | 2 | 1 | 2 | 1 | 0 | 2 | 1 | 0 | 0 | 1 | 0 | 1 | 2 |
| $c_2$ | 1 | 0 | 2 | 1 | 2 | 0 | 2 | 1 | 0 | 2 | 0 | 1 | 1 | 0 | 0 | 1 | 2 |
| $c_3$ | 2 | 0 | 1 | 0 | 2 | 1 | 0 | 1 | 2 | 1 | 2 | 0 | 1 | 2 | 0 | 0 | 1 |
| $c_4$ | 2 | 0 | 1 | 1 | 2 | 0 | 0 | 1 | 2 | 0 | 2 | 0 | 1 | 2 | 1 | 1 | 0 |

TABLE S42: All two-valued states of $H_{21,13}$. The last column gives an odd cycle in the complement of each nonextendable support.

| $j$ | support of $s_j$ | extendable | odd cycle |
|---|---|---|---|
| 1 | $\{1,4,8,14,15,21,22,23\}$ | no | $(7,11,3,2,5,6,13)$ |
| 2 | $\{1,5,7,14,18,22,23,24\}$ | no | $(4,11,3,2,8,9,10)$ |
| 3 | $\{1,5,9,11,18,21,22\}$ | yes | – |
| 4 | $\{1,5,9,14,15,18,21,22,24\}$ | no | $(11,3,2,8,7)$ |
| 5 | $\{1,6,8,11,18,21,23\}$ | yes | – |
| 6 | $\{1,6,8,14,15,18,21,23,24\}$ | no | $(11,3,2,5,4)$ |
| 7 | $\{2,4,7,14,19,20,22,23\}$ | no | $(8,9,10,1,13,6,5)$ |
| 8 | $\{2,4,9,13,14,15,19\}$ | yes | – |
| 9 | $\{2,4,9,14,15,19,20,21,22\}$ | no | $(11,3,1,13,7)$ |
| 10 | $\{2,6,7,10,14,20,24\}$ | yes | – |
| 11 | $\{2,6,7,14,18,19,20,23,24\}$ | no | $(11,3,1,10,4)$ |
| 12 | $\{2,6,9,11,18,19,20,21\}$ | no | $(5,4,10,1,13,7,8)$ |
| 13 | $\{2,6,9,14,15,18,19,20,21,24\}$ | no | $(11,3,1,10,4)$ |
| 14 | $\{3,4,8,13,15,19,23\}$ | yes | – |
| 15 | $\{3,4,8,15,19,20,21,22,23\}$ | no | $(5,2,1,13,6)$ |
| 16 | $\{3,5,7,10,20,22,24\}$ | yes | – |
| 17 | $\{3,5,7,18,19,20,22,23,24\}$ | no | $(8,2,1,10,9)$ |
| 18 | $\{3,5,9,13,15,18,19,24\}$ | no | $(7,8,2,1,10,4,11)$ |
| 19 | $\{3,5,9,15,18,19,20,21,22,24\}$ | no | $(8,2,1,13,7)$ |
| 20 | $\{3,6,8,10,15,20,21,24\}$ | no | $(4,5,2,1,13,7,11)$ |
| 21 | $\{3,6,8,15,18,19,20,21,23,24\}$ | no | $(5,2,1,10,4)$ |

TABLE S43. All strong three-colorings of $H_{21,13}$ up to color permutation. The anchor is $(2, 8, 5) = (0, 1, 2)$.

| $i$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 13 | 14 | 15 | 18 | 19 | 20 | 21 | 22 | 23 | 24 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $c_1$ | 1 | 0 | 2 | 0 | 2 | 1 | 2 | 1 | 0 | 2 | 1 | 0 | 0 | 0 | 1 | 0 | 2 | 1 | 2 | 1 | 2 |
| $c_2$ | 2 | 0 | 1 | 1 | 2 | 0 | 0 | 1 | 2 | 0 | 2 | 1 | 0 | 1 | 2 | 1 | 0 | 2 | 2 | 1 | 0 |

TABLE S44. The nine chromatic signatures of $H_{21,13}$. Columns give $c_1$ and rows give $c_2$ for the two normalized colorings in Table S43. Each cell is one inseparability class: its vertices have the same color in every admissible coloring. Ordinary two-valued states distinguish every pair, including vertices in the same cell.

| $c_2 \backslash c_1$ | 0 | 1 | 2 |
|---|---|---|---|
| 2 | $\{9\}$ | $\{1, 11, 18, 21\}$ | $\{5, 22\}$ |
| 1 | $\{4, 13, 15, 19\}$ | $\{8, 23\}$ | $\{3\}$ |
| 0 | $\{2, 14\}$ | $\{6\}$ | $\{7, 10, 20, 24\}$ |

TABLE S45: Complete state profiles in hexadecimal, and all Born probabilities for the 17-ray preparation in Eq. (26). The probabilities are also reproduced by Eq. (27).

| $i$ | $H_{17,10}$ profile | $H_{21,13}$ profile | $p_i$ on $H_{17,10}$ |
|---|---|---|---|
| 1 | 3F | 3F | 1/12 |
| 2 | FC0 | 1FC0 | 1/6 |
| 3 | 3F000 | 1FE000 | 3/4 |
| 4 | 10C1 | 61C1 | 0 |
| 5 | E00E | 7800E | 2/3 |
| 6 | 30F30 | 181E30 | 1/3 |
| 7 | 6342 | 18642 | 3/4 |
| 8 | 31031 | 186031 | 1/6 |
| 9 | 8C8C | 6198C | 1/12 |
| 10 | 12100 | 88200 | 8/9 |
| 11 | 414 | 814 | 0 |
| 13 | - | 22080 | – |
| 14 | BEB | 17EB | 1/4 |
| 15 | 398A9 | 1E71A9 | 1/4 |
| 18 | 2CE3E | 171C3E | 1/9 |
| 19 | 2DEC0 | 177DC0 | 1/36 |
| 20 | - | 1DDF40 | – |
| 21 | - | 1C593D | – |
| 22 | - | 5C14F | – |
| 23 | 25273 | 116473 | 1/36 |
| 24 | 3EB2A | 1F962A | 1 |

TABLE S46. Valid states and exact weights reproducing two $H_{17,10}$ preparations: the five-state mixture for $P_{24}$ (upper block, Eq. (27)) and the four-state mixture for $P_{z_3}$, the preparation along $z_3 = r_5$ (lower block). State numbering is that of Table S40. Every context equation and every Born marginal are checked with rational arithmetic.

| State | Weight | Support | Extendable? |
|---|---|---|---|
| Preparation $P_{24}$, nonextendable weight $\frac{1}{9}$ | | | |
| $s_4$ | $\frac{1}{12}$ | $\{1,5,9,14,15,18,24\}$ | no |
| $s_9$ | $\frac{5}{36}$ | $\{2,6,7,10,14,24\}$ | yes |
| $s_{10}$ | $\frac{1}{36}$ | $\{2,6,7,14,18,19,23,24\}$ | no |
| $s_{14}$ | $\frac{7}{12}$ | $\{3,5,7,10,24\}$ | yes |
| $s_{17}$ | $\frac{1}{6}$ | $\{3,6,8,10,15,24\}$ | yes |
| Preparation $P_{z_3}$, nonextendable weight $\frac{1}{3}$ | | | |
| $s_3$ | $\frac{1}{3}$ | $\{1,5,9,11,18\}$ | yes |
| $s_4$ | $\frac{1}{6}$ | $\{1,5,9,14,15,18,24\}$ | no |
| $s_{14}$ | $\frac{1}{3}$ | $\{3,5,7,10,24\}$ | yes |
| $s_{15}$ | $\frac{1}{6}$ | $\{3,5,7,18,19,23,24\}$ | no |

TABLE S47. Exact Born probabilities $p_i = (r_{24}\cdot r_i)^2/(6\|r_i\|^2)$ for $P_{24}$ and $p'_i = (r_5\cdot r_i)^2/\|r_i\|^2$ for $P_{z_3}$, also reproduced by the two mixtures of Table S46.

| $i$ | $p_i$ | $p'_i$ | $i$ | $p_i$ | $p'_i$ |
|---|---|---|---|---|---|
| 1 | $\frac{1}{12}$ | $\frac{1}{2}$ | 10 | $\frac{8}{9}$ | $\frac{1}{3}$ |
| 2 | $\frac{1}{6}$ | 0 | 11 | 0 | $\frac{1}{3}$ |
| 3 | $\frac{3}{4}$ | $\frac{1}{2}$ | 14 | $\frac{1}{4}$ | $\frac{1}{6}$ |
| 4 | 0 | 0 | 15 | $\frac{1}{4}$ | $\frac{1}{6}$ |
| 5 | $\frac{2}{3}$ | 1 | 18 | $\frac{1}{9}$ | $\frac{2}{3}$ |
| 6 | $\frac{1}{3}$ | 0 | 19 | $\frac{1}{36}$ | $\frac{1}{6}$ |
| 7 | $\frac{3}{4}$ | $\frac{1}{2}$ | 23 | $\frac{1}{36}$ | $\frac{1}{6}$ |
| 8 | $\frac{1}{6}$ | 0 | 24 | 1 | $\frac{2}{3}$ |
| 9 | $\frac{1}{12}$ | $\frac{1}{2}$ | | | |

### S2.2. State conventions for the additional gadgets

The state tables below are local to each labeled configuration. Supports are in lexicographic order; E is yes precisely for color-class supports. The three-dimensional pentagon and $H_{21,13}$ use the complete earlier tables and are not repeated. The four-dimensional apex-only states are included explicitly. All ordinary state families listed here are unital and separating.

### S2.3. All states of $B_{13,7}$

TABLE S48: All two-valued states of $B_{13,7}$. E indicates extendability.

| $j$ | Support | E | $j$ | Support | E |
|---|---|---|---|---|---|
| 1 | $\{1,5,7,14,23\}$ | yes | 8 | $\{2,9,11,19\}$ | yes |
| 2 | $\{1,5,9,11\}$ | yes | 9 | $\{2,9,14,15,19\}$ | yes |
| 3 | $\{1,5,9,14,15\}$ | no | 10 | $\{3,5,7,10\}$ | yes |
| 4 | $\{1,8,11,23\}$ | yes | 11 | $\{3,5,7,19,23\}$ | no |
| 5 | $\{1,8,14,15,23\}$ | yes | 12 | $\{3,5,9,15,19\}$ | yes |
| 6 | $\{2,7,10,14\}$ | yes | 13 | $\{3,8,10,15\}$ | yes |
| 7 | $\{2,7,14,19,23\}$ | yes | 14 | $\{3,8,15,19,23\}$ | yes |

### S2.4. All states of $D_{16,9}$

TABLE S49: All two-valued states of $D_{16,9}$. E indicates extendability.

| $j$ | Support | E | $j$ | Support | E |
|---|---|---|---|---|---|
| 1 | $\{1,4,8,14,15,23\}$ | yes | 12 | $\{2,6,7,10,14,24\}$ | yes |
| 2 | $\{1,5,7,10,14,24\}$ | yes | 13 | $\{2,6,7,14,18,23,24\}$ | no |
| 3 | $\{1,5,7,14,18,23,24\}$ | no | 14 | $\{2,6,9,11,18\}$ | yes |
| 4 | $\{1,5,9,11,18\}$ | yes | 15 | $\{2,6,9,14,15,18,24\}$ | no |
| 5 | $\{1,5,9,14,15,18,24\}$ | no | 16 | $\{3,4,8,15,23\}$ | yes |
| 6 | $\{1,6,8,10,11\}$ | yes | 17 | $\{3,5,7,10,24\}$ | yes |
| 7 | $\{1,6,8,10,14,15,24\}$ | no | 18 | $\{3,5,7,18,23,24\}$ | yes |
| 8 | $\{1,6,8,11,18,23\}$ | yes | 19 | $\{3,5,9,15,18,24\}$ | yes |
| 9 | $\{1,6,8,14,15,18,23,24\}$ | no | 20 | $\{3,6,8,10,15,24\}$ | yes |
| 10 | $\{2,4,7,14,23\}$ | yes | 21 | $\{3,6,8,15,18,23,24\}$ | no |
| 11 | $\{2,4,9,14,15\}$ | yes | | | |

### S2.5. All states of $I_{16,9}$

TABLE S50: All two-valued states of $I_{16,9}$. E indicates extendability.

| $j$ | Support | E | $j$ | Support | E |
|---|---|---|---|---|---|
| 1 | $\{1,4,5,7,14,23\}$ | yes | 12 | $\{2,7,14,18,19,23,24\}$ | no |
| 2 | $\{1,4,5,9,14,15\}$ | no | 13 | $\{2,9,11,18,19\}$ | yes |
| 3 | $\{1,4,8,14,15,23\}$ | yes | 14 | $\{2,9,14,15,18,19,24\}$ | no |
| 4 | $\{1,5,7,14,18,23,24\}$ | no | 15 | $\{3,4,5,7,19,23\}$ | no |
| 5 | $\{1,5,9,11,18\}$ | yes | 16 | $\{3,4,5,9,15,19\}$ | yes |
| 6 | $\{1,5,9,14,15,18,24\}$ | no | 17 | $\{3,4,8,15,19,23\}$ | yes |
| 7 | $\{1,8,11,18,23\}$ | yes | 18 | $\{3,5,7,10,24\}$ | yes |
| 8 | $\{1,8,14,15,18,23,24\}$ | no | 19 | $\{3,5,7,18,19,23,24\}$ | no |
| 9 | $\{2,4,7,14,19,23\}$ | yes | 20 | $\{3,5,9,15,18,19,24\}$ | no |
| 10 | $\{2,4,9,14,15,19\}$ | yes | 21 | $\{3,8,10,15,24\}$ | yes |
| 11 | $\{2,7,10,14,24\}$ | yes | 22 | $\{3,8,15,18,19,23,24\}$ | no |

### S2.6. All states of $T_{19,11}$

TABLE S51: All two-valued states of $T_{19,11}$. E indicates extendability.

| $j$ | Support | E | $j$ | Support | E |
|---|---|---|---|---|---|
| 1 | $\{1,4,8,14,15,23,28\}$ | yes | 14 | $\{2,6,7,14,18,23,24\}$ | no |
| 2 | $\{1,5,7,10,14,24,28\}$ | yes | 15 | $\{2,6,9,11,18,27\}$ | yes |
| 3 | $\{1,5,7,14,18,23,24\}$ | no | 16 | $\{2,6,9,14,15,18,24\}$ | no |
| 4 | $\{1,5,9,11,18,27\}$ | yes | 17 | $\{3,4,8,15,23,26\}$ | yes |
| 5 | $\{1,5,9,14,15,18,24\}$ | no | 18 | $\{3,4,8,15,23,27,28\}$ | no |
| 6 | $\{1,6,8,10,11,26\}$ | yes | 19 | $\{3,5,7,10,24,26\}$ | yes |
| 7 | $\{1,6,8,10,11,27,28\}$ | no | 20 | $\{3,5,7,10,24,27,28\}$ | no |
| 8 | $\{1,6,8,10,14,15,24,28\}$ | no | 21 | $\{3,5,7,18,23,24,27\}$ | yes |
| 9 | $\{1,6,8,11,18,23,27\}$ | yes | 22 | $\{3,5,9,15,18,24,27\}$ | yes |
| 10 | $\{1,6,8,14,15,18,23,24\}$ | no | 23 | $\{3,6,8,10,15,24,26\}$ | yes |
| 11 | $\{2,4,7,14,23,28\}$ | yes | 24 | $\{3,6,8,10,15,24,27,28\}$ | no |
| 12 | $\{2,4,9,14,15,28\}$ | yes | 25 | $\{3,6,8,15,18,23,24,27\}$ | no |
| 13 | $\{2,6,7,10,14,24,28\}$ | yes | | | |

## S2.7. All states of $W_{11,5}^{(4)}$

TABLE S52: All two-valued states of $W_{11,5}^{(4)}$. E indicates extendability.

| $j$ | Support | E | $j$ | Support | E |
|---|---|---|---|---|---|
| 1 | $\{0\}$ | yes | 7 | $\{2,8,9,10\}$ | yes |
| 2 | $\{1,3,9\}$ | yes | 8 | $\{3,5,6\}$ | yes |
| 3 | $\{1,4,7\}$ | yes | 9 | $\{3,6,9,10\}$ | yes |
| 4 | $\{1,7,8,9\}$ | yes | 10 | $\{4,6,7,10\}$ | yes |
| 5 | $\{2,4,10\}$ | yes | 11 | $\{5,6,7,8\}$ | yes |
| 6 | $\{2,5,8\}$ | yes | 12 | $\{6,7,8,9,10\}$ | no |

## S2.8. All states of $W_{16,6}^{(4)}$

TABLE S53: All two-valued states of $W_{16,6}^{(4)}$. E indicates extendability.

| $j$ | Support | E | $j$ | Support | E |
|---|---|---|---|---|---|
| 1 | $\{0,7,9,10\}$ | yes | 37 | $\{2,8,12,14,15\}$ | yes |
| 2 | $\{0,7,9,15\}$ | yes | 38 | $\{3,5,6\}$ | yes |
| 3 | $\{0,7,10,14\}$ | yes | 39 | $\{3,6,9,10,11\}$ | yes |
| 4 | $\{0,7,14,15\}$ | yes | 40 | $\{3,6,9,10,12\}$ | yes |
| 5 | $\{0,9,10,13\}$ | yes | 41 | $\{3,6,9,11,15\}$ | yes |
| 6 | $\{0,9,13,15\}$ | yes | 42 | $\{3,6,9,12,15\}$ | yes |
| 7 | $\{0,10,13,14\}$ | yes | 43 | $\{3,6,10,11,14\}$ | yes |
| 8 | $\{0,13,14,15\}$ | yes | 44 | $\{3,6,10,12,14\}$ | yes |
| 9 | $\{1,3,9,11\}$ | yes | 45 | $\{3,6,11,14,15\}$ | yes |
| 10 | $\{1,3,9,12\}$ | yes | 46 | $\{3,6,12,14,15\}$ | yes |
| 11 | $\{1,3,11,14\}$ | yes | 47 | $\{4,6,7,10,11\}$ | yes |
| 12 | $\{1,3,12,14\}$ | yes | 48 | $\{4,6,7,10,12\}$ | yes |
| 13 | $\{1,4,7,11\}$ | yes | 49 | $\{4,6,7,11,15\}$ | yes |
| 14 | $\{1,4,7,12\}$ | yes | 50 | $\{4,6,7,12,15\}$ | yes |
| 15 | $\{1,4,11,13\}$ | yes | 51 | $\{4,6,10,11,13\}$ | yes |
| 16 | $\{1,4,12,13\}$ | yes | 52 | $\{4,6,10,12,13\}$ | yes |
| 17 | $\{1,7,8,9,11\}$ | yes | 53 | $\{4,6,11,13,15\}$ | yes |
| 18 | $\{1,7,8,9,12\}$ | yes | 54 | $\{4,6,12,13,15\}$ | yes |
| 19 | $\{1,7,8,11,14\}$ | yes | 55 | $\{5,6,7,8\}$ | yes |
| 20 | $\{1,7,8,12,14\}$ | yes | 56 | $\{5,6,8,13\}$ | yes |
| 21 | $\{1,8,9,11,13\}$ | yes | 57 | $\{6,7,8,9,10,11\}$ | no |
| 22 | $\{1,8,9,12,13\}$ | yes | 58 | $\{6,7,8,9,10,12\}$ | no |
| 23 | $\{1,8,11,13,14\}$ | yes | 59 | $\{6,7,8,9,11,15\}$ | no |
| 24 | $\{1,8,12,13,14\}$ | yes | 60 | $\{6,7,8,9,12,15\}$ | no |
| 25 | $\{2,4,10,11\}$ | yes | 61 | $\{6,7,8,10,11,14\}$ | no |
| 26 | $\{2,4,10,12\}$ | yes | 62 | $\{6,7,8,10,12,14\}$ | no |
| 27 | $\{2,4,11,15\}$ | yes | 63 | $\{6,7,8,11,14,15\}$ | no |
| 28 | $\{2,4,12,15\}$ | yes | 64 | $\{6,7,8,12,14,15\}$ | no |
| 29 | $\{2,5,8\}$ | yes | 65 | $\{6,8,9,10,11,13\}$ | no |
| 30 | $\{2,8,9,10,11\}$ | yes | 66 | $\{6,8,9,10,12,13\}$ | no |
| 31 | $\{2,8,9,10,12\}$ | yes | 67 | $\{6,8,9,11,13,15\}$ | no |
| 32 | $\{2,8,9,11,15\}$ | yes | 68 | $\{6,8,9,12,13,15\}$ | no |
| 33 | $\{2,8,9,12,15\}$ | yes | 69 | $\{6,8,10,11,13,14\}$ | no |
| 34 | $\{2,8,10,11,14\}$ | yes | 70 | $\{6,8,10,12,13,14\}$ | no |
| 35 | $\{2,8,10,12,14\}$ | yes | 71 | $\{6,8,11,13,14,15\}$ | no |
| 36 | $\{2,8,11,14,15\}$ | yes | 72 | $\{6,8,12,13,14,15\}$ | no |

### S2.9. All states of $B^{(4)}_{14,7}$

TABLE S54: All two-valued states of $B^{(4)}_{14,7}$. E indicates extendability.

| $j$ | Support | E | $j$ | Support | E |
|---|---|---|---|---|---|
| 1 | $\{0\}$ | yes | 9 | $\{2,9,11,19\}$ | yes |
| 2 | $\{1,5,7,14,23\}$ | yes | 10 | $\{2,9,14,15,19\}$ | yes |
| 3 | $\{1,5,9,11\}$ | yes | 11 | $\{3,5,7,10\}$ | yes |
| 4 | $\{1,5,9,14,15\}$ | no | 12 | $\{3,5,7,19,23\}$ | no |
| 5 | $\{1,8,11,23\}$ | yes | 13 | $\{3,5,9,15,19\}$ | yes |
| 6 | $\{1,8,14,15,23\}$ | yes | 14 | $\{3,8,10,15\}$ | yes |
| 7 | $\{2,7,10,14\}$ | yes | 15 | $\{3,8,15,19,23\}$ | yes |
| 8 | $\{2,7,14,19,23\}$ | yes | | | |

### S2.10. All states of $L^{(4)}_{18,7}$

TABLE S55: All two-valued states of $L^{(4)}_{18,7}$. E indicates extendability.

| $j$ | Support | E | $j$ | Support | E |
|---|---|---|---|---|---|
| 1 | $\{1,6,9,13,16\}$ | yes | 35 | $\{3,5,9,13,18\}$ | yes |
| 2 | $\{1,6,9,14,16\}$ | yes | 36 | $\{3,5,9,14,17\}$ | yes |
| 3 | $\{1,6,10,13,16\}$ | yes | 37 | $\{3,5,9,14,18\}$ | yes |
| 4 | $\{1,6,10,14,16\}$ | yes | 38 | $\{3,5,10,13,17\}$ | yes |
| 5 | $\{1,7,9,13,17\}$ | yes | 39 | $\{3,5,10,13,18\}$ | yes |
| 6 | $\{1,7,9,13,18\}$ | yes | 40 | $\{3,5,10,14,17\}$ | yes |
| 7 | $\{1,7,9,14,17\}$ | yes | 41 | $\{3,5,10,14,18\}$ | yes |
| 8 | $\{1,7,9,14,18\}$ | yes | 42 | $\{3,6,9,11\}$ | yes |
| 9 | $\{1,7,10,13,17\}$ | yes | 43 | $\{3,6,9,12,13\}$ | yes |
| 10 | $\{1,7,10,13,18\}$ | yes | 44 | $\{3,6,9,12,14\}$ | yes |
| 11 | $\{1,7,10,14,17\}$ | yes | 45 | $\{3,6,10,11\}$ | yes |
| 12 | $\{1,7,10,14,18\}$ | yes | 46 | $\{3,6,10,12,13\}$ | yes |
| 13 | $\{1,8,13,15\}$ | yes | 47 | $\{3,6,10,12,14\}$ | yes |
| 14 | $\{1,8,13,16,17\}$ | yes | 48 | $\{3,8,11,17\}$ | yes |
| 15 | $\{1,8,13,16,18\}$ | yes | 49 | $\{3,8,11,18\}$ | yes |
| 16 | $\{1,8,14,15\}$ | yes | 50 | $\{3,8,12,13,17\}$ | yes |
| 17 | $\{1,8,14,16,17\}$ | yes | 51 | $\{3,8,12,13,18\}$ | yes |
| 18 | $\{1,8,14,16,18\}$ | yes | 52 | $\{3,8,12,14,17\}$ | yes |
| 19 | $\{2,5,9,15\}$ | yes | 53 | $\{3,8,12,14,18\}$ | yes |
| 20 | $\{2,5,9,16,17\}$ | yes | 54 | $\{4,5,13,15\}$ | yes |
| 21 | $\{2,5,9,16,18\}$ | yes | 55 | $\{4,5,13,16,17\}$ | yes |
| 22 | $\{2,5,10,15\}$ | yes | 56 | $\{4,5,13,16,18\}$ | yes |
| 23 | $\{2,5,10,16,17\}$ | yes | 57 | $\{4,5,14,15\}$ | yes |
| 24 | $\{2,5,10,16,18\}$ | yes | 58 | $\{4,5,14,16,17\}$ | yes |
| 25 | $\{2,6,9,12,16\}$ | yes | 59 | $\{4,5,14,16,18\}$ | yes |
| 26 | $\{2,6,10,12,16\}$ | yes | 60 | $\{4,6,11,16\}$ | yes |
| 27 | $\{2,7,9,12,17\}$ | yes | 61 | $\{4,6,12,13,16\}$ | yes |
| 28 | $\{2,7,9,12,18\}$ | yes | 62 | $\{4,6,12,14,16\}$ | yes |
| 29 | $\{2,7,10,12,17\}$ | yes | 63 | $\{4,7,11,17\}$ | yes |
| 30 | $\{2,7,10,12,18\}$ | yes | 64 | $\{4,7,11,18\}$ | yes |
| 31 | $\{2,8,12,15\}$ | yes | 65 | $\{4,7,12,13,17\}$ | yes |
| 32 | $\{2,8,12,16,17\}$ | yes | 66 | $\{4,7,12,13,18\}$ | yes |
| 33 | $\{2,8,12,16,18\}$ | yes | 67 | $\{4,7,12,14,17\}$ | yes |
| 34 | $\{3,5,9,13,17\}$ | yes | 68 | $\{4,7,12,14,18\}$ | yes |

### S2.11. All states of $D^{(4)}_{17,9}$

TABLE S56: All two-valued states of $D^{(4)}_{17,9}$. E indicates extendability.

| $j$ | Support | E | $j$ | Support | E |
|---|---|---|---|---|---|
| 1 | $\{0\}$ | yes | 12 | $\{2,4,9,14,15\}$ | yes |
| 2 | $\{1,4,8,14,15,23\}$ | yes | 13 | $\{2,6,7,10,14,24\}$ | yes |
| 3 | $\{1,5,7,10,14,24\}$ | yes | 14 | $\{2,6,7,14,18,23,24\}$ | no |
| 4 | $\{1,5,7,14,18,23,24\}$ | no | 15 | $\{2,6,9,11,18\}$ | yes |
| 5 | $\{1,5,9,11,18\}$ | yes | 16 | $\{2,6,9,14,15,18,24\}$ | no |
| 6 | $\{1,5,9,14,15,18,24\}$ | no | 17 | $\{3,4,8,15,23\}$ | yes |
| 7 | $\{1,6,8,10,11\}$ | yes | 18 | $\{3,5,7,10,24\}$ | yes |
| 8 | $\{1,6,8,10,14,15,24\}$ | no | 19 | $\{3,5,7,18,23,24\}$ | yes |
| 9 | $\{1,6,8,11,18,23\}$ | yes | 20 | $\{3,5,9,15,18,24\}$ | yes |
| 10 | $\{1,6,8,14,15,18,23,24\}$ | no | 21 | $\{3,6,8,10,15,24\}$ | yes |
| 11 | $\{2,4,7,14,23\}$ | yes | 22 | $\{3,6,8,15,18,23,24\}$ | no |

### S2.12. All states of $L^{(4)}_{22,9}$

TABLE S57: All two-valued states of $L^{(4)}_{22,9}$. E indicates extendability.

| $j$ | Support | E | $j$ | Support | E |
|---|---|---|---|---|---|
| 1 | $\{1,6,9,13,16,21\}$ | yes | 80 | $\{2,8,12,16,17,20,21\}$ | no |
| 2 | $\{1,6,9,13,16,22\}$ | yes | 81 | $\{2,8,12,16,17,20,22\}$ | no |
| 3 | $\{1,6,9,14,16,19\}$ | yes | 82 | $\{2,8,12,16,18,19\}$ | yes |
| 4 | $\{1,6,9,14,16,20,21\}$ | no | 83 | $\{2,8,12,16,18,20,21\}$ | no |
| 5 | $\{1,6,9,14,16,20,22\}$ | no | 84 | $\{2,8,12,16,18,20,22\}$ | no |
| 6 | $\{1,6,10,13,16,21\}$ | yes | 85 | $\{3,5,9,13,17,21\}$ | yes |
| 7 | $\{1,6,10,13,16,22\}$ | yes | 86 | $\{3,5,9,13,17,22\}$ | yes |
| 8 | $\{1,6,10,14,16,19\}$ | yes | 87 | $\{3,5,9,13,18,21\}$ | yes |
| 9 | $\{1,6,10,14,16,20,21\}$ | no | 88 | $\{3,5,9,13,18,22\}$ | yes |
| 10 | $\{1,6,10,14,16,20,22\}$ | no | 89 | $\{3,5,9,14,17,19\}$ | yes |
| 11 | $\{1,7,9,13,17,21\}$ | yes | 90 | $\{3,5,9,14,17,20,21\}$ | no |
| 12 | $\{1,7,9,13,17,22\}$ | yes | 91 | $\{3,5,9,14,17,20,22\}$ | no |
| 13 | $\{1,7,9,13,18,21\}$ | yes | 92 | $\{3,5,9,14,18,19\}$ | yes |
| 14 | $\{1,7,9,13,18,22\}$ | yes | 93 | $\{3,5,9,14,18,20,21\}$ | no |
| 15 | $\{1,7,9,14,17,19\}$ | yes | 94 | $\{3,5,9,14,18,20,22\}$ | no |
| 16 | $\{1,7,9,14,17,20,21\}$ | no | 95 | $\{3,5,10,13,17,21\}$ | yes |
| 17 | $\{1,7,9,14,17,20,22\}$ | no | 96 | $\{3,5,10,13,17,22\}$ | yes |
| 18 | $\{1,7,9,14,18,19\}$ | yes | 97 | $\{3,5,10,13,18,21\}$ | yes |
| 19 | $\{1,7,9,14,18,20,21\}$ | no | 98 | $\{3,5,10,13,18,22\}$ | yes |
| 20 | $\{1,7,9,14,18,20,22\}$ | no | 99 | $\{3,5,10,14,17,19\}$ | yes |
| 21 | $\{1,7,10,13,17,21\}$ | yes | 100 | $\{3,5,10,14,17,20,21\}$ | no |
| 22 | $\{1,7,10,13,17,22\}$ | yes | 101 | $\{3,5,10,14,17,20,22\}$ | no |
| 23 | $\{1,7,10,13,18,21\}$ | yes | 102 | $\{3,5,10,14,18,19\}$ | yes |
| 24 | $\{1,7,10,13,18,22\}$ | yes | 103 | $\{3,5,10,14,18,20,21\}$ | no |
| 25 | $\{1,7,10,14,17,19\}$ | yes | 104 | $\{3,5,10,14,18,20,22\}$ | no |
| 26 | $\{1,7,10,14,17,20,21\}$ | no | 105 | $\{3,6,9,11,20\}$ | yes |
| 27 | $\{1,7,10,14,17,20,22\}$ | no | 106 | $\{3,6,9,12,13,21\}$ | yes |
| 28 | $\{1,7,10,14,18,19\}$ | yes | 107 | $\{3,6,9,12,13,22\}$ | yes |
| 29 | $\{1,7,10,14,18,20,21\}$ | no | 108 | $\{3,6,9,12,14,19\}$ | yes |
| 30 | $\{1,7,10,14,18,20,22\}$ | no | 109 | $\{3,6,9,12,14,20,21\}$ | no |
| 31 | $\{1,8,13,16,17,21\}$ | yes | 110 | $\{3,6,9,12,14,20,22\}$ | no |
| 32 | $\{1,8,13,16,17,22\}$ | yes | 111 | $\{3,6,10,11,20\}$ | yes |
| 33 | $\{1,8,13,16,18,21\}$ | yes | 112 | $\{3,6,10,12,13,21\}$ | yes |
| 34 | $\{1,8,13,16,18,22\}$ | yes | 113 | $\{3,6,10,12,13,22\}$ | yes |
| 35 | $\{1,8,14,15,21\}$ | yes | 114 | $\{3,6,10,12,14,19\}$ | yes |
| 36 | $\{1,8,14,15,22\}$ | yes | 115 | $\{3,6,10,12,14,20,21\}$ | no |
| 37 | $\{1,8,14,16,17,19\}$ | yes | 116 | $\{3,6,10,12,14,20,22\}$ | no |
| 38 | $\{1,8,14,16,17,20,21\}$ | no | 117 | $\{3,8,11,17,20\}$ | yes |
| 39 | $\{1,8,14,16,17,20,22\}$ | no | 118 | $\{3,8,11,18,20\}$ | yes |

| $j$ | Support | E | $j$ | Support | E |
|---|---|---|---|---|---|
| 40 | $\{1,8,14,16,18,19\}$ | yes | 119 | $\{3,8,12,13,17,21\}$ | yes |
| 41 | $\{1,8,14,16,18,20,21\}$ | no | 120 | $\{3,8,12,13,17,22\}$ | yes |
| 42 | $\{1,8,14,16,18,20,22\}$ | no | 121 | $\{3,8,12,13,18,21\}$ | yes |
| 43 | $\{2,5,9,15,21\}$ | yes | 122 | $\{3,8,12,13,18,22\}$ | yes |
| 44 | $\{2,5,9,15,22\}$ | yes | 123 | $\{3,8,12,14,17,19\}$ | yes |
| 45 | $\{2,5,9,16,17,19\}$ | yes | 124 | $\{3,8,12,14,17,20,21\}$ | no |
| 46 | $\{2,5,9,16,17,20,21\}$ | no | 125 | $\{3,8,12,14,17,20,22\}$ | no |
| 47 | $\{2,5,9,16,17,20,22\}$ | no | 126 | $\{3,8,12,14,18,19\}$ | yes |
| 48 | $\{2,5,9,16,18,19\}$ | yes | 127 | $\{3,8,12,14,18,20,21\}$ | no |
| 49 | $\{2,5,9,16,18,20,21\}$ | no | 128 | $\{3,8,12,14,18,20,22\}$ | no |
| 50 | $\{2,5,9,16,18,20,22\}$ | no | 129 | $\{4,5,13,16,17,21\}$ | yes |
| 51 | $\{2,5,10,15,21\}$ | yes | 130 | $\{4,5,13,16,17,22\}$ | yes |
| 52 | $\{2,5,10,15,22\}$ | yes | 131 | $\{4,5,13,16,18,21\}$ | yes |
| 53 | $\{2,5,10,16,17,19\}$ | yes | 132 | $\{4,5,13,16,18,22\}$ | yes |
| 54 | $\{2,5,10,16,17,20,21\}$ | no | 133 | $\{4,5,14,15,21\}$ | yes |
| 55 | $\{2,5,10,16,17,20,22\}$ | no | 134 | $\{4,5,14,15,22\}$ | yes |
| 56 | $\{2,5,10,16,18,19\}$ | yes | 135 | $\{4,5,14,16,17,19\}$ | yes |
| 57 | $\{2,5,10,16,18,20,21\}$ | no | 136 | $\{4,5,14,16,17,20,21\}$ | no |
| 58 | $\{2,5,10,16,18,20,22\}$ | no | 137 | $\{4,5,14,16,17,20,22\}$ | no |
| 59 | $\{2,6,9,12,16,19\}$ | yes | 138 | $\{4,5,14,16,18,19\}$ | yes |
| 60 | $\{2,6,9,12,16,20,21\}$ | no | 139 | $\{4,5,14,16,18,20,21\}$ | no |
| 61 | $\{2,6,9,12,16,20,22\}$ | no | 140 | $\{4,5,14,16,18,20,22\}$ | no |
| 62 | $\{2,6,10,12,16,19\}$ | yes | 141 | $\{4,6,11,16,20\}$ | yes |
| 63 | $\{2,6,10,12,16,20,21\}$ | no | 142 | $\{4,6,12,13,16,21\}$ | yes |
| 64 | $\{2,6,10,12,16,20,22\}$ | no | 143 | $\{4,6,12,13,16,22\}$ | yes |
| 65 | $\{2,7,9,12,17,19\}$ | yes | 144 | $\{4,6,12,14,16,19\}$ | yes |
| 66 | $\{2,7,9,12,17,20,21\}$ | no | 145 | $\{4,6,12,14,16,20,21\}$ | no |
| 67 | $\{2,7,9,12,17,20,22\}$ | no | 146 | $\{4,6,12,14,16,20,22\}$ | no |
| 68 | $\{2,7,9,12,18,19\}$ | yes | 147 | $\{4,7,11,17,20\}$ | yes |
| 69 | $\{2,7,9,12,18,20,21\}$ | no | 148 | $\{4,7,11,18,20\}$ | yes |
| 70 | $\{2,7,9,12,18,20,22\}$ | no | 149 | $\{4,7,12,13,17,21\}$ | yes |
| 71 | $\{2,7,10,12,17,19\}$ | yes | 150 | $\{4,7,12,13,17,22\}$ | yes |
| 72 | $\{2,7,10,12,17,20,21\}$ | no | 151 | $\{4,7,12,13,18,21\}$ | yes |
| 73 | $\{2,7,10,12,17,20,22\}$ | no | 152 | $\{4,7,12,13,18,22\}$ | yes |
| 74 | $\{2,7,10,12,18,19\}$ | yes | 153 | $\{4,7,12,14,17,19\}$ | yes |
| 75 | $\{2,7,10,12,18,20,21\}$ | no | 154 | $\{4,7,12,14,17,20,21\}$ | no |
| 76 | $\{2,7,10,12,18,20,22\}$ | no | 155 | $\{4,7,12,14,17,20,22\}$ | no |
| 77 | $\{2,8,12,15,21\}$ | yes | 156 | $\{4,7,12,14,18,19\}$ | yes |
| 78 | $\{2,8,12,15,22\}$ | yes | 157 | $\{4,7,12,14,18,20,21\}$ | no |
| 79 | $\{2,8,12,16,17,19\}$ | yes | 158 | $\{4,7,12,14,18,20,22\}$ | no |

### S2.13. Complete normalized colorings of the gadget panels

Each color word gives one representative under global color permutation; the vertices and the anchor convention are specified separately for each table. The full labeled family is recovered by all palette permutations. Equal color labels across different colorings carry no extra constraint.

### S2.14. Colorings of $P_5$

The columns of each word correspond, in order, to vertices $1, 2, 3, 4, 5, 6, 7, 8, 9, 10$. The anchor context $(1, 6, 2)$ has colors $(0, \ldots, 2)$.

TABLE S58: Every normalized coloring of $P_5$.

| $j$ | Color word | $j$ | Color word |
|---|---|---|---|
| 1 | 0201211201 | 4 | 0210210211 |
| 2 | 0202111102 | 5 | 0212110002 |
| 3 | 0210110222 | | |

### S2.15. Colorings of $B_{13,7}$

The columns of each word correspond, in order, to vertices $1, 2, 3, 5, 7, 8, 9, 10, 11, 14, 15, 19, 23$. The anchor context $(2, 8, 5)$ has colors $(0, \ldots, 2)$.

TABLE S59: Every normalized coloring of $B_{13,7}$.

| $j$ | Color word | $j$ | Color word |
|---|---|---|---|
| 1 | 1022012010221 | 4 | 2012012020111 |
| 2 | 1022210201101 | 5 | 2012012120100 |
| 3 | 1022210210001 | 6 | 2012210102102 |

### S2.16. Colorings of $D_{16,9}$

The columns of each word correspond, in order, to vertices $1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 14, 15, 18, 23, 24$. The anchor context $(2, 8, 5)$ has colors $(0, \ldots, 2)$.

TABLE S60: Every normalized coloring of $D_{16,9}$.

| $j$ | Color word | $j$ | Color word |
|---|---|---|---|
| 1 | 1020210121102202 | 5 | 2010210121201201 |
| 2 | 1020212101100222 | 6 | 2011200120201210 |
| 3 | 1020212102100112 | 7 | 2011202102021012 |
| 4 | 1021202102011012 | | |

### S2.17. Colorings of $I_{16,9}$

The columns of each word correspond, in order, to vertices $1, 2, 3, 4, 5, 7, 8, 9, 10, 11, 14, 15, 18, 19, 23, 24$. The anchor context $(2, 8, 5)$ has colors $(0, \ldots, 2)$.

TABLE S61: Every normalized coloring of $I_{16,9}$.

| $j$ | Color word | $j$ | Color word |
|---|---|---|---|
| 1 | 1020221021001012 | 4 | 2010201212012001 |
| 2 | 1021221020110012 | 5 | 2011201202012110 |
| 3 | 1022201201021210 | 6 | 2012221010210021 |

### S2.18. Colorings of $T_{19,11}$

The columns of each word correspond, in order, to vertices $1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 14, 15, 18, 23, 24, 26, 27, 28$. The anchor context $(2, 8, 5)$ has colors $(0, \ldots, 2)$.

TABLE S62: Every normalized coloring of $T_{19,11}$.

| $j$ | Color word | $j$ | Color word |
|---|---|---|---|
| 1 | 1020210121102202120 | 5 | 2010210121201201120 |
| 2 | 1020212101100222120 | 6 | 2011200120201210120 |
| 3 | 1020212102100112210 | 7 | 2011202102021012102 |
| 4 | 1021202102011012201 | | |

### S2.19. Colorings of $H_{21,13}$

The columns of each word correspond, in order, to vertices $1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15, 18, 19, 20, 21, 22, 23, 24$. The anchor context $(2, 8, 5)$ has colors $(0, \ldots, 2)$.

TABLE S63: Every normalized coloring of $H_{21,13}$.

| $j$ | Color word | $j$ | Color word |
|---|---|---|---|
| 1 | 102021210210001021212 | 2 | 201120012021012102210 |

### S2.20. Colorings of $W_{11,5}^{(4)}$

The columns of each word correspond, in order, to vertices $0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10$. The anchor context $(0, 1, 6, 2)$ has colors $(0, \ldots, 3)$.

TABLE S64: Every normalized coloring of $W_{11,5}^{(4)}$.

| $j$ | Color word | $j$ | Color word |
|---|---|---|---|
| 1 | 01312322312 | 4 | 01321321322 |
| 2 | 01313222213 | 5 | 01323221113 |
| 3 | 01321221333 | | |

### S2.21. Colorings of $W_{16,6}^{(4)}$

The columns of each word correspond, in order, to vertices $0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15$. The anchor context $(2, 0, 6, 1)$ has colors $(0, \ldots, 3)$.

TABLE S65: Every normalized coloring of $W_{16,6}^{(4)}$.

| $j$ | Color word | $j$ | Color word |
|---|---|---|---|
| 1 | 1302302101123322 | 41 | 1302022331003131 |
| 2 | 1302302101132322 | 42 | 1302022331030131 |
| 3 | 1302302101223321 | 43 | 1302022331103130 |
| 4 | 1302302101232321 | 44 | 1302022331130130 |
| 5 | 1302302102123312 | 45 | 1302022333003111 |
| 6 | 1302302102132312 | 46 | 1302022333030111 |
| 7 | 1302302102223311 | 47 | 1302022333103110 |
| 8 | 1302302102232311 | 48 | 1302022333130110 |
| 9 | 1302302301123122 | 49 | 1302322100003311 |
| 10 | 1302302301132122 | 50 | 1302322100030311 |
| 11 | 1302302301223121 | 51 | 1302322100103310 |
| 12 | 1302302301232121 | 52 | 1302322100130310 |
| 13 | 1302302302123112 | 53 | 1302322101003301 |

| $j$ | Color word | $j$ | Color word |
|---|---|---|---|
| 14 | 1302302302132112 | 54 | 1302322101030301 |
| 15 | 1302302302223111 | 55 | 1302322101103300 |
| 16 | 1302302302232111 | 56 | 1302322101130300 |
| 17 | 1303202101123232 | 57 | 1302322300003111 |
| 18 | 1303202101132232 | 58 | 1302322300030111 |
| 19 | 1303202101223231 | 59 | 1302322300103110 |
| 20 | 1303202101232231 | 60 | 1302322300130110 |
| 21 | 1303202103123212 | 61 | 1302322301003101 |
| 22 | 1303202103132212 | 62 | 1302322301030101 |
| 23 | 1303202103223211 | 63 | 1302322301103100 |
| 24 | 1303202103232211 | 64 | 1302322301130100 |
| 25 | 1303202201123132 | 65 | 1303022121003231 |
| 26 | 1303202201132132 | 66 | 1303022121030231 |
| 27 | 1303202201223131 | 67 | 1303022121103230 |
| 28 | 1303202201232131 | 68 | 1303022121130230 |
| 29 | 1303202203123112 | 69 | 1303022123003211 |
| 30 | 1303202203132112 | 70 | 1303022123030211 |
| 31 | 1303202203223111 | 71 | 1303022123103210 |
| 32 | 1303202203232111 | 72 | 1303022123130210 |
| 33 | 1302022131003331 | 73 | 1303022221003131 |
| 34 | 1302022131030331 | 74 | 1303022221030131 |
| 35 | 1302022131103330 | 75 | 1303022221103130 |
| 36 | 1302022131130330 | 76 | 1303022221130130 |
| 37 | 1302022133003311 | 77 | 1303022223003111 |
| 38 | 1302022133030311 | 78 | 1303022223030111 |
| 39 | 1302022133103310 | 79 | 1303022223103110 |
| 40 | 1302022133130310 | 80 | 1303022223130110 |

### S2.22. Colorings of $B^{(4)}_{14,7}$

The columns of each word correspond, in order, to vertices $0, 1, 2, 3, 5, 7, 8, 9, 10, 11, 14, 15, 19, 23$. The anchor context $(0, 1, 2, 3)$ has colors $(0, \ldots, 3)$.

TABLE S66: Every normalized coloring of $B^{(4)}_{14,7}$.

| $j$ | Color word | $j$ | Color word |
|---|---|---|---|
| 1 | 01233213212331 | 4 | 01231132321321 |
| 2 | 01233312312221 | 5 | 01231231212333 |
| 3 | 01233312321121 | 6 | 01231231312322 |

### S2.23. Colorings of $L^{(4)}_{18,7}$

The columns of each word correspond, in order, to vertices $1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18$. The anchor context $(1, 2, 4, 3)$ has colors $(0, \ldots, 3)$.

TABLE S67: Every normalized coloring of $L^{(4)}_{18,7}$.

| $j$ | Color word | $j$ | Color word |
|---|---|---|---|
| 1 | 013212030123031203 | 73 | 013210230132021023 |
| 2 | 013212030123031230 | 74 | 013210230132021032 |
| 3 | 013212030123301203 | 75 | 013210230132201023 |
| 4 | 013212030123301230 | 76 | 013210230132201032 |
| 5 | 013212031023031203 | 77 | 013210231032021023 |
| 6 | 013212031023031230 | 78 | 013210231032021032 |
| 7 | 013212031023301203 | 79 | 013210231032201023 |
| 8 | 013212031023301230 | 80 | 013210231032201032 |
| 9 | 013210230123031023 | 81 | 013213201332020112 |

| $j$ | Color word | $j$ | Color word |
|---|---|---|---|
| 10 | 013210230123031032 | 82 | 013213201332020121 |
| 11 | 013210230123301023 | 83 | 013213201332200112 |
| 12 | 013210230123301032 | 84 | 013213201332200121 |
| 13 | 013210231023031023 | 85 | 013213203132020112 |
| 14 | 013210231023031032 | 86 | 013213203132020121 |
| 15 | 013210231023301023 | 87 | 013213203132200112 |
| 16 | 013210231023301032 | 88 | 013213203132200121 |
| 17 | 013213201323030112 | 89 | 013213201332021002 |
| 18 | 013213201323030121 | 90 | 013213201332021020 |
| 19 | 013213201323300112 | 91 | 013213201332201002 |
| 20 | 013213201323300121 | 92 | 013213201332201020 |
| 21 | 013213203123030112 | 93 | 013213203132021002 |
| 22 | 013213203123030121 | 94 | 013213203132021020 |
| 23 | 013213203123300112 | 95 | 013213203132201002 |
| 24 | 013213203123300121 | 96 | 013213203132201020 |
| 25 | 013213201323031002 | 97 | 013221030131022103 |
| 26 | 013213201323031020 | 98 | 013221030131022130 |
| 27 | 013213201323301002 | 99 | 013221030131202103 |
| 28 | 013213201323301020 | 100 | 013221030131202130 |
| 29 | 013213203123031002 | 101 | 013221031031022103 |
| 30 | 013213203123031020 | 102 | 013221031031022130 |
| 31 | 013213203123301002 | 103 | 013221031031202103 |
| 32 | 013213203123301020 | 104 | 013221031031202130 |
| 33 | 013231201321030123 | 105 | 013223010331021202 |
| 34 | 013231201321030132 | 106 | 013223010331021220 |
| 35 | 013231201321300123 | 107 | 013223010331201202 |
| 36 | 013231201321300132 | 108 | 013223010331201220 |
| 37 | 013231203121030123 | 109 | 013223013031021202 |
| 38 | 013231203121030132 | 110 | 013223013031021220 |
| 39 | 013231203121300123 | 111 | 013223013031201202 |
| 40 | 013231203121300132 | 112 | 013223013031201220 |
| 41 | 013232101321030213 | 113 | 013223010331022101 |
| 42 | 013232101321030231 | 114 | 013223010331022110 |
| 43 | 013232101321300213 | 115 | 013223010331202101 |
| 44 | 013232101321300231 | 116 | 013223010331202110 |
| 45 | 013232103121030213 | 117 | 013223013031022101 |
| 46 | 013232103121030231 | 118 | 013223013031022110 |
| 47 | 013232103121300213 | 119 | 013223013031202101 |
| 48 | 013232103121300231 | 120 | 013223013031202110 |
| 49 | 013230210321031023 | 121 | 013220130131022013 |
| 50 | 013230210321031032 | 122 | 013220130131022031 |
| 51 | 013230210321301023 | 123 | 013220130131202013 |
| 52 | 013230210321301032 | 124 | 013220130131202031 |
| 53 | 013230213021031023 | 125 | 013220131031022013 |
| 54 | 013230213021031032 | 126 | 013220131031022031 |
| 55 | 013230213021301023 | 127 | 013220131031202013 |
| 56 | 013230213021301032 | 128 | 013220131031202031 |
| 57 | 013232010321031203 | 129 | 013223101331020212 |
| 58 | 013232010321031230 | 130 | 013223101331020221 |
| 59 | 013232010321301203 | 131 | 013223101331200212 |
| 60 | 013232010321301230 | 132 | 013223101331200221 |
| 61 | 013232013021031203 | 133 | 013223103131020212 |
| 62 | 013232013021031230 | 134 | 013223103131020221 |
| 63 | 013232013021301203 | 135 | 013223103131200212 |
| 64 | 013232013021301230 | 136 | 013223103131200221 |
| 65 | 013212030132021203 | 137 | 013223101331022001 |
| 66 | 013212030132021230 | 138 | 013223101331022010 |
| 67 | 013212030132201203 | 139 | 013223101331202001 |
| 68 | 013212030132201230 | 140 | 013223101331202010 |
| 69 | 013212031032021203 | 141 | 013223103131022001 |
| 70 | 013212031032021230 | 142 | 013223103131022010 |
| 71 | 013212031032201203 | 143 | 013223103131202001 |

| $j$ | Color word | $j$ | Color word |
|---|---|---|---|
| 72 | 013212031032201230 | 144 | 013223103131202010 |

## S2.24. Colorings of $D^{(4)}_{17,9}$

The columns of each word correspond, in order, to vertices $0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 14, 15, 18, 23, 24$. The anchor context $(0, 1, 2, 3)$ has colors $(0, \ldots, 3)$.

TABLE S68: Every normalized coloring of $D^{(4)}_{17,9}$.

| $j$ | Color word | $j$ | Color word |
|---|---|---|---|
| 1 | 01232132313123123 | 5 | 01233122312123132 |
| 2 | 01232312131123323 | 6 | 01231323123211213 |
| 3 | 01232313121122333 | 7 | 01233121321213231 |
| 4 | 01232313123122113 | | |

## S2.25. Colorings of $L^{(4)}_{22,9}$

The columns of each word correspond, in order, to vertices $1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22$. The anchor context $(1, 2, 4, 3)$ has colors $(0, \ldots, 3)$.

TABLE S69: Every normalized coloring of $L^{(4)}_{22,9}$.

| $j$ | Color word | $j$ | Color word |
|---|---|---|---|
| 1 | 0132120301230312033201 | 109 | 0132120310320212032301 |
| 2 | 0132120301230312033210 | 110 | 0132120310320212032310 |
| 3 | 0132120301230312303201 | 111 | 0132120310320212302301 |
| 4 | 0132120301230312303210 | 112 | 0132120310320212302310 |
| 5 | 0132120310230312033201 | 113 | 0132120301322012030312 |
| 6 | 0132120310230312033210 | 114 | 0132120301322012030321 |
| 7 | 0132120310230312303201 | 115 | 0132120301322012300312 |
| 8 | 0132120310230312303210 | 116 | 0132120301322012300321 |
| 9 | 0132120301233012030213 | 117 | 0132120310322012030312 |
| 10 | 0132120301233012030231 | 118 | 0132120310322012030321 |
| 11 | 0132120301233012300213 | 119 | 0132120310322012300312 |
| 12 | 0132120301233012300231 | 120 | 0132120310322012300321 |
| 13 | 0132120310233012030213 | 121 | 0132132013322001121302 |
| 14 | 0132120310233012030231 | 122 | 0132132013322001121320 |
| 15 | 0132120310233012300213 | 123 | 0132132013322001211302 |
| 16 | 0132120310233012300231 | 124 | 0132132013322001211320 |
| 17 | 0132132013233001121203 | 125 | 0132132031322001121302 |
| 18 | 0132132013233001121230 | 126 | 0132132031322001121320 |
| 19 | 0132132013233001211203 | 127 | 0132132031322001211302 |
| 20 | 0132132013233001211230 | 128 | 0132132031322001211320 |
| 21 | 0132132031233001121203 | 129 | 0132102301320210232301 |
| 22 | 0132132031233001121230 | 130 | 0132102301320210232310 |
| 23 | 0132132031233001211203 | 131 | 0132102301320210322301 |
| 24 | 0132132031233001211230 | 132 | 0132102301320210322310 |
| 25 | 0132102301230310233201 | 133 | 0132102310320210232301 |
| 26 | 0132102301230310233210 | 134 | 0132102310320210232310 |
| 27 | 0132102301230310323201 | 135 | 0132102310320210322301 |
| 28 | 0132102301230310323210 | 136 | 0132102310320210322310 |
| 29 | 0132102310230310233201 | 137 | 0132102301322010230312 |
| 30 | 0132102310230310233210 | 138 | 0132102301322010230321 |
| 31 | 0132102310230310323201 | 139 | 0132102301322010320312 |
| 32 | 0132102310230310323210 | 140 | 0132102301322010320321 |
| 33 | 0132102301233010230213 | 141 | 0132102310322010230312 |
| 34 | 0132102301233010230231 | 142 | 0132102310322010230321 |

| $j$ | Color word | $j$ | Color word |
|---|---|---|---|
| 35 | 013210230123301032021 3 | 143 | 01321023103220103203 12 |
| 36 | 013210230123301032023 1 | 144 | 01321023103220103203 21 |
| 37 | 013210231023301023021 3 | 145 | 01321320133202100223 01 |
| 38 | 013210231023301023023 1 | 146 | 01321320133202100223 10 |
| 39 | 013210231023301032021 3 | 147 | 01321320133202102023 01 |
| 40 | 013210231023301032023 1 | 148 | 01321320133202102023 10 |
| 41 | 013213201323031002320 1 | 149 | 01321320313202100223 01 |
| 42 | 013213201323031002321 0 | 150 | 01321320313202100223 10 |
| 43 | 013213201323031020320 1 | 151 | 01321320313202102023 01 |
| 44 | 013213201323031020321 0 | 152 | 01321320313202102023 10 |
| 45 | 013213203123031002320 1 | 153 | 01321320133220100203 12 |
| 46 | 013213203123031002321 0 | 154 | 01321320133220100203 21 |
| 47 | 013213203123031020320 1 | 155 | 01321320133220102003 12 |
| 48 | 013213203123031020321 0 | 156 | 01321320133220102003 21 |
| 49 | 013213201323301002021 3 | 157 | 01321320313220100203 12 |
| 50 | 013213201323301002023 1 | 158 | 01321320313220100203 21 |
| 51 | 013213201323301020021 3 | 159 | 01321320313220102003 12 |
| 52 | 013213201323301020023 1 | 160 | 01321320313220102003 21 |
| 53 | 013213203123301002021 3 | 161 | 01322301033102120223 01 |
| 54 | 013213203123301002023 1 | 162 | 01322301033102120223 10 |
| 55 | 013213203123301020021 3 | 163 | 01322301033102122023 01 |
| 56 | 013213203123301020023 1 | 164 | 01322301033102122023 10 |
| 57 | 013231201321300123120 3 | 165 | 01322301303102120223 01 |
| 58 | 013231201321300123123 0 | 166 | 01322301303102120223 10 |
| 59 | 013231201321300132120 3 | 167 | 01322301303102122023 01 |
| 60 | 013231201321300132123 0 | 168 | 01322301303102122023 10 |
| 61 | 013231203121300123120 3 | 169 | 01322301033120120203 12 |
| 62 | 013231203121300123123 0 | 170 | 01322301033120120203 21 |
| 63 | 013231203121300132120 3 | 171 | 01322301033120122003 12 |
| 64 | 013231203121300132123 0 | 172 | 01322301033120122003 21 |
| 65 | 013232101321300213120 3 | 173 | 01322301303120120203 12 |
| 66 | 013232101321300213123 0 | 174 | 01322301303120120203 21 |
| 67 | 013232101321300231120 3 | 175 | 01322301303120122003 12 |
| 68 | 013232101321300231123 0 | 176 | 01322301303120122003 21 |
| 69 | 013232103121300213120 3 | 177 | 01322103013102210313 02 |
| 70 | 013232103121300213123 0 | 178 | 01322103013102210313 20 |
| 71 | 013232103121300231120 3 | 179 | 01322103013102213013 02 |
| 72 | 013232103121300231123 0 | 180 | 01322103013102213013 20 |
| 73 | 013230210321031023320 1 | 181 | 01322103103102210313 02 |
| 74 | 013230210321031023321 0 | 182 | 01322103103102210313 20 |
| 75 | 013230210321031032320 1 | 183 | 01322103103102213013 02 |
| 76 | 013230210321031032321 0 | 184 | 01322103103102213013 20 |
| 77 | 013230213021031023320 1 | 185 | 01322301033102211013 02 |
| 78 | 013230213021031023321 0 | 186 | 01322301033102211013 20 |
| 79 | 013230213021031032320 1 | 187 | 01322301033102211013 02 |
| 80 | 013230213021031032321 0 | 188 | 01322301033102211013 20 |
| 81 | 013230210321301023021 3 | 189 | 01322301303102211013 02 |
| 82 | 013230210321301023023 1 | 190 | 01322301303102211013 20 |
| 83 | 013230210321301032021 3 | 191 | 01322301303102211013 02 |
| 84 | 013230210321301032023 1 | 192 | 01322301303102211013 20 |
| 85 | 013230213021301023021 3 | 193 | 01322310133120021213 02 |
| 86 | 013230213021301023023 1 | 194 | 01322310133120021213 20 |
| 87 | 013230213021301032021 3 | 195 | 01322310133120022113 02 |
| 88 | 013230213021301032023 1 | 196 | 01322310133120022113 20 |
| 89 | 013232010321031203320 1 | 197 | 01322310313120021213 02 |
| 90 | 013232010321031203321 0 | 198 | 01322310313120021213 20 |
| 91 | 013232010321031230320 1 | 199 | 01322310313120022113 02 |
| 92 | 013232010321031230321 0 | 200 | 01322310313120022113 20 |
| 93 | 013232013021031203320 1 | 201 | 01322013013102201313 02 |
| 94 | 013232013021031203321 0 | 202 | 01322013013102201313 20 |
| 95 | 013232013021031230320 1 | 203 | 01322013013102203113 02 |
| 96 | 013232013021031230321 0 | 204 | 01322013013102203113 20 |

| $j$ | Color word | $j$ | Color word |
|---|---|---|---|
| 97 | 0132320103213012030213 | 205 | 013220131031022013 1302 |
| 98 | 0132320103213012030231 | 206 | 013220131031022013 1320 |
| 99 | 0132320103213012300213 | 207 | 013220131031022031 1302 |
| 100 | 0132320103213012300231 | 208 | 013220131031022031 1320 |
| 101 | 0132320130213012030213 | 209 | 013223101331022001 1302 |
| 102 | 0132320130213012030231 | 210 | 013223101331022001 1320 |
| 103 | 0132320130213012300213 | 211 | 013223101331022010 1302 |
| 104 | 0132320130213012300231 | 212 | 013223101331022010 1320 |
| 105 | 0132120301320212032301 | 213 | 013223103131022001 1302 |
| 106 | 0132120301320212032310 | 214 | 013223103131022001 1320 |
| 107 | 0132120301320212302301 | 215 | 013223103131022010 1302 |
| 108 | 0132120301320212302310 | 216 | 013223103131022010 1320 |

*Exhaustive classification of chromatic identities.* For a chromatically inseparable pair $(u, v)$, form the two directed witness sets

$$W_{10}(u,v) = \{j : s_j(u) = 1,\ s_j(v) = 0\},$$
$$W_{01}(u,v) = \{j : s_j(u) = 0,\ s_j(v) = 1\}.$$

Both sets are nonempty exactly for a two-way purely chromatic identity. If only $W_{10}$ is empty, $u \Rightarrow v$ holds and its converse fails; if only $W_{01}$ is empty, the implication is reversed. If both are empty, the pair is inseparable under all ordinary states. Any member of either witness set is automatically nonextendable, since every color-class state has equal values on a chromatically inseparable pair.

The following Python standard-library procedure reconstructs $H_{21,13}$ directly from its 13 contexts in Table S3, enumerates all binary assignments, and reconstructs normalized strong colorings as partitions into three disjoint state supports. Lexicographic support order agrees with Table S42. It prints both complete directed witness sets for every chromatic identity, rather than only the representatives in Table S70.

TABLE S70. Both directed truth-pattern witnesses for all 15 two-way purely chromatic pairs of $H_{21,13}$ and both displayed pairs of $T_{19,11}$. For $H_{21,13}$, $s_j$ is the state in Table S42; for $T_{19,11}$, entries give supports. Every witness is nonextendable and satisfies every context equation.

| Configuration | Pair $(u, v)$ | $(s(u), s(v)) = (1, 0)$ | $(s(u), s(v)) = (0, 1)$ |
|---|---|---|---|
| $T_{19,11}$ | $(14, 28)$ | $\{1, 5, 7, 14, 18, 23, 24\}$ | $\{1, 6, 8, 10, 11, 27, 28\}$ |
| $T_{19,11}$ | $(18, 27)$ | $\{1, 5, 7, 14, 18, 23, 24\}$ | $\{1, 6, 8, 10, 11, 27, 28\}$ |
| $H_{21,13}$ | $(1, 11)$ | $s_1$ | $s_{12}$ |
| $H_{21,13}$ | $(1, 18)$ | $s_1$ | $s_{11}$ |
| $H_{21,13}$ | $(1, 21)$ | $s_2$ | $s_9$ |
| $H_{21,13}$ | $(2, 14)$ | $s_{12}$ | $s_1$ |
| $H_{21,13}$ | $(4, 13)$ | $s_1$ | $s_{18}$ |
| $H_{21,13}$ | $(4, 15)$ | $s_7$ | $s_4$ |
| $H_{21,13}$ | $(4, 19)$ | $s_1$ | $s_{11}$ |
| $H_{21,13}$ | $(5, 22)$ | $s_{18}$ | $s_1$ |
| $H_{21,13}$ | $(7, 10)$ | $s_2$ | $s_{20}$ |
| $H_{21,13}$ | $(7, 20)$ | $s_2$ | $s_9$ |
| $H_{21,13}$ | $(7, 24)$ | $s_7$ | $s_4$ |
| $H_{21,13}$ | $(8, 23)$ | $s_{20}$ | $s_2$ |
| $H_{21,13}$ | $(15, 19)$ | $s_1$ | $s_7$ |
| $H_{21,13}$ | $(18, 21)$ | $s_2$ | $s_1$ |
| $H_{21,13}$ | $(20, 24)$ | $s_7$ | $s_2$ |

```
from itertools import combinations
E = [(1,2,3),(7,8,9),(4,5,6),(2,8,5),
     (3,11,14),(7,11,15),(4,11,24),
     (9,10,23),(1,10,19),(4,10,18),
     (1,13,20),(7,13,21),(6,13,22)]
V = sorted(set().union(*map(set, E)))
M = [sum(1 << V.index(v) for v in e) for e in E]
S = sorted(tuple(v for i,v in enumerate(V) if b >> i & 1)
           for b in range(1 << len(V))
           if all((b & m).bit_count() == 1 for m in M))
S = list(map(frozenset, S))
C = []
for blocks in combinations(S, 3):
    if any(a & b for a,b in combinations(blocks, 2)):
        continue
    blocks = [next(s for s in blocks if a in s) for a in (2,8,5)]
    C.append({v:k for k,s in enumerate(blocks) for v in s})
assert (len(S), len(C)) == (21, 2)
counts = [0, 0, 0]
for u,v in combinations(V, 2):
    if not all(c[u] == c[v] for c in C):
        continue
    W10 = [j for j,s in enumerate(S, 1) if u in s and v not in s]
    W01 = [j for j,s in enumerate(S, 1) if v in s and u not in s]
    k = 0 if W10 and W01 else 1 if W10 or W01 else 2
    counts[k] += 1
    print((u,v), W10, W01)
assert counts == [15, 6, 0]
```

TABLE S71: All ten connected, edge-faithful, colorable restrictions with a forced color identity in the specified inventory.

| retained diagonals | rays | contexts | colorings | inseparable pairs |
|---|---|---|---|---|
| $\{10,11\}$ | 17 | 10 | 4 | $(10,24)$, $(11,18)$ |
| $\{10,12\}$ | 17 | 10 | 4 | $(10,16)$, $(12,23)$ |
| $\{10,13\}$ | 17 | 10 | 4 | $(10,20)$, $(13,19)$ |
| $\{11,12\}$ | 17 | 10 | 4 | $(11,25)$, $(12,14)$ |
| $\{11,13\}$ | 17 | 10 | 4 | $(11,21)$, $(13,15)$ |
| $\{12,13\}$ | 17 | 10 | 4 | $(12,22)$, $(13,17)$ |
| $\{10,11,12\}$ | 21 | 13 | 2 | 21 pairs |
| $\{10,11,13\}$ | 21 | 13 | 2 | 21 pairs |
| $\{10,12,13\}$ | 21 | 13 | 2 | 21 pairs |
| $\{11,12,13\}$ | 21 | 13 | 2 | 21 pairs |

### S2.26. An explicit state certificate for the completed Yu–Oh hypergraph

The first four contexts of Eq. (15) force exactly one of vertices 2, 5, and 8 to have value one. In the triad containing that coordinate vertex, both other vertices have value zero. In each of the other two triads, one of the two remaining vertices has value one. There are therefore $3 \cdot 2^2 = 12$ possible supports $T_k$ on vertices $1, \ldots, 9$, listed in Table S72.

For each such support, exactly one of vertices $10, 11, 12, 13$ has no orthogonal neighbor in $T_k$; call it $\eta_k$. The other three are forced to zero, while $s(\eta_k)$ may be either zero or one. This gives precisely 24 assignments on the original 13 vertices. Every completion vertex $w$ belongs to one context $\{i, j, w\}$ and has the uniquely determined value

$$s(w) = 1 - s(i) - s(j). \tag{S7}$$

The listed assignments have $s(i) + s(j) \leq 1$ on every original orthogonal pair, so these extensions are valid two-valued states. Conversely, every state of the completed hypergraph restricts to one of these assignments.

For a compact separation certificate, encode the profile of vertex $i$ as the nonnegative integer

$$m_i = \sum_{k=1}^{12} \sum_{b=0}^{1} s_{k,b}(i)\, 2^{2(k-1)+b}.$$

TABLE S72. The 24 states are indexed by $(k, b)$, where $1 \leq k \leq 12$ and $b \in \{0, 1\}$. Their support on vertices $1, \ldots, 13$ is $T_k$ when $b = 0$ and $T_k \cup \{\eta_k\}$ when $b = 1$. Equation (S7) determines all remaining values.

| $k$ | $T_k$ | $\eta_k$ |
|---|---|---|
| 1 | $\{1,4,8\}$ | 12 |
| 2 | $\{1,5,7\}$ | 12 |
| 3 | $\{1,5,9\}$ | 11 |
| 4 | $\{1,6,8\}$ | 11 |
| 5 | $\{2,4,7\}$ | 12 |
| 6 | $\{2,4,9\}$ | 13 |
| 7 | $\{2,6,7\}$ | 10 |
| 8 | $\{2,6,9\}$ | 11 |
| 9 | $\{3,4,8\}$ | 13 |
| 10 | $\{3,5,7\}$ | 10 |
| 11 | $\{3,5,9\}$ | 13 |
| 12 | $\{3,6,8\}$ | 10 |

In hexadecimal notation these 25 integers are

| $i$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $m_i$ | 0000FF | 00FF00 | FF0000 | 030F03 | 3C003C |

| $i$ | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|
| $m_i$ | C0F0C0 | 0C330C | C300C3 | 30CC30 | 882000 |

| $i$ | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|
| $m_i$ | 0080A0 | 00020A | 220800 | 007F5F | F34C53 |

| $i$ | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|
| $m_i$ | CF31C5 | 3F0D35 | 74D0FC | 77DF00 | DDF700 |

| $i$ | 21 | 22 | 23 | 24 | 25 |
|---|---|---|---|---|---|
| $m_i$ | D1C4F3 | 1D073F | 4713CF | FC705C | 00FDF5 |

(S8)

All 25 integers are nonzero and pairwise distinct, certifying unitality and separation. Notably, the Hamming weights (number of states in which $s(i) = 1$) partition the vertices into three classes: 8 for $i \in \{1, \ldots, 9\}$, 3 for $i \in \{10, \ldots, 13\}$, and 13 for the completion vertices $i \in \{14, \ldots, 25\}$ (with $8 + 3 + 13 = 24$ in each completion context), in contrast to the uniform state weight 2 across all 15 vertices of $G_{32}$ and consistent with the fractional chromatic–clique gap $\chi_f(\Gamma(H_{\mathrm{YO}}^+)) = 35/11 > 3 = \omega(\Gamma(H_{\mathrm{YO}}^+))$.

### S2.27. Hesse $H_{37}$: all states and separating profiles

The canonical states $s_1, \ldots, s_{153}$ are listed in Table S74. A two-valued coloring is a state with exactly one value one per context. Vertex numbers are those of Appendix S1.14 and Fig. 4: vertices $1, \ldots, 16$ form the core, and $17, \ldots, 37$ are completions. We list every state, a minimum separating family, and every vertex profile.

#### *S2.27.1. State parametrization and all 27 branches*

Choose $t = (t_0, t_1, t_2) \in \mathbb{Z}_3^3$ and $\varepsilon \in \{0, 1\}$. Put

$$s(M_{ak}) = \mathbf{1}_{k=t_a}, \qquad s(z) = \varepsilon, \tag{S9}$$

$$U(t) = \mathbb{Z}_3 \setminus \{t_0, t_1, t_2\}, \qquad W(t) = \mathbb{Z}_3 \setminus \{t_a - a : a \in \mathbb{Z}_3\}. \tag{S10}$$

For $\varepsilon = 0$, choose the values of $P_k$ freely for $k \in U(t)$, with other $P$ values zero. For $\varepsilon = 1$, all $P$ values are zero. Independently, choose the values of $Q_k$ freely for $k \in W(t)$, with other $Q$ values zero. All completion values are forced:

$$s(A_k) = 1 - \varepsilon - s(P_k), \tag{S11}$$

$$s(B_{ak}) = 1 - s(M_{ak}) - s(P_k), \tag{S12}$$

$$s(C_{ak}) = 1 - s(M_{a,a+k}) - s(Q_k). \tag{S13}$$

Enumerate $t$ lexicographically with $0 < 1 < 2$, then $\varepsilon = 1, 0$. Within each branch, list the available $P$ bits in increasing $k$, followed by the available $Q$ bits in increasing $k$, and order the resulting bit strings with 1 before 0. This rule fixes all state indices used in the text, tables, and verification data. Table S73 gives the index ranges of all 27 branches.

TABLE S73: All 27 state branches. The sets $U(t), W(t)$ identify the free core bits. The two choices of $\varepsilon$ give $N_0 = 2^{|U|+|W|}$ and $N_1 = 2^{|W|}$ states; their index ranges refer to Table S74.

| $t$ | $U(t)$ | $W(t)$ | $N_0$ | $r$ for $\varepsilon = 0$ | $N_1$ | $r$ for $\varepsilon = 1$ |
|---|---|---|---|---|---|---|
| $(0,0,0)$ | $\{1,2\}$ | $\varnothing$ | 4 | 2–5 | 1 | 1 |
| $(0,0,1)$ | $\{2\}$ | $\{1\}$ | 4 | 8–11 | 2 | 6–7 |
| $(0,0,2)$ | $\{1\}$ | $\{1\}$ | 4 | 14–17 | 2 | 12–13 |
| $(0,1,0)$ | $\{2\}$ | $\{2\}$ | 4 | 20–23 | 2 | 18–19 |
| $(0,1,1)$ | $\{2\}$ | $\{1\}$ | 4 | 26–29 | 2 | 24–25 |
| $(0,1,2)$ | $\varnothing$ | $\{1,2\}$ | 4 | 34–37 | 4 | 30–33 |
| $(0,2,0)$ | $\{1\}$ | $\{2\}$ | 4 | 40–43 | 2 | 38–39 |
| $(0,2,1)$ | $\varnothing$ | $\varnothing$ | 1 | 45 | 1 | 44 |
| $(0,2,2)$ | $\{1\}$ | $\{2\}$ | 4 | 48–51 | 2 | 46–47 |
| $(1,0,0)$ | $\{2\}$ | $\{0\}$ | 4 | 54–57 | 2 | 52–53 |
| $(1,0,1)$ | $\{2\}$ | $\{0\}$ | 4 | 60–63 | 2 | 58–59 |
| $(1,0,2)$ | $\varnothing$ | $\varnothing$ | 1 | 65 | 1 | 64 |
| $(1,1,0)$ | $\{2\}$ | $\{2\}$ | 4 | 68–71 | 2 | 66–67 |
| $(1,1,1)$ | $\{0,2\}$ | $\varnothing$ | 4 | 73–76 | 1 | 72 |
| $(1,1,2)$ | $\{0\}$ | $\{2\}$ | 4 | 79–82 | 2 | 77–78 |
| $(1,2,0)$ | $\varnothing$ | $\{0,2\}$ | 4 | 87–90 | 4 | 83–86 |
| $(1,2,1)$ | $\{0\}$ | $\{0\}$ | 4 | 93–96 | 2 | 91–92 |
| $(1,2,2)$ | $\{0\}$ | $\{2\}$ | 4 | 99–102 | 2 | 97–98 |
| $(2,0,0)$ | $\{1\}$ | $\{0\}$ | 4 | 105–108 | 2 | 103–104 |
| $(2,0,1)$ | $\varnothing$ | $\{0,1\}$ | 4 | 113–116 | 4 | 109–112 |
| $(2,0,2)$ | $\{1\}$ | $\{1\}$ | 4 | 119–122 | 2 | 117–118 |
| $(2,1,0)$ | $\varnothing$ | $\varnothing$ | 1 | 124 | 1 | 123 |
| $(2,1,1)$ | $\{0\}$ | $\{1\}$ | 4 | 127–130 | 2 | 125–126 |
| $(2,1,2)$ | $\{0\}$ | $\{1\}$ | 4 | 133–136 | 2 | 131–132 |
| $(2,2,0)$ | $\{1\}$ | $\{0\}$ | 4 | 139–142 | 2 | 137–138 |
| $(2,2,1)$ | $\{0\}$ | $\{0\}$ | 4 | 145–148 | 2 | 143–144 |
| $(2,2,2)$ | $\{0,1\}$ | $\varnothing$ | 4 | 150–153 | 1 | 149 |
| Total | | | 99 | | 54 | |

### *S2.27.2. Canonical two-valued colorings: all 153 states*

Table S74 explicitly lists all 153 states. The last column is the complete binary vector in vertex order $1, \ldots, 37$, separated after the 16 core entries. Every row has context sum one in all 24 contexts. The support-size distribution is

$$\begin{array}{r|cccccc} m & 12 & 13 & 14 & 15 & 16 & 18 \\ \#\{s : |\operatorname{supp} s| = m\} & 6 & 18 & 27 & 24 & 51 & 27 \end{array} \tag{S14}$$

TABLE S74: Canonical enumeration of all 153 two-valued states, in the canonical order defined above. The support is split into core and completion vertices. The last column is $(s_r(1), \ldots, s_r(16)\,;\, s_r(17), \ldots, s_r(37))$.

| $r$ | Core support | Completion support | $\lvert\operatorname{supp} s_r\rvert$ | 37-bit state vector |
|---|---|---|---|---|
| 1 | $\{1,2,5,8\}$ | $\{21,22,24,25,27,28,30,31,32,33,35,37\}$ | 16 | 1100100100000000 0000110110110111110101 |
| 2 | $\{2,5,8,12,13\}$ | $\{17,30,31,32,33,35,37\}$ | 12 | 0100100100011000 1000000000000111110101 |
| 3 | $\{2,5,8,12\}$ | $\{17,19,22,25,28,30,31,32,33,35,37\}$ | 15 | 0100100100010000 1010010010011110101 |
| 4 | $\{2,5,8,13\}$ | $\{17,18,21,24,27,30,31,32,33,35,37\}$ | 15 | 0100100100001000 1100100100100111110101 |
| 5 | $\{2,5,8\}$ | $\{17,18,19,21,22,24,25,27,28,30,31,32,33,35,37\}$ | 18 | 0100100100000000 1110110110110111110101 |
| 6 | $\{1,2,5,9,15\}$ | $\{21,22,24,25,26,28,31,32,35\}$ | 14 | 1100100010000010 0000110111010011100100 |
| 7 | $\{1,2,5,9\}$ | $\{21,22,24,25,26,28,30,31,32,33,35,36\}$ | 16 | 1100100010000000 0000110111010111100110 |
| 8 | $\{2,5,9,13,15\}$ | $\{17,18,21,24,26,31,32,35\}$ | 13 | 0100100010001010 1100100101000011100100 |
| 9 | $\{2,5,9,13\}$ | $\{17,18,21,24,26,30,31,32,33,35,36\}$ | 15 | 0100100010001000 1100100101000011110110 |
| 10 | $\{2,5,9,15\}$ | $\{17,18,19,21,22,24,25,26,28,31,32,35\}$ | 16 | 0100100010000010 1110110111010011100100 |
| 11 | $\{2,5,9\}$ | $\{17,18,19,21,22,24,25,26,28,30,31,32,33,35,36\}$ | 18 | 0100100010000000 1110110111010111110110 |
| 12 | $\{1,2,5,10,15\}$ | $\{21,22,24,25,26,27,31,32,37\}$ | 14 | 1100100001000010 0000110111100011100001 |
| 13 | $\{1,2,5,10\}$ | $\{21,22,24,25,26,27,30,31,32,33,36,37\}$ | 16 | 1100100001000000 0000110111100111110011 |
| 14 | $\{2,5,10,12,15\}$ | $\{17,19,22,25,26,31,32,37\}$ | 13 | 0100100001010010 1010010011000011100001 |
| 15 | $\{2,5,10,12\}$ | $\{17,19,22,25,26,30,31,32,33,36,37\}$ | 15 | 0100100001010000 1010010011000111110011 |
| 16 | $\{2,5,10,15\}$ | $\{17,18,19,21,22,24,25,26,27,31,32,37\}$ | 16 | 0100100001000010 1110110111100011100001 |
| 17 | $\{2,5,10\}$ | $\{17,18,19,21,22,24,25,26,27,30,31,32,33,36,37\}$ | 18 | 0100100001000000 1110110111100111110011 |
| 18 | $\{1,2,6,8,16\}$ | $\{21,22,23,25,27,28,30,33,35\}$ | 14 | 1100010100000001 0000111010110101010100 |
| 19 | $\{1,2,6,8\}$ | $\{21,22,23,25,27,28,30,31,33,34,35,37\}$ | 16 | 1100010100000000 0000111010111011011101 |
| 20 | $\{2,6,8,13,16\}$ | $\{17,18,21,23,27,30,33,35\}$ | 13 | 0100010100001001 1100101000100101010100 |
| 21 | $\{2,6,8,13\}$ | $\{17,18,21,23,27,30,31,33,34,35,37\}$ | 15 | 0100010100001000 1100101000100111011101 |
| 22 | $\{2,6,8,16\}$ | $\{17,18,19,21,22,23,25,27,28,30,33,35\}$ | 16 | 0100010100000001 1110111010110101010100 |
| 23 | $\{2,6,8\}$ | $\{17,18,19,21,22,23,25,27,28,30,31,33,34,35,37\}$ | 18 | 0100010100000000 1110111010110111011101 |
| 24 | $\{1,2,6,9,15\}$ | $\{21,22,23,25,26,28,31,34,35\}$ | 14 | 1100010010000010 0000111010100100110 0 |
| 25 | $\{1,2,6,9\}$ | $\{21,22,23,25,26,28,30,31,33,34,35,36\}$ | 16 | 1100010010000000 0000111010110111011110 |
| 26 | $\{2,6,9,13,15\}$ | $\{17,18,21,23,26,31,34,35\}$ | 13 | 0100010010001010 1100101001000010011 00 |
| 27 | $\{2,6,9,13\}$ | $\{17,18,21,23,26,30,31,33,34,35,36\}$ | 15 | 0100010010001000 1100101001000111011110 |
| 28 | $\{2,6,9,15\}$ | $\{17,18,19,21,22,23,25,26,28,31,34,35\}$ | 16 | 0100010010000010 1110111010100100110 0 |
| 29 | $\{2,6,9\}$ | $\{17,18,19,21,22,23,25,26,28,30,31,33,34,35,36\}$ | 18 | 0100010010000000 1110111010110111011110 |
| 30 | $\{1,2,6,10,15,16\}$ | $\{21,22,23,25,26,27\}$ | 12 | 1100010001000011 0000111011100000000000 |
| 31 | $\{1,2,6,10,15\}$ | $\{21,22,23,25,26,27,31,34,37\}$ | 14 | 1100010001000010 0000111011100010010010 |
| 32 | $\{1,2,6,10,16\}$ | $\{21,22,23,25,26,27,30,33,36\}$ | 14 | 1100010001000001 0000111011100100100100 |
| 33 | $\{1,2,6,10\}$ | $\{21,22,23,25,26,27,30,31,33,34,36,37\}$ | 16 | 1100010001000000 0000111011100111011011 |
| 34 | $\{2,6,10,15,16\}$ | $\{17,18,19,21,22,23,25,26,27\}$ | 14 | 0100010001000011 1110111011100000000000 |
| 35 | $\{2,6,10,15\}$ | $\{17,18,19,21,22,23,25,26,27,31,34,37\}$ | 16 | 0100010001000010 1110111011100010010010 |
| 36 | $\{2,6,10,16\}$ | $\{17,18,19,21,22,23,25,26,27,30,33,36\}$ | 16 | 0100010001000001 1110111011100100100100 |
| 37 | $\{2,6,10\}$ | $\{17,18,19,21,22,23,25,26,27,30,31,33,34,36,37\}$ | 18 | 0100010001000000 1110111011100111011011 |
| 38 | $\{1,2,7,8,16\}$ | $\{21,22,23,24,27,28,30,32,35\}$ | 14 | 1100001100000001 0000111100110101010100 |
| 39 | $\{1,2,7,8\}$ | $\{21,22,23,24,27,28,30,31,32,34,35,37\}$ | 16 | 1100001100000000 0000111100111011101101 |
| 40 | $\{2,7,8,12,16\}$ | $\{17,19,22,23,28,30,32,35\}$ | 13 | 0100001100010001 1010011000010101010100 |
| 41 | $\{2,7,8,12\}$ | $\{17,19,22,23,28,30,31,32,34,35,37\}$ | 15 | 0100001100010000 1010011000010111011101 |
| 42 | $\{2,7,8,16\}$ | $\{17,18,19,21,22,23,24,27,28,30,32,35\}$ | 16 | 0100001100000001 1110111100110101010100 |
| 43 | $\{2,7,8\}$ | $\{17,18,19,21,22,23,24,27,28,30,31,32,34,35,37\}$ | 18 | 0100001100000000 1110111100111011101101 |
| 44 | $\{1,2,7,9\}$ | $\{21,22,23,24,26,28,30,31,32,34,35,36\}$ | 16 | 1100001010000000 0000111101010111011110 |
| 45 | $\{2,7,9\}$ | $\{17,18,19,21,22,23,24,26,28,30,31,32,34,35,36\}$ | 18 | 0100001010000000 1110111101010111011110 |
| 46 | $\{1,2,7,10,16\}$ | $\{21,22,23,24,26,27,30,32,36\}$ | 14 | 1100001001000001 0000111101100101001010 |
| 47 | $\{1,2,7,10\}$ | $\{21,22,23,24,26,27,30,31,32,34,36,37\}$ | 16 | 1100001001000000 0000111101100111011011 |
| 48 | $\{2,7,10,12,16\}$ | $\{17,19,22,23,26,30,32,36\}$ | 13 | 0100001001010001 1010011001000101010 0 |
| 49 | $\{2,7,10,12\}$ | $\{17,19,22,23,26,30,31,32,34,36,37\}$ | 15 | 0100001001010000 1010011001000111011011 |
| 50 | $\{2,7,10,16\}$ | $\{17,18,19,21,22,23,24,26,27,30,32,36\}$ | 16 | 0100001001000001 1110111101100101001010 |
| 51 | $\{2,7,10\}$ | $\{17,18,19,21,22,23,24,26,27,30,31,32,34,36,37\}$ | 18 | 0100001001000000 1110111101100111011011 |
| 52 | $\{1,3,5,8,14\}$ | $\{20,22,24,25,27,28,31,33,37\}$ | 14 | 1010100100000100 0001010110110010100 01 |
| 53 | $\{1,3,5,8\}$ | $\{20,22,24,25,27,28,29,31,32,33,35,37\}$ | 16 | 1010100100000000 0001010110111011101 01 |
| 54 | $\{3,5,8,13,14\}$ | $\{17,18,20,24,27,31,33,37\}$ | 13 | 0010100100001100 1101000100100010100 01 |
| 55 | $\{3,5,8,13\}$ | $\{17,18,20,24,27,29,31,32,33,35,37\}$ | 15 | 0010100100001000 1101000100101011101 01 |
| 56 | $\{3,5,8,14\}$ | $\{17,18,19,20,22,24,25,27,28,31,33,37\}$ | 16 | 0010100100000100 1111010110110010100 01 |
| 57 | $\{3,5,8\}$ | $\{17,18,19,20,22,24,25,27,28,29,31,32,33,35,37\}$ | 18 | 0010100100000000 1111010110111011101 01 |
| 58 | $\{1,3,5,9,14\}$ | $\{20,22,24,25,26,28,31,33,36\}$ | 14 | 1010100010000100 0001010111010010100 10 |
| 59 | $\{1,3,5,9\}$ | $\{20,22,24,25,26,28,29,31,32,33,35,36\}$ | 16 | 1010100010000000 0001010111011011101 10 |
| 60 | $\{3,5,9,13,14\}$ | $\{17,18,20,24,26,31,33,36\}$ | 13 | 0010100010001100 1101000101000010100 10 |
| 61 | $\{3,5,9,13\}$ | $\{17,18,20,24,26,29,31,32,33,35,36\}$ | 15 | 0010100010001000 1101000101001011101 10 |
| 62 | $\{3,5,9,14\}$ | $\{17,18,19,20,22,24,25,26,28,31,33,36\}$ | 16 | 0010100010000100 1111010111010010100 10 |
| 63 | $\{3,5,9\}$ | $\{17,18,19,20,22,24,25,26,28,29,31,32,33,35,36\}$ | 18 | 0010100010000000 1111010111011011101 10 |
| 64 | $\{1,3,5,10\}$ | $\{20,22,24,25,26,27,29,31,32,33,36,37\}$ | 16 | 1010100001000000 0001010111101011101 11 |
| 65 | $\{3,5,10\}$ | $\{17,18,19,20,22,24,25,26,27,29,31,32,33,36,37\}$ | 18 | 0010100001000000 1111010111101011101 11 |
| 66 | $\{1,3,6,8,16\}$ | $\{20,22,23,25,27,28,29,33,35\}$ | 14 | 1010010100000001 0001011010110010100 |
| 67 | $\{1,3,6,8\}$ | $\{20,22,23,25,27,28,29,31,33,34,35,37\}$ | 16 | 1010010100000000 0001011010111011011 01 |

*Continued on the next page.*

TABLE S74: Canonical enumeration of all 153 two-valued states, in the canonical order defined above. The support is split into core and completion vertices. The last column is $(s_r(1), \ldots, s_r(16)\,;\, s_r(17), \ldots, s_r(37))$. (continued).

| $r$ | Core support | Completion support | $\lvert\operatorname{supp} s_r\rvert$ | 37-bit state vector |
|---|---|---|---|---|
| 68 | $\{3,6,8,13,16\}$ | $\{17,18,20,23,27,29,33,35\}$ | 13 | 0010010100001001 1101001000101000010100 |
| 69 | $\{3,6,8,13\}$ | $\{17,18,20,23,27,29,31,33,34,35,37\}$ | 15 | 0010010100001000 1101001000101011101 |
| 70 | $\{3,6,8,16\}$ | $\{17,18,19,20,22,23,25,27,28,29,33,35\}$ | 16 | 0010010100000001 1111011010111000101000 |
| 71 | $\{3,6,8\}$ | $\{17,18,19,20,22,23,25,27,28,29,31,33,34,35,37\}$ | 18 | 0010010100000000 1111011010111010111101 |
| 72 | $\{1,3,6,9\}$ | $\{20,22,23,25,26,28,29,31,33,34,35,36\}$ | 16 | 1010010010000000 0001011011011010111110 |
| 73 | $\{3,6,9,11,13\}$ | $\{18,29,31,33,34,35,36\}$ | 12 | 0010010010101000 0100000000001010111110 |
| 74 | $\{3,6,9,11\}$ | $\{18,19,22,25,28,29,31,33,34,35,36\}$ | 15 | 0010010010100000 0110010010011010111110 |
| 75 | $\{3,6,9,13\}$ | $\{17,18,20,23,26,29,31,33,34,35,36\}$ | 15 | 0010010010001000 1101001000101011110 |
| 76 | $\{3,6,9\}$ | $\{17,18,19,20,22,23,25,26,28,29,31,33,34,35,36\}$ | 18 | 0010010010000000 1111011011011011110 |
| 77 | $\{1,3,6,10,16\}$ | $\{20,22,23,25,26,27,29,33,36\}$ | 14 | 1010010001000001 0001011011101000010010 |
| 78 | $\{1,3,6,10\}$ | $\{20,22,23,25,26,27,29,31,33,34,36,37\}$ | 16 | 1010010001000000 0001011011010101011011 |
| 79 | $\{3,6,10,11,16\}$ | $\{18,19,22,25,27,29,33,36\}$ | 13 | 0010010001100001 0110010010101000010010 |
| 80 | $\{3,6,10,11\}$ | $\{18,19,22,25,27,29,31,33,34,36,37\}$ | 15 | 0010010001100000 0110010010101011011 |
| 81 | $\{3,6,10,16\}$ | $\{17,18,19,20,22,23,25,26,27,29,33,36\}$ | 16 | 0010010001000001 1111011011101000010010 |
| 82 | $\{3,6,10\}$ | $\{17,18,19,20,22,23,25,26,27,29,31,33,34,36,37\}$ | 18 | 0010010001000000 1111011011101010111011 |
| 83 | $\{1,3,7,8,14,16\}$ | $\{20,22,23,24,27,28\}$ | 12 | 1010001100000101 0001011100110000000000 |
| 84 | $\{1,3,7,8,14\}$ | $\{20,22,23,24,27,28,31,34,37\}$ | 14 | 1010001100000100 0001011100110010010001 |
| 85 | $\{1,3,7,8,16\}$ | $\{20,22,23,24,27,28,29,32,35\}$ | 14 | 1010001100000001 0001011100111001001000 |
| 86 | $\{1,3,7,8\}$ | $\{20,22,23,24,27,28,29,31,32,34,35,37\}$ | 16 | 1010001100000000 0001011100111011011010 |
| 87 | $\{3,7,8,14,16\}$ | $\{17,18,19,20,22,23,24,27,28\}$ | 14 | 0010001100000101 1111011100110000000000 |
| 88 | $\{3,7,8,14\}$ | $\{17,18,19,20,22,23,24,27,28,31,34,37\}$ | 16 | 0010001100000100 1111011100110010010001 |
| 89 | $\{3,7,8,16\}$ | $\{17,18,19,20,22,23,24,27,28,29,32,35\}$ | 16 | 0010001100000001 1111011100111001001000 |
| 90 | $\{3,7,8\}$ | $\{17,18,19,20,22,23,24,27,28,29,31,32,34,35,37\}$ | 18 | 0010001100000000 1111011100111011011010 |
| 91 | $\{1,3,7,9,14\}$ | $\{20,22,23,24,26,28,31,34,36\}$ | 14 | 1010001010000100 0001011010100100101000 |
| 92 | $\{1,3,7,9\}$ | $\{20,22,23,24,26,28,29,31,32,34,35,36\}$ | 16 | 1010001010000000 0001011010110110111010 |
| 93 | $\{3,7,9,11,14\}$ | $\{18,19,22,24,28,31,34,36\}$ | 13 | 0010001010100100 0110010000100100101000 |
| 94 | $\{3,7,9,11\}$ | $\{18,19,22,24,28,29,31,32,34,35,36\}$ | 15 | 0010001010100000 0110010000110110111010 |
| 95 | $\{3,7,9,14\}$ | $\{17,18,19,20,22,23,24,26,28,31,34,36\}$ | 16 | 0010001010000100 1111011010010010001000 |
| 96 | $\{3,7,9\}$ | $\{17,18,19,20,22,23,24,26,28,29,31,32,34,35,36\}$ | 18 | 0010001010000000 1111011010110110111010 |
| 97 | $\{1,3,7,10,16\}$ | $\{20,22,23,24,26,27,29,32,36\}$ | 14 | 1010001001000001 0001011011010010001000 |
| 98 | $\{1,3,7,10\}$ | $\{20,22,23,24,26,27,29,31,32,34,36,37\}$ | 16 | 1010001001000000 0001011011010110111011 |
| 99 | $\{3,7,10,11,16\}$ | $\{18,19,22,24,27,29,32,36\}$ | 13 | 0010001001100001 0110010000100100100010 |
| 100 | $\{3,7,10,11\}$ | $\{18,19,22,24,27,29,31,32,34,36,37\}$ | 15 | 0010001001100000 0110010000101011011011 |
| 101 | $\{3,7,10,16\}$ | $\{17,18,19,20,22,23,24,26,27,29,32,36\}$ | 16 | 0010001001000001 1111011011010010001000 |
| 102 | $\{3,7,10\}$ | $\{17,18,19,20,22,23,24,26,27,29,31,32,34,36,37\}$ | 18 | 0010001001000000 1111011011010110111011 |
| 103 | $\{1,4,5,8,14\}$ | $\{20,21,24,25,27,28,30,33,37\}$ | 14 | 1001100100000100 0001100110110100100001 |
| 104 | $\{1,4,5,8\}$ | $\{20,21,24,25,27,28,29,30,32,33,35,37\}$ | 16 | 1001100100000000 0001100110111101101010 |
| 105 | $\{4,5,8,12,14\}$ | $\{17,19,20,25,28,30,33,37\}$ | 13 | 0001100100010100 1011000010010100100001 |
| 106 | $\{4,5,8,12\}$ | $\{17,19,20,25,28,29,30,32,33,35,37\}$ | 15 | 0001100100010000 1011000010011101101010 |
| 107 | $\{4,5,8,14\}$ | $\{17,18,19,20,21,24,25,27,28,30,33,37\}$ | 16 | 0001100100000100 1111100110110100100001 |
| 108 | $\{4,5,8\}$ | $\{17,18,19,20,21,24,25,27,28,29,30,32,33,35,37\}$ | 18 | 0001100100000000 1111100110111101101010 |
| 109 | $\{1,4,5,9,14,15\}$ | $\{20,21,24,25,26,28\}$ | 12 | 1001100010000110 0001100110100000000000 |
| 110 | $\{1,4,5,9,14\}$ | $\{20,21,24,25,26,28,30,33,36\}$ | 14 | 1001100010000100 0001100110101001001000 |
| 111 | $\{1,4,5,9,15\}$ | $\{20,21,24,25,26,28,29,32,35\}$ | 14 | 1001100010000010 0001100111011001001000 |
| 112 | $\{1,4,5,9\}$ | $\{20,21,24,25,26,28,29,30,32,33,35,36\}$ | 16 | 1001100010000000 0001100111011101101010 |
| 113 | $\{4,5,9,14,15\}$ | $\{17,18,19,20,21,24,25,26,28\}$ | 14 | 0001100010000110 1111100111010000000000 |
| 114 | $\{4,5,9,14\}$ | $\{17,18,19,20,21,24,25,26,28,30,33,36\}$ | 16 | 0001100010000100 1111100111010100100010 |
| 115 | $\{4,5,9,15\}$ | $\{17,18,19,20,21,24,25,26,28,29,32,35\}$ | 16 | 0001100010000010 1111100111011001001000 |
| 116 | $\{4,5,9\}$ | $\{17,18,19,20,21,24,25,26,28,29,30,32,33,35,36\}$ | 18 | 0001100010000000 1111100111011101101010 |
| 117 | $\{1,4,5,10,15\}$ | $\{20,21,24,25,26,27,29,32,37\}$ | 14 | 1001100001000010 0001100111010010001000 |
| 118 | $\{1,4,5,10\}$ | $\{20,21,24,25,26,27,29,30,32,33,36,37\}$ | 16 | 1001100001000000 0001100111011011011010 |
| 119 | $\{4,5,10,12,15\}$ | $\{17,19,20,25,26,29,32,37\}$ | 13 | 0001100001010010 1011000011001001000010 |
| 120 | $\{4,5,10,12\}$ | $\{17,19,20,25,26,29,30,32,33,36,37\}$ | 15 | 0001100001010000 1011000011001101101011 |
| 121 | $\{4,5,10,15\}$ | $\{17,18,19,20,21,24,25,26,27,29,32,37\}$ | 16 | 0001100001000010 1111100111010010001000 |
| 122 | $\{4,5,10\}$ | $\{17,18,19,20,21,24,25,26,27,29,30,32,33,36,37\}$ | 18 | 0001100001000000 1111100111011011010011 |
| 123 | $\{1,4,6,8\}$ | $\{20,21,23,25,27,28,29,30,33,34,35,37\}$ | 16 | 1001010100000000 0001101010111100111101 |
| 124 | $\{4,6,8\}$ | $\{17,18,19,20,21,23,25,27,28,29,30,33,34,35,37\}$ | 18 | 0001010100000000 1111101010111100111101 |
| 125 | $\{1,4,6,9,15\}$ | $\{20,21,23,25,26,28,29,34,35\}$ | 14 | 1001010010000010 0001101010110000110000 |
| 126 | $\{1,4,6,9\}$ | $\{20,21,23,25,26,28,29,30,33,34,35,36\}$ | 16 | 1001010010000000 0001101010111001111010 |
| 127 | $\{4,6,9,11,15\}$ | $\{18,19,21,25,28,29,34,35\}$ | 13 | 0001010010100010 0110100010011000011000 |
| 128 | $\{4,6,9,11\}$ | $\{18,19,21,25,28,29,30,33,34,35,36\}$ | 15 | 0001010010100000 0110100010011101111000 |
| 129 | $\{4,6,9,15\}$ | $\{17,18,19,20,21,23,25,26,28,29,34,35\}$ | 16 | 0001010010000010 1111101011011000011000 |
| 130 | $\{4,6,9\}$ | $\{17,18,19,20,21,23,25,26,28,29,30,33,34,35,36\}$ | 18 | 0001010010000000 1111101010110111001111 |
| 131 | $\{1,4,6,10,15\}$ | $\{20,21,23,25,26,27,29,34,37\}$ | 14 | 1001010001000010 0001101011010000010001 |
| 132 | $\{1,4,6,10\}$ | $\{20,21,23,25,26,27,29,30,33,34,36,37\}$ | 16 | 1001010001000000 0001101011011011011011 |
| 133 | $\{4,6,10,11,15\}$ | $\{18,19,21,25,27,29,34,37\}$ | 13 | 0001010001100010 0110100010101000010001 |

*Continued on the next page.*

TABLE S74: Canonical enumeration of all 153 two-valued states, in the canonical order defined above. The support is split into core and completion vertices. The last column is $(s_r(1), \ldots, s_r(16)\,;\, s_r(17), \ldots, s_r(37))$. (continued).

| $r$ | Core support | Completion support | $\lvert\operatorname{supp} s_r\rvert$ | 37-bit state vector |
|---|---|---|---|---|
| 134 | $\{4, 6, 10, 11\}$ | $\{18, 19, 21, 25, 27, 29, 30, 33, 34, 36, 37\}$ | 15 | 0001010001100000 0110100010101100 11011 |
| 135 | $\{4, 6, 10, 15\}$ | $\{17, 18, 19, 20, 21, 23, 25, 26, 27, 29, 34, 37\}$ | 16 | 0001010001000010 1111101011101000 01001 |
| 136 | $\{4, 6, 10\}$ | $\{17, 18, 19, 20, 21, 23, 25, 26, 27, 29, 30, 33, 34, 36, 37\}$ | 18 | 0001010001000000 1111101011101100 11011 |
| 137 | $\{1, 4, 7, 8, 14\}$ | $\{20, 21, 23, 24, 27, 28, 30, 34, 37\}$ | 14 | 1001001100000100 0001101100110100 01001 |
| 138 | $\{1, 4, 7, 8\}$ | $\{20, 21, 23, 24, 27, 28, 29, 30, 32, 34, 35, 37\}$ | 16 | 1001001100000000 0001101100111110 10101 |
| 139 | $\{4, 7, 8, 12, 14\}$ | $\{17, 19, 20, 23, 28, 30, 34, 37\}$ | 13 | 0001001100010100 1011001000010100 01001 |
| 140 | $\{4, 7, 8, 12\}$ | $\{17, 19, 20, 23, 28, 29, 30, 32, 34, 35, 37\}$ | 15 | 0001001100010000 1011001000011101 01101 |
| 141 | $\{4, 7, 8, 14\}$ | $\{17, 18, 19, 20, 21, 23, 24, 27, 28, 30, 34, 37\}$ | 16 | 0001001100000100 1111101100110100 01001 |
| 142 | $\{4, 7, 8\}$ | $\{17, 18, 19, 20, 21, 23, 24, 27, 28, 29, 30, 32, 34, 35, 37\}$ | 18 | 0001001100000000 1111101100111101 01101 |
| 143 | $\{1, 4, 7, 9, 14\}$ | $\{20, 21, 23, 24, 26, 28, 30, 34, 36\}$ | 14 | 1001001010000100 0001101101010100 01010 |
| 144 | $\{1, 4, 7, 9\}$ | $\{20, 21, 23, 24, 26, 28, 29, 30, 32, 34, 35, 36\}$ | 16 | 1001001010000000 0001101101011101 01110 |
| 145 | $\{4, 7, 9, 11, 14\}$ | $\{18, 19, 21, 24, 28, 30, 34, 36\}$ | 13 | 0001001010100100 0110100100010100 01010 |
| 146 | $\{4, 7, 9, 11\}$ | $\{18, 19, 21, 24, 28, 29, 30, 32, 34, 35, 36\}$ | 15 | 0001001010100000 0110100100011101 01110 |
| 147 | $\{4, 7, 9, 14\}$ | $\{17, 18, 19, 20, 21, 23, 24, 26, 28, 30, 34, 36\}$ | 16 | 0001001010000100 1111101101010100 01010 |
| 148 | $\{4, 7, 9\}$ | $\{17, 18, 19, 20, 21, 23, 24, 26, 28, 29, 30, 32, 34, 35, 36\}$ | 18 | 0001001010000000 1111101101011101 01110 |
| 149 | $\{1, 4, 7, 10\}$ | $\{20, 21, 23, 24, 26, 27, 29, 30, 32, 34, 36, 37\}$ | 16 | 1001001001000000 0001101101011010 01011 |
| 150 | $\{4, 7, 10, 11, 12\}$ | $\{19, 29, 30, 32, 34, 36, 37\}$ | 12 | 0001001001110000 0010000000011010 10011 |
| 151 | $\{4, 7, 10, 11\}$ | $\{18, 19, 21, 24, 27, 29, 30, 32, 34, 36, 37\}$ | 15 | 0001001001100000 0110100100101101 01011 |
| 152 | $\{4, 7, 10, 12\}$ | $\{17, 19, 20, 23, 26, 29, 30, 32, 34, 36, 37\}$ | 15 | 0001001001010000 1011001001001101 01011 |
| 153 | $\{4, 7, 10\}$ | $\{17, 18, 19, 20, 21, 23, 24, 26, 27, 29, 30, 32, 34, 36, 37\}$ | 18 | 0001001001000000 1111101101101101 01011 |

### *S2.27.3. A minimum six-state unital, separating family*

Define $\Omega_6 = \{\sigma_1, \ldots, \sigma_6\}$ in the following order:

$$(\sigma_1, \ldots, \sigma_6) = (s_3, s_{31}, s_{75}, s_{89}, s_{113}, s_{151}). \tag{S15}$$

Table S75 prints their complete supports. Table S76 gives all 37 profiles $(\sigma_1(i), \ldots, \sigma_6(i))$ and their active state sets. Every profile is nonzero and no two coincide, so $\Omega_6$ is unital and separating. Since five states supply at most $2^5 - 1 = 31 < 37$ nonzero binary profiles, no smaller unital, separating family can exist.

TABLE S75: Complete supports and state vectors of the minimum unital, separating family $\Omega_6$. The full state indices are those of Table S74.

| State | Full index | Complete support | Weight | 37-bit vector |
|---|---|---|---|---|
| $\sigma_1$ | $s_3$ | $\{2, 5, 8, 12, 17, 19, 22, 25, 28, 30, 31, 32, 33, 35, 37\}$ | 15 | 0100100100010000 1010010010011110101 |
| $\sigma_2$ | $s_{31}$ | $\{1, 2, 6, 10, 15, 21, 22, 23, 25, 26, 27, 31, 34, 37\}$ | 14 | 1100010001000010 0000111011100010010 01 |
| $\sigma_3$ | $s_{75}$ | $\{3, 6, 9, 13, 17, 18, 20, 23, 26, 29, 31, 33, 34, 35, 36\}$ | 15 | 0010010010001000 1101001001001011011 10 |
| $\sigma_4$ | $s_{89}$ | $\{3, 7, 8, 16, 17, 18, 19, 20, 22, 23, 24, 27, 28, 29, 32, 35\}$ | 16 | 0010001100000001 1111011100111001001 00 |
| $\sigma_5$ | $s_{113}$ | $\{4, 5, 9, 14, 15, 17, 18, 19, 20, 21, 24, 25, 26, 28\}$ | 14 | 0001100010000110 1111100111010000000 00 |
| $\sigma_6$ | $s_{151}$ | $\{4, 7, 10, 11, 18, 19, 21, 24, 27, 29, 30, 32, 34, 36, 37\}$ | 15 | 0001001001100000 0110100100101101011 |

TABLE S76: All 37 vertex profiles on $\Omega_6$, in two panels. Here $X_i^{(6)} = \{j : \sigma_j(i) = 1\}$, $W_6 = |X_i^{(6)}|$, and $W_{153}$ is the weight across the full state space. The six-bit rows are nonzero and pairwise distinct.

| $i$ | Ray | $X_i^{(6)}$ | Profile | $W_6$ | $W_{153}$ | $i$ | Ray | $X_i^{(6)}$ | Profile | $W_6$ | $W_{153}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | $z$ | $\{2\}$ | 010000 | 1 | 54 | 20 | $B_{00}$ | $\{3,4,5\}$ | 001110 | 3 | 86 |
| 2 | $M_{00}$ | $\{1,2\}$ | 110000 | 2 | 51 | 21 | $B_{01}$ | $\{2,5,6\}$ | 010011 | 3 | 86 |
| 3 | $M_{01}$ | $\{3,4\}$ | 001100 | 2 | 51 | 22 | $B_{02}$ | $\{1,2,4\}$ | 110100 | 3 | 86 |
| 4 | $M_{02}$ | $\{5,6\}$ | 000011 | 2 | 51 | 23 | $B_{10}$ | $\{2,3,4\}$ | 011100 | 3 | 86 |
| 5 | $M_{10}$ | $\{1,5\}$ | 100010 | 2 | 51 | 24 | $B_{11}$ | $\{4,5,6\}$ | 000111 | 3 | 86 |
| 6 | $M_{11}$ | $\{2,3\}$ | 011000 | 2 | 51 | 25 | $B_{12}$ | $\{1,2,5\}$ | 110010 | 3 | 86 |
| 7 | $M_{12}$ | $\{4,6\}$ | 000101 | 2 | 51 | 26 | $B_{20}$ | $\{2,3,5\}$ | 011010 | 3 | 86 |
| 8 | $M_{20}$ | $\{1,4\}$ | 100100 | 2 | 51 | 27 | $B_{21}$ | $\{2,4,6\}$ | 010101 | 3 | 86 |
| 9 | $M_{21}$ | $\{3,5\}$ | 001010 | 2 | 51 | 28 | $B_{22}$ | $\{1,4,5\}$ | 100110 | 3 | 86 |
| 10 | $M_{22}$ | $\{2,6\}$ | 010001 | 2 | 51 | 29 | $C_{00}$ | $\{3,4,6\}$ | 001101 | 3 | 76 |
| 11 | $P_0$ | $\{6\}$ | 000001 | 1 | 16 | 30 | $C_{01}$ | $\{1,6\}$ | 100001 | 2 | 76 |
| 12 | $P_1$ | $\{1\}$ | 100000 | 1 | 16 | 31 | $C_{02}$ | $\{1,2,3\}$ | 111000 | 3 | 76 |
| 13 | $P_2$ | $\{3\}$ | 001000 | 1 | 16 | 32 | $C_{10}$ | $\{1,4,6\}$ | 100101 | 3 | 76 |
| 14 | $Q_0$ | $\{5\}$ | 000010 | 1 | 26 | 33 | $C_{11}$ | $\{1,3\}$ | 101000 | 2 | 76 |
| 15 | $Q_1$ | $\{2,5\}$ | 010010 | 2 | 26 | 34 | $C_{12}$ | $\{2,3,6\}$ | 011001 | 3 | 76 |
| 16 | $Q_2$ | $\{4\}$ | 000100 | 1 | 26 | 35 | $C_{20}$ | $\{1,3,4\}$ | 101100 | 3 | 76 |
| 17 | $A_0$ | $\{1,3,4,5\}$ | 101110 | 4 | 83 | 36 | $C_{21}$ | $\{3,6\}$ | 001001 | 2 | 76 |
| 18 | $A_1$ | $\{3,4,5,6\}$ | 001111 | 4 | 83 | 37 | $C_{22}$ | $\{1,2,6\}$ | 110001 | 3 | 76 |
| 19 | $A_2$ | $\{1,4,5,6\}$ | 100111 | 4 | 83 | | | | | | |

### *S2.27.4. All full 153-state vertex profiles*

Table S77 prints the transpose of the complete state table: the full 153-bit profile of every vertex, in three consecutive blocks of 51 states. Thus it also enumerates the complete active state set $X_i = \{s_r : s_r(i) = 1\}$ for every vertex: a one in position $r$ means exactly $s_r \in X_i$. The row weights are 54 for $z$, 51 for $M$, 16 for $P$, 26 for $Q$, 83 for $A$, 86 for $B$, and 76 for $C$.

TABLE S77: Complete 153-bit vertex profiles, with consecutive 51-bit blocks. Every row is nonzero and all 37 rows are distinct.

| $i$ | Ray | $W_{153}$ | $s_1 \dots s_{51}$ | $s_{52} \dots s_{102}$ | $s_{103} \dots s_{153}$ |
|---|---|---|---|---|---|
| 1 | $z$ | 54 | 100001100001100001100001100001111000011000010110000 | 110000110000101100001000011000011110000110000110000 | 110000111100001100001011000011000011000011000010000 |
| 2 | $M_{00}$ | 51 | 111111111111111111111111111111111111111111111111111 | 000000000000000000000000000000000000000000000000000 | 000000000000000000000000000000000000000000000000000 |
| 3 | $M_{01}$ | 51 | 000000000000000000000000000000000000000000000000000 | 111111111111111111111111111111111111111111111111111 | 000000000000000000000000000000000000000000000000000 |
| 4 | $M_{02}$ | 51 | 000000000000000000000000000000000000000000000000000 | 000000000000000000000000000000000000000000000000000 | 111111111111111111111111111111111111111111111111111 |
| 5 | $M_{10}$ | 51 | 111111111111111100000000000000000000000000000000000 | 111111111111100000000000000000000000000000000000000 | 111111111111111111000000000000000000000000000000000 |
| 6 | $M_{11}$ | 51 | 000000000000000011111111111111111111000000000000000 | 000000000000011111111111111111000000000000000000000 | 000000000000000000111111111111100000000000000000000 |
| 7 | $M_{12}$ | 51 | 000000000000000000000000000000000111111111111111111 | 000000000000000000000000000000011111111111111111111 | 000000000000000000000000000000011111111111111111111 |
| 8 | $M_{20}$ | 51 | 111110000000000011111100000000000001111110000000000 | 111111000000011111100000000011111111000000000000000 | 111111000000000000011000000000001111110000000000000 |
| 9 | $M_{21}$ | 51 | 000001111110000000000011111100000000000011000000000 | 000000111111000000011111000000000000011111100000000 | 000000111111100000001111110000000000001111110000000 |
| 10 | $M_{22}$ | 51 | 000000000001111110000000000011111111000000001111111 | 000000000001100000000011111100000000000001111111111 | 000000000000011111100000001111111000000000011111111 |
| 11 | $P_0$ | 16 | 000000000000000000000000000000000000000000000000000 | 000000000000000000001100001100000000001100001100000 | 000000000000000000000011000011000000000110000110000 |
| 12 | $P_1$ | 16 | 011000000000110000000000000000000000011000000110000 | 000000000000000000000000000000000000000000000000000 | 001100000000000110000000000000000011000000001010000 |
| 13 | $P_2$ | 16 | 010100011000000000110000110000000000000000000000000 | 001100001100000011000101000000000000000000000000000 | 000000000000000000000000000000000000000000000000000 |
| 14 | $Q_0$ | 26 | 000000000000000000000000000000000000000000000000000 | 101010101010000000000000000000011001100101010000000 | 101010110011000000000000000000000010101010101000000 |
| 15 | $Q_1$ | 26 | 000001010101010100000001010101100110000000000000000 | 000000000000000000000000000000000000000000000000000 | 000000101010101010100010101010101000000000000000000 |
| 16 | $Q_2$ | 26 | 000000000000000001010100000001010101010101000101010 | 000000000000001010100000010101010101010000000101010 | 000000000000000000000000000000000000000000000000000 |
| 17 | $A_0$ | 83 | 011110011110011100111001111001111000011110011110101001111 | 001111001111010011110001100001100001111000011000011 | 001111000011110011110100001100001100011001111000011100011 |
| 18 | $A_1$ | 83 | 000110011110000110011110011110000111100001101000011 | 001111001111010011110111001110000111100111001111 | 000011000011110000110100111001111000011001111001101 |
| 19 | $A_2$ | 83 | 001010000110011110000110000110000111100111010011111 | 000011000011010000110010100111000011110011110011111 | 001111000011110011110100111001111000111001111011111 |
| 20 | $B_{00}$ | 86 | 000000000000000000000000000000000000000000000000000 | 111111111111111111111111111001111001111111111100111110011 | 111111111111111111111111111001111001111001111111111001110011 |
| 21 | $B_{01}$ | 86 | 100111111111100111111111111111111111111001111110011 | 000000000000000000000000000000000000000000000000000000000 | 110011111111111001111111111111111001111111110101 |
| 22 | $B_{02}$ | 86 | 101011100111111111100111100111111111111111111111111 | 110011100111111001110101111111111111111111111111111 | 000000000000000000000000000000000000000000000000000000000 |
| 23 | $B_{10}$ | 86 | 000000000000000011111111111111111111111111111111111 | 000000000000001111111100111001111111111001111001111 | 000000000000000000000111001111001111111111001110011 |
| 24 | $B_{11}$ | 86 | 100111111111001100000000000000000001100111111011 | 111111111111100000000000000011111111111111111111 | 110011111111111001100000000000001100111111110101 |
| 25 | $B_{12}$ | 86 | 101011100111111111100111100111111111100000000000000 | 110011100111111001110101111111000000000000000000 | 111111111111111111111111111111111100000000000000000 |
| 26 | $B_{20}$ | 86 | 000001111111111100000011111111111111000000111111 | 000000111111110000010011100110000000110011110011 | 000000111111111111100110011100110000011001110011 |
| 27 | $B_{21}$ | 86 | 100110000001100111111110000001111111100100111011 | 111111000000111111100000111111111000000111111111 | 110011000000011001111000001111111001100000010101 |
| 28 | $B_{22}$ | 86 | 101011100110000011001110011000000011111111000000 | 110011100110011001110101000001111111111111000000 | 111111111111100000011111111000000111111111100000 |
| 29 | $C_{00}$ | 76 | 000000000000000000000000000000000000000000000000000 | 010101010101111111111111111111111001100110101011111 | 010101001100111111111111111111111010101010101111111 |
| 30 | $C_{01}$ | 76 | 111110101010101011111111010101001100111111111111111 | 000000000000000000000000000000000000000000000000000 | 111111010101010101011110101010101011111111111111111 |
| 31 | $C_{02}$ | 76 | 111111111111111110101011111110101010101011010101 | 111111111111101010111110101010101011111110101 | 000000000000000000000000000000000000000000000000000 |
| 32 | $C_{10}$ | 76 | 111111111111111100000000000000000001111111111111 | 010101010101110000000000000000001100110101011111 | 010101001100111111110000000000000010101010111111 |
| 33 | $C_{11}$ | 76 | 111110101010101011111110101010011001100000000000 | 111111111111111111111111111111000000000000000000 | 111111010101010101011101010101010100000000000000 |
| 34 | $C_{12}$ | 76 | 000000000000000010101111110101010101011010101 | 000000000000010101111110101010101011111110101 | 000000000000000000011111111111111111111111111111 |
| 35 | $C_{20}$ | 76 | 111111111110000011111111111100000000111111110000 | 010101010100111111111110000000011001101010100000 | 010101001100110000011111110000000010101010100000 |
| 36 | $C_{21}$ | 76 | 000000101010101000000010101001100110000011111111 | 000000111111110000011111111110000000111111111111 | 000000010101010101010001010101010100000011111111111 |
| 37 | $C_{22}$ | 76 | 111110000001111110101010000000101010101010100010101 | 111111000000110101010000001010101010100000010101 | 111111000000001111111110000011111111111110000011111 |

### *S2.27.5. The six-state double cover*

The fractional-coloring certificate uses a second six-state family,

$$\mathcal{R}_6 = \{s_2, s_{30}, s_{73}, s_{99}, s_{109}, s_{139}\}. \tag{S16}$$

Table S78 specifies these states in the branch parametrization. Each of the 37 vertices occurs in exactly two of their supports. Thus weights 1/2 give a fractional coloring of total weight three, and uniform weights 1/6 reproduce the maximally mixed probabilities 1/3 for every ray. The family $\Omega_6$ in Eq. (S15) supplies the separating certificate; $\mathcal{R}_6$ supplies the double-cover certificate.

TABLE S78: Explicit six-state double cover $\mathcal{R}_6$. Omitted $P, Q$ values are zero; completion values follow from the state equations.

| $j$ | State | $t$ | $\varepsilon$ | Active $P$ subscripts | Active $Q$ subscripts |
|---|---|---|---|---|---|
| 1 | $s_2$ | $(0,0,0)$ | 0 | $\{1,2\}$ | $\varnothing$ |
| 2 | $s_{30}$ | $(0,1,2)$ | 1 | $\varnothing$ | $\{1,2\}$ |
| 3 | $s_{73}$ | $(1,1,1)$ | 0 | $\{0,2\}$ | $\varnothing$ |
| 4 | $s_{99}$ | $(1,2,2)$ | 0 | $\{0\}$ | $\{2\}$ |
| 5 | $s_{109}$ | $(2,0,1)$ | 1 | $\varnothing$ | $\{0,1\}$ |
| 6 | $s_{139}$ | $(2,2,0)$ | 0 | $\{1\}$ | $\{0\}$ |

### *S2.27.6. Order determination and exact verification*

The companion file `sources/hesse/hesse37-canonical-evaluations.txt` records the 24 numbered contexts and all 153 binary state vectors. The parametrization in this appendix defines their canonical order independently of any external program.

Extend the canonical states to the 76 events of $L(H_{37})$ as in Sec. V A. If $X_x = \{r : s_r(x) = 1\}$, then the full profiles satisfy

$$X_x \subseteq X_y \quad \Longleftrightarrow \quad x \leq y. \tag{S17}$$

The verifier checks all $76^2 = 5{,}776$ ordered event pairs against the incidence order in Eq. (S27). Whenever $x \not\leq y$, the companion file `sources/hesse/hesse37-order-witnesses.csv` gives the least canonical index $r$ with $s_r(x) = 1$ and $s_r(y) = 0$. Its event labels `a1`,...,`a37` denote atoms and `c1`,...,`c37` their complements; `0` and `1` denote the bounds. Every one of the 594 distinct nonorthogonal atom pairs also has a joint-one state. Thus the full family is order determining; the six-state family $\Omega_6$ supplies the minimum unital, separating vertex representation in Eq. (S26). The distinct minima for event separation and order determination are certified in Appendix S3.3.

The verifier `hesse37-verify.py` uses exact rational arithmetic in $\mathbb{Q}(\zeta)$, with no numerical tolerance or external solver. It checks distinct rays, all 666 Hermitian inner products, the 24 bases, all graph triangles, linearity, the chromatic number and core criticality, all 153 states, the minimum separating family, both profile tables, the maximum squared ray overlap, the six-state double cover, and all event-order comparisons. It also compares the reconstructed state vectors with the archived enumeration. The accompanying machine-readable data give the same numbering and context ordering as the figure and these appendices. From the extracted joint project directory, run

```
python verification/verify-integrated.py
```

## S2.28. Four-dimensional $H_{\mathrm{YO}}^{(4)}$ and $H_{65}$: states and profiles

This appendix provides the complete enumeration of all two-valued states on the four-dimensional examples $H_{\mathrm{YO}}^{(4)}$ and $H_{65}$: the 25 two-valued states of the non-linear warm-up hypergraph $H_{\mathrm{YO}}^{(4)}$ (Appendix S2.28.1), and the complete branch-by-branch parametrization of all $24 \times 2^{14} = 393{,}216$ two-valued states—together with the 48 canonical unital and separating states $s_{\alpha,\varepsilon}$ and the 65 vertex state-profiles—of the canonical linear hypergraph $H_{65}$ (Appendix S2.28.2).

### *S2.28.1. Complete enumeration of the* 25 *two-valued states on* $H^{(4)}_{\mathrm{YO}}$

Table S79 lists all 25 two-valued states $s_1, \dots, s_{25} : V_{26} \to \{0,1\}$ on $H^{(4)}_{\mathrm{YO}} = (V_{26}, E^{(4)}_{16})$. States $s_1, \dots, s_{24}$ have $s_r(26) = 0$ and correspond to the 24 two-valued states of the completed Yu–Oh hypergraph $H^{+}_{\mathrm{YO}}$ in $\mathbb{R}^3$ (grouped into 12 pairs of states sharing the same support on the three coordinate contexts $\{1,2,3\}$, $\{4,5,6\}$, $\{7,8,9\}$), while $s_{25}$ is the unique apex state with support $\{26\}$. Table S80 displays the transposed $26 \times 25$ vertex-by-state binary matrix, listing for each vertex $i \in \{1, \dots, 26\}$ its state weight $\sum_{r=1}^{25} s_r(i) \in \{1, 3, 8, 13\}$, the set of states $A_i = \{s_r : s_r(i) = 1\}$, and its 25-bit profile $(s_1(i), \dots, s_{25}(i))$.

Inspection of Table S80 highlights three structural properties of the state space $\mathcal{S}(H^{(4)}_{\mathrm{YO}})$:

1. *Unitality and atom separation:* Every row of Table S80 has positive weight (1, 3, 8, or 13) and all 26 binary rows are distinct ($A_i \neq A_j$ for all $1 \le i < j \le 26$), so $\mathcal{S}(H^{(4)}_{\mathrm{YO}})$ is nonempty, unital, and separating on atoms.

2. *Exclusion of simultaneous* 1*s on the non-orthogonal triad atoms* $\{10, 11, 12, 13\}$*:* Although the four Yu–Oh triad rays $v_{10}, v_{11}, v_{12}, v_{13}$ are pairwise non-orthogonal ($v_i \cdot v_j = \pm 1 \neq 0$ for $10 \le i < j \le 13$), their active state sets in Table S80 are pairwise disjoint:

$$\begin{aligned} A_{10} &= \{s_{13}, s_{19}, s_{23}\}, \quad A_{11} = \{s_5, s_7, s_{15}\}, \\ A_{12} &= \{s_1, s_3, s_9\}, \qquad A_{13} = \{s_{11}, s_{17}, s_{21}\}. \end{aligned} \tag{S18}$$

   Consequently, $\sum_{i=10}^{13} s(i) \le 1$ holds for every $s \in \mathcal{S}(H^{(4)}_{\mathrm{YO}})$, and no two-valued state assigns $(1,1)$ to any of the $\binom{4}{2} = 6$ non-orthogonal pairs $\{10,11\}$, $\{10,12\}$, $\{10,13\}$, $\{11,12\}$, $\{11,13\}$, $\{12,13\}$.

3. *Failure of order-determinacy:* Even though $A_i \neq A_j$ for all $i \neq j$, the state space $\mathcal{S}(H^{(4)}_{\mathrm{YO}})$ is not order-determining on atoms or compound propositions[4]. In particular, each triad atom $t \in \{10, 11, 12, 13\}$ (*weight* 3) is strictly dominated across all 25 states ($A_t \subsetneq A_w$, i.e., $s(t) \le s(w)$ for all $s \in \mathcal{S}(H^{(4)}_{\mathrm{YO}})$) by three non-orthogonal completion atoms $w \in \{14, \dots, 25\}$ (*weight* 13):

$$\begin{aligned} A_{10} &\subsetneq A_{16},\, A_{20},\, A_{24}, \quad A_{11} \subsetneq A_{18},\, A_{21},\, A_{25}, \\ A_{12} &\subsetneq A_{14},\, A_{22},\, A_{23}, \quad A_{13} \subsetneq A_{15},\, A_{17},\, A_{19}, \end{aligned} \tag{S19}$$

   together with compound-event inclusions such as $A_{12} \subsetneq A_3 \dot\cup A_{14}$, $A_{11} \subsetneq A_3 \dot\cup A_{25}$, and their unions with the apex set $A_{26} = \{s_{25}\}$.

TABLE S79: Complete enumeration of the 25 two-valued states $s_1, \dots, s_{25} : V_{26} \to \{0,1\}$ on the non-linear warm-up hypergraph $H^{(4)}_{\mathrm{YO}} = (V_{26}, E^{(4)}_{16})$, showing for each state its support on the 13-ray Yu–Oh core $\{1, \dots, 13\}$, its forced support on $\{14, \dots, 26\}$, its total support size $|S_r|$, and its 26-bit binary vector $(s_r(1 \dots 13)\, s_r(14 \dots 25)\, s_r(26))$.

| State | Core support | Completion/apex support | $\lvert S_r\rvert$ | 26-bit state vector |
|---|---|---|---|---|
| $s_1$ | $\{1,4,8,12\}$ | $\{14,15,21,22,23\}$ | 9 | 1001000100010 110000011100 0 |
| $s_2$ | $\{1,4,8\}$ | $\{14,15,16,17,21,22,23,25\}$ | 11 | 1001000100000 111100011101 0 |
| $s_3$ | $\{1,5,7,12\}$ | $\{14,18,22,23,24\}$ | 9 | 1000101000010 100010001110 0 |
| $s_4$ | $\{1,5,7\}$ | $\{14,16,17,18,22,23,24,25\}$ | 11 | 1000101000000 101110001111 0 |
| $s_5$ | $\{1,5,9,11\}$ | $\{17,18,21,22,25\}$ | 9 | 1000100010100 000110011001 0 |
| $s_6$ | $\{1,5,9\}$ | $\{14,15,17,18,21,22,24,25\}$ | 11 | 1000100010000 110110011011 0 |
| $s_7$ | $\{1,6,8,11\}$ | $\{16,18,21,23,25\}$ | 9 | 1000010100100 001010010101 0 |
| $s_8$ | $\{1,6,8\}$ | $\{14,15,16,18,21,23,24,25\}$ | 11 | 1000010100000 111010010111 0 |
| $s_9$ | $\{2,4,7,12\}$ | $\{14,19,20,22,23\}$ | 9 | 0101001000010 100001101100 0 |
| $s_{10}$ | $\{2,4,7\}$ | $\{14,16,17,19,20,22,23,25\}$ | 11 | 0101001000000 101101101101 0 |
| $s_{11}$ | $\{2,4,9,13\}$ | $\{14,15,17,19,25\}$ | 9 | 0101000010001 110101000001 0 |
| $s_{12}$ | $\{2,4,9\}$ | $\{14,15,17,19,20,21,22,25\}$ | 11 | 0101000010000 110101111001 0 |
| $s_{13}$ | $\{2,6,7,10\}$ | $\{14,16,20,24,25\}$ | 9 | 0100011001000 101000100011 0 |
| $s_{14}$ | $\{2,6,7\}$ | $\{14,16,18,19,20,23,24,25\}$ | 11 | 0100011000000 101011100111 0 |
| $s_{15}$ | $\{2,6,9,11\}$ | $\{18,19,20,21,25\}$ | 9 | 0100010010100 000011110001 0 |

*Continued on the next page.*

TABLE S79: All 25 states of $H_{\mathrm{YO}}^{(4)}$ (continued).

| State | Core support | Completion/apex support | $\lvert S_r\rvert$ | 26-bit state vector |
|---|---|---|---|---|
| $s_{16}$ | $\{2,6,9\}$ | $\{14,15,18,19,20,21,24,25\}$ | 11 | 0100010010000 1100111100110 |
| $s_{17}$ | $\{3,4,8,13\}$ | $\{15,16,17,19,23\}$ | 9 | 0011000100001 0111010001000 |
| $s_{18}$ | $\{3,4,8\}$ | $\{15,16,17,19,20,21,22,23\}$ | 11 | 0011000100000 0111011111000 |
| $s_{19}$ | $\{3,5,7,10\}$ | $\{16,17,20,22,24\}$ | 9 | 0010101001000 0011001010100 |
| $s_{20}$ | $\{3,5,7\}$ | $\{16,17,18,19,20,22,23,24\}$ | 11 | 0010101000000 0011111011100 |
| $s_{21}$ | $\{3,5,9,13\}$ | $\{15,17,18,19,24\}$ | 9 | 0010100010001 0101110000100 |
| $s_{22}$ | $\{3,5,9\}$ | $\{15,17,18,19,20,21,22,24\}$ | 11 | 0010100010000 0101111110100 |
| $s_{23}$ | $\{3,6,8,10\}$ | $\{15,16,20,21,24\}$ | 9 | 0010010101000 0110001100100 |
| $s_{24}$ | $\{3,6,8\}$ | $\{15,16,18,19,20,21,23,24\}$ | 11 | 0010010100000 0110111101100 |
| $s_{25}$ | $\emptyset$ | $\{26\}$ | 1 | 0000000000000 0000000000001 |

TABLE S80: Vertex state profiles across the 25 two-valued states $s_1,\dots,s_{25}$ of $H_{\mathrm{YO}}^{(4)}$, showing for each vertex $i \in \{1,\dots,26\}$ its state weight $\sum_{r=1}^{25} s_r(i)$, the active state indices $\{r : s_r(i) = 1\}$, and the 25-bit profile $(s_{1\dots12}(i)\, s_{13\dots24}(i)\, s_{25}(i))$. All 26 rows are nonzero and pairwise distinct.

| Vertex $i$ | Weight $\sum_r s_r(i)$ | Active states $\{r \in \{1,\dots,25\} : s_r(i) = 1\}$ | 25-bit profile $(s_{1\dots12}(i)\, s_{13\dots24}(i)\, s_{25}(i))$ |
|---|---|---|---|
| 1 | 8 | $\{1,2,3,4,5,6,7,8\}$ | 1111111100000 0000000000000 |
| 2 | 8 | $\{9,10,11,12,13,14,15,16\}$ | 0000000011110 1111000000000 |
| 3 | 8 | $\{17,18,19,20,21,22,23,24\}$ | 0000000000000 0000111111110 |
| 4 | 8 | $\{1,2,9,10,11,12,17,18\}$ | 1100000011110 0000110000000 |
| 5 | 8 | $\{3,4,5,6,19,20,21,22\}$ | 0011110000000 0000001111000 |
| 6 | 8 | $\{7,8,13,14,15,16,23,24\}$ | 0000001100000 1111000000110 |
| 7 | 8 | $\{3,4,9,10,13,14,19,20\}$ | 0011000011000 1100001100000 |
| 8 | 8 | $\{1,2,7,8,17,18,23,24\}$ | 1100001100000 0000110000110 |
| 9 | 8 | $\{5,6,11,12,15,16,21,22\}$ | 0000110000110 0011000011000 |
| 10 | 3 | $\{13,19,23\}$ | 0000000000000 1000001000100 |
| 11 | 3 | $\{5,7,15\}$ | 0000101000000 0010000000000 |
| 12 | 3 | $\{1,3,9\}$ | 1010000010000 0000000000000 |
| 13 | 3 | $\{11,17,21\}$ | 0000000000100 0000100010000 |
| 14 | 13 | $\{1,2,3,4,6,8,9,10,11,12,13,14,16\}$ | 1111010111110 1101000000000 |
| 15 | 13 | $\{1,2,6,8,11,12,16,17,18,21,22,23,24\}$ | 1100010100110 0001110011110 |
| 16 | 13 | $\{2,4,7,8,10,13,14,17,18,19,20,23,24\}$ | 0101001101000 1100111100110 |
| 17 | 13 | $\{2,4,5,6,10,11,12,17,18,19,20,21,22\}$ | 0101110001110 0000111111000 |
| 18 | 13 | $\{3,4,5,6,7,8,14,15,16,20,21,22,24\}$ | 0011111100000 1110001110100 |
| 19 | 13 | $\{9,10,11,12,14,15,16,17,18,20,21,22,24\}$ | 0000000011110 1111101110100 |
| 20 | 13 | $\{9,10,12,13,14,15,16,18,19,20,22,23,24\}$ | 0000000011010 1111011101110 |
| 21 | 13 | $\{1,2,5,6,7,8,12,15,16,18,22,23,24\}$ | 1100111100010 0011010001110 |
| 22 | 13 | $\{1,2,3,4,5,6,9,10,12,18,19,20,22\}$ | 1111110011010 0000011101000 |
| 23 | 13 | $\{1,2,3,4,7,8,9,10,14,17,18,20,24\}$ | 1111001111000 1001101000100 |
| 24 | 13 | $\{3,4,6,8,13,14,16,19,20,21,22,23,24\}$ | 0011010100001 1010011111110 |
| 25 | 13 | $\{2,4,5,6,7,8,10,11,12,13,14,15,16\}$ | 0101111101111 1111000000000 |
| 26 | 1 | $\{25\}$ | 0000000000000 0000000000001 |

### *S2.28.2. Complete parametrization of all* 393,216 *two-valued states and the* 48 *canonical separating states on* $H_{65}$

Every two-valued state $s \in \mathcal{S}(H_{65})$ on the canonical linear hypergraph $H_{65} = (V_{65}, E_{29})$ is uniquely indexed by a pair $(\alpha, \boldsymbol{\varepsilon})$ consisting of:

1. a branch index $\alpha \in \{1,\dots,24\}$ selecting one of the 24 four-element core state supports $S_\alpha \subset V_{\mathrm{core}} = \{1,\dots,17\}$; and
2. a 14-bit binary vector $\boldsymbol{\varepsilon} = (\varepsilon_m)_{m \in \mathcal{F}_\alpha} \in \{0,1\}^{14}$ indexed by the 14 *free completion contexts*

$$\mathcal{F}_\alpha = \big\{ m \in \{1,\dots,24\} : S_\alpha \cap \{u_m, v_m\} = \emptyset \big\},$$

whose complement is the set of 10 *hit completion contexts* $\mathcal{H}_\alpha = \{1,\dots,24\} \setminus \mathcal{F}_\alpha$.

Explicitly, the state $s_{\alpha,\boldsymbol{\varepsilon}} : V_{65} \to \{0,1\}$ takes values

$$\begin{aligned} s_{\alpha,\boldsymbol{\varepsilon}}(i) &= \mathbf{1}_{i\in S_\alpha} \qquad (1 \le i \le 17), \\ \big(s_{\alpha,\boldsymbol{\varepsilon}}(2m+16),\, s_{\alpha,\boldsymbol{\varepsilon}}(2m+17)\big) &= \begin{cases} (0,0), & m \in \mathcal{H}_\alpha, \\ (1-\varepsilon_m,\, \varepsilon_m), & m \in \mathcal{F}_\alpha. \end{cases} \end{aligned} \tag{S20}$$

Because $|\mathcal{F}_\alpha| = 14$ for every $\alpha \in \{1,\dots,24\}$, each branch $\alpha$ yields $2^{14} = 16{,}384$ states $s_{\alpha,\boldsymbol{\varepsilon}}$, giving $|\mathcal{S}(H_{65})| = 24 \times 2^{14} = 393{,}216$ two-valued states in total. Setting all 14 free bits equal to $\varepsilon \in \{0,1\}$ ($\varepsilon_m = \varepsilon$ for all $m \in \mathcal{F}_\alpha$) yields the 48 *canonical states* $s_{\alpha,0}, s_{\alpha,1}$ of Eq. (S29), which have support size $|S_\alpha| + 14 = 18$:

$$\begin{aligned} \mathrm{supp}(s_{\alpha,0}) &= S_\alpha \cup \big\{2m+16 : m \in \mathcal{F}_\alpha\big\}, \\ \mathrm{supp}(s_{\alpha,1}) &= S_\alpha \cup \big\{2m+17 : m \in \mathcal{F}_\alpha\big\}. \end{aligned} \tag{S21}$$

Table S81 enumerates all 24 branches $\alpha = 1,\dots,24$, specifying $S_\alpha$, the 10 hit context indices $\mathcal{H}_\alpha \subset \{1,\dots,24\}$, the 14 free context indices $\mathcal{F}_\alpha \subset \{1,\dots,24\}$, and the 24-bit free-context indicator string. Table S82 lists the resulting state profiles and weights for all 65 vertices of $H_{65}$ across both the 48 canonical states $\{s_{\alpha,\varepsilon}\}$ and the full space of 393,216 states: across the 48 canonical states $(s_{1,0}, s_{1,1}, \dots, s_{24,0}, s_{24,1})$, each '1' in row $i$ of Table S82 expands to '(1,1)' for core vertices $i \in \{1,\dots,17\}$, to '(1,0)' for first completion vertices $w_{m,1} = 2m+16$, and to '(0,1)' for second completion vertices $w_{m,2} = 2m+17$ (and each '0' expands to '(0,0)'), making all 65 canonical 48-bit vertex profiles nonzero and pairwise distinct by inspection.

TABLE S81: Complete branch-by-branch parametrization of the $24 \times 2^{14} = 393{,}216$ two-valued states $s_{\alpha,\boldsymbol{\varepsilon}}$ and the 48 canonical states $s_{\alpha,0}, s_{\alpha,1}$ on $H_{65}$ via Eqs. (S20)–(S21). For each $\alpha \in \{1,\dots,24\}$, the table lists the 4-element core support $S_\alpha$, the 10 hit completion context indices $m \in \mathcal{H}_\alpha$ where $(s(2m+16), s(2m+17)) = (0,0)$ is forced on $E_m$, the 14 free completion context indices $m \in \mathcal{F}_\alpha$ where $(s(2m+16), s(2m+17)) = (1-\varepsilon_m, \varepsilon_m)$ with $\varepsilon_m \in \{0,1\}$ on $E_m$, and the 24-bit indicator string $(\mathbf{1}_{1\in\mathcal{F}_\alpha} \dots \mathbf{1}_{12\in\mathcal{F}_\alpha}\, \mathbf{1}_{13\in\mathcal{F}_\alpha} \dots \mathbf{1}_{24\in\mathcal{F}_\alpha})$.

| Branch | Core support $S_\alpha$ | Hit contexts $\mathcal{H}_\alpha$ | Free contexts $\mathcal{F}_\alpha$ | Free indicator $m = 1\dots24$ |
|---|---|---|---|---|
| $S_1$ | {1,5,9,16} | {3,7,9,10,11,13,19,21,22,23} | {1,2,4,5,6,8,12,14,15,16,17,18,20,24} | 110111010001 011111010001 |
| $S_2$ | {1,5,10,14} | {3,5,6,7,11,13,17,18,19,23} | {1,2,4,8,9,10,12,14,15,16,20,21,22,24} | 110100011101 011100011101 |
| $S_3$ | {1,6,9,16} | {3,7,9,10,11,15,17,21,22,24} | {1,2,4,5,6,8,12,13,14,16,18,19,20,23} | 110111010001 110101110010 |
| $S_4$ | {1,6,11,12} | {1,2,3,7,11,13,14,15,17,24} | {4,5,6,8,9,10,12,16,18,19,20,21,22,23} | 000111011101 000101111110 |
| $S_5$ | {1,7,10,14} | {3,5,6,7,11,16,17,18,20,21} | {1,2,4,8,9,10,12,13,14,15,19,22,23,24} | 110100011101 111000100111 |
| $S_6$ | {1,7,11,12} | {1,2,3,7,11,13,14,16,20,21} | {4,5,6,8,9,10,12,15,17,18,19,22,23,24} | 000111011101 001011100111 |
| $S_7$ | {2,5,9,16} | {1,5,9,10,12,13,19,21,22,23} | {2,3,4,6,7,8,11,14,15,16,17,18,20,24} | 011101110010 011111010001 |
| $S_8$ | {2,5,11,13} | {1,3,4,5,12,13,15,16,19,23} | {2,6,7,8,9,10,11,14,17,18,20,21,22,24} | 010001111110 010011011101 |
| $S_9$ | {2,6,9,16} | {1,5,9,10,12,15,17,21,22,24} | {2,3,4,6,7,8,11,13,14,16,18,19,20,23} | 011101110010 110101110010 |
| $S_{10}$ | {2,6,10,15} | {1,5,7,8,12,15,17,19,20,24} | {2,3,4,6,9,10,11,13,14,16,18,21,22,23} | 011101001110 110101001110 |
| $S_{11}$ | {2,8,10,15} | {1,5,7,8,12,14,18,19,20,22} | {2,3,4,6,9,10,11,13,15,16,17,21,23,24} | 011101001110 101110001011 |
| $S_{12}$ | {2,8,11,13} | {1,3,4,5,12,14,15,16,18,22} | {2,6,7,8,9,10,11,13,17,19,20,21,23,24} | 010001111110 100010111011 |
| $S_{13}$ | {3,5,10,14} | {2,5,6,8,9,13,17,18,19,23} | {1,3,4,7,10,11,12,14,15,16,20,21,22,24} | 101100100111 011100011101 |
| $S_{14}$ | {3,5,11,13} | {2,3,4,8,9,13,15,16,19,23} | {1,5,6,7,10,11,12,14,17,18,20,21,22,24} | 100011100111 010011011101 |
| $S_{15}$ | {3,7,9,17} | {2,8,9,11,12,16,20,21,23,24} | {1,3,4,5,6,7,10,13,14,15,17,18,19,22} | 101111100100 111011100100 |
| $S_{16}$ | {3,7,10,14} | {2,5,6,8,9,16,17,18,20,21} | {1,3,4,7,10,11,12,13,14,15,19,22,23,24} | 101100100111 111000100111 |
| $S_{17}$ | {3,8,9,17} | {2,8,9,11,12,14,18,22,23,24} | {1,3,4,5,6,7,10,13,15,16,17,19,20,21} | 101111100100 101110111000 |
| $S_{18}$ | {3,8,11,13} | {2,3,4,8,9,14,15,16,18,22} | {1,5,6,7,10,11,12,13,17,19,20,21,23,24} | 100011100111 100010111011 |
| $S_{19}$ | {4,6,10,15} | {4,6,7,8,10,15,17,19,20,24} | {1,2,3,5,9,11,12,13,14,16,18,21,22,23} | 111010001011 110101001110 |
| $S_{20}$ | {4,6,11,12} | {1,2,4,6,10,13,14,15,17,24} | {3,5,7,8,9,11,12,16,18,19,20,21,22,23} | 001010111011 000101111110 |
| $S_{21}$ | {4,7,9,17} | {4,6,10,11,12,16,20,21,23,24} | {1,2,3,5,7,8,9,13,14,15,17,18,19,22} | 111010111000 111011100100 |
| $S_{22}$ | {4,7,11,12} | {1,2,4,6,10,13,14,16,20,21} | {3,5,7,8,9,11,12,15,17,18,19,22,23,24} | 001010111011 001011100111 |
| $S_{23}$ | {4,8,9,17} | {4,6,10,11,12,14,18,22,23,24} | {1,2,3,5,7,8,9,13,15,16,17,19,20,21} | 111010111000 101110111000 |
| $S_{24}$ | {4,8,10,15} | {4,6,7,8,10,14,18,19,20,22} | {1,2,3,5,9,11,12,13,15,16,17,21,23,24} | 111010001011 101110001011 |

TABLE S82: Vertex state profiles and weights for all 65 vertices of $H_{65}$, showing for each core vertex $i \in \{1,\dots,17\}$ and each completion pair $(w_{m,1}, w_{m,2}) = (2m+16, 2m+17)$ $(m = 1,\dots,24)$ the number of active branches $W_{24}$, the weight $W_{48} = \sum_{\alpha,\varepsilon} s_{\alpha,\varepsilon}(i)$ across the 48 canonical states, the weight $W_{393,216} = \sum_{s\in\mathcal{S}(H_{65})} s(i)$ across all 393,216 states, the active branch indices $\alpha \in \{1,\dots,24\}$ (equal to $Z(e_m)$ for $(w_{m,1}, w_{m,2})$), and the 24-bit branch profile across $S_1,\dots,S_{24}$ (in which each `1` expands to `11` for core vertices $i$, to `10` for $w_{m,1}$, and to `01` for $w_{m,2}$ across the 48 canonical states).

| Vertex / pair | $W_{24}$ | $W_{48}$ | $W_{393,216}$ | Active branches $\alpha$ | 24-bit branch profile |
|---|---|---|---|---|---|
| Core 1 | 6 | 12 | 98,304 | {1, 2, 3, 4, 5, 6} | 11111100 00000000 00000000 |
| Core 2 | 6 | 12 | 98,304 | {7, 8, 9, 10, 11, 12} | 00000011 11110000 00000000 |
| Core 3 | 6 | 12 | 98,304 | {13, 14, 15, 16, 17, 18} | 00000000 00001111 11000000 |
| Core 4 | 6 | 12 | 98,304 | {19, 20, 21, 22, 23, 24} | 00000000 00000000 00111111 |
| Core 5 | 6 | 12 | 98,304 | {1, 2, 7, 8, 13, 14} | 11000011 00001100 00000000 |
| Core 6 | 6 | 12 | 98,304 | {3, 4, 9, 10, 19, 20} | 00110000 11000000 00110000 |
| Core 7 | 6 | 12 | 98,304 | {5, 6, 15, 16, 21, 22} | 00001100 00000011 00001100 |
| Core 8 | 6 | 12 | 98,304 | {11, 12, 17, 18, 23, 24} | 00000000 00110000 11000011 |
| Core 9 | 8 | 16 | 131,072 | {1, 3, 7, 9, 15, 17, 21, 23} | 10100010 10000010 10001010 |
| Core 10 | 8 | 16 | 131,072 | {2, 5, 10, 11, 13, 16, 19, 24} | 01001000 01101001 00100001 |
| Core 11 | 8 | 16 | 131,072 | {4, 6, 8, 12, 14, 18, 20, 22} | 00010101 00010100 01010100 |
| Core 12 | 4 | 8 | 65,536 | {4, 6, 20, 22} | 00010100 00000000 00010100 |
| Core 13 | 4 | 8 | 65,536 | {8, 12, 14, 18} | 00000001 00010100 01000000 |
| Core 14 | 4 | 8 | 65,536 | {2, 5, 13, 16} | 01001000 00001001 00000000 |
| Core 15 | 4 | 8 | 65,536 | {10, 11, 19, 24} | 00000000 01100000 00100001 |
| Core 16 | 4 | 8 | 65,536 | {1, 3, 7, 9} | 10100010 10000000 00000000 |
| Core 17 | 4 | 8 | 65,536 | {15, 17, 21, 23} | 00000000 00000010 10001010 |
| $E_1$ : (18, 19) | 14 | 14 | 114,688 | {1, 2, 3, 5, 13, 14, 15, 16, 17, 18, 19, 21, 23, 24} | 11101000 00001111 11101011 |
| $E_2$ : (20, 21) | 14 | 14 | 114,688 | {1, 2, 3, 5, 7, 8, 9, 10, 11, 12, 19, 21, 23, 24} | 11101011 11110000 00101011 |
| $E_3$ : (22, 23) | 14 | 14 | 114,688 | {7, 9, 10, 11, 13, 15, 16, 17, 19, 20, 21, 22, 23, 24} | 00000010 11101011 10111111 |
| $E_4$ : (24, 25) | 14 | 14 | 114,688 | {1, 2, 3, 4, 5, 6, 7, 9, 10, 11, 13, 15, 16, 17} | 11111110 11101011 10000000 |
| $E_5$ : (26, 27) | 14 | 14 | 114,688 | {1, 3, 4, 6, 14, 15, 17, 18, 19, 20, 21, 22, 23, 24} | 10110100 00000110 11111111 |
| $E_6$ : (28, 29) | 14 | 14 | 114,688 | {1, 3, 4, 6, 7, 8, 9, 10, 11, 12, 14, 15, 17, 18} | 10110111 11110110 11000000 |
| $E_7$ : (30, 31) | 14 | 14 | 114,688 | {7, 8, 9, 12, 13, 14, 15, 16, 17, 18, 20, 21, 22, 23} | 00000011 10011111 11011110 |
| $E_8$ : (32, 33) | 14 | 14 | 114,688 | {1, 2, 3, 4, 5, 6, 7, 8, 9, 12, 20, 21, 22, 23} | 11111111 10010000 00011110 |
| $E_9$ : (34, 35) | 14 | 14 | 114,688 | {2, 4, 5, 6, 8, 10, 11, 12, 19, 20, 21, 22, 23, 24} | 01011101 01110000 00111111 |
| $E_{10}$ : (36, 37) | 14 | 14 | 114,688 | {2, 4, 5, 6, 8, 10, 11, 12, 13, 14, 15, 16, 17, 18} | 01011101 01111111 11000000 |
| $E_{11}$ : (38, 39) | 14 | 14 | 114,688 | {7, 8, 9, 10, 11, 12, 13, 14, 16, 18, 19, 20, 22, 24} | 00000011 11111101 01110101 |
| $E_{12}$ : (40, 41) | 14 | 14 | 114,688 | {1, 2, 3, 4, 5, 6, 13, 14, 16, 18, 19, 20, 22, 24} | 11111100 00001101 01110101 |
| $E_{13}$ : (42, 43) | 14 | 14 | 114,688 | {3, 5, 9, 10, 11, 12, 15, 16, 17, 18, 19, 21, 23, 24} | 00101000 11110011 11101011 |
| $E_{14}$ : (44, 45) | 14 | 14 | 114,688 | {1, 2, 3, 5, 7, 8, 9, 10, 13, 14, 15, 16, 19, 21} | 11101011 11001111 00101000 |
| $E_{15}$ : (46, 47) | 14 | 14 | 114,688 | {1, 2, 5, 6, 7, 11, 13, 15, 16, 17, 21, 22, 23, 24} | 11001110 00101011 10001111 |
| $E_{16}$ : (48, 49) | 14 | 14 | 114,688 | {1, 2, 3, 4, 7, 9, 10, 11, 13, 17, 19, 20, 23, 24} | 11110010 11101000 10110011 |
| $E_{17}$ : (50, 51) | 14 | 14 | 114,688 | {1, 6, 7, 8, 11, 12, 14, 15, 17, 18, 21, 22, 23, 24} | 10000111 00110110 11001111 |
| $E_{18}$ : (52, 53) | 14 | 14 | 114,688 | {1, 3, 4, 6, 7, 8, 9, 10, 14, 15, 19, 20, 21, 22} | 10110111 11000110 00111100 |
| $E_{19}$ : (54, 55) | 14 | 14 | 114,688 | {3, 4, 5, 6, 9, 12, 15, 16, 17, 18, 20, 21, 22, 23} | 00111100 10010011 11011110 |
| $E_{20}$ : (56, 57) | 14 | 14 | 114,688 | {1, 2, 3, 4, 7, 8, 9, 12, 13, 14, 17, 18, 20, 23} | 11110011 10011100 11010010 |
| $E_{21}$ : (58, 59) | 14 | 14 | 114,688 | {2, 4, 8, 10, 11, 12, 13, 14, 17, 18, 19, 20, 23, 24} | 01010001 01111100 11110011 |
| $E_{22}$ : (60, 61) | 14 | 14 | 114,688 | {2, 4, 5, 6, 8, 10, 13, 14, 15, 16, 19, 20, 21, 22} | 01011101 01001111 00111100 |
| $E_{23}$ : (62, 63) | 14 | 14 | 114,688 | {3, 4, 5, 6, 9, 10, 11, 12, 16, 18, 19, 20, 22, 24} | 00111100 11110001 01110101 |
| $E_{24}$ : (64, 65) | 14 | 14 | 114,688 | {1, 2, 5, 6, 7, 8, 11, 12, 13, 14, 16, 18, 22, 24} | 11001111 00111101 01000101 |

### S2.29. Parent-state and deletion certificates

The 49-ray, 87-ray, 40-ray, and 72-ray parents have no context states. The two smaller 68-ray parents are treated in Appendix S2.30.1. The first and last two have the independently verified complete bases printed in Appendix S1.16; the 87-ray parent contains the known 69-ray KS lineage. The 70-ray single-apex lift of the completed Cabello parent instead has the unique state supported at its apex.

For the real Yu–Oh parent, the intermediate 33 rays admit 192 assignments to the twenty bases alone and none after its sixteen additional pair exclusions. The sixteen missing pairs are uniquely completed by rays 34–49. The x-axis-plane extension by rays 26–29 alone has 72 basis assignments; all 24 original states extend. Including its eight additional pair exclusions leaves eight extensions, all with $s(2) = 1$. The z-axis-plane case is identical, with $s(5) = 1$. The two added bases on either axis each exclude twelve of the sixteen original axis-zero states when their further pair exclusions are included; eight original states are excluded by both, leaving none. Thus the axis forcing uses the

complete orthogonality constraints, not the added bases alone.

TABLE S83. Context-state counts after deleting each single context of the real 49-ray parent. Every count is positive. Context numbering is Table S21. Deleting a completion context 21–36 leaves its fresh ray in no context; that ray is counted as an unconstrained free vertex, which doubles the count. If the orthogonalities of the deleted context are kept as geometric exclusions, deletions 21–36 leave no state, while deletions 1–20 leave 1, 2, or 4 states.

| Deleted context | States | Deleted context | States | Deleted context | States |
|---|---|---|---|---|---|
| 1 | 12 | 13 | 2 | 25 | 2 |
| 2 | 4 | 14 | 4 | 26 | 2 |
| 3 | 4 | 15 | 4 | 27 | 2 |
| 4 | 32 | 16 | 16 | 28 | 2 |
| 5 | 2 | 17 | 4 | 29 | 2 |
| 6 | 2 | 18 | 4 | 30 | 2 |
| 7 | 4 | 19 | 4 | 31 | 2 |
| 8 | 4 | 20 | 4 | 32 | 2 |
| 9 | 12 | 21 | 2 | 33 | 2 |
| 10 | 4 | 22 | 2 | 34 | 2 |
| 11 | 4 | 23 | 2 | 35 | 2 |
| 12 | 2 | 24 | 2 | 36 | 2 |

All complete states of $H_{65}$ are additionally supplied as `data/h65-all-states.csv.gz`. Each row gives its branch index, the fourteen-bit choice index within that branch, and its 65-bit state vector in vertex order. There are exactly $24 \times 2^{14} = 393{,}216$ data rows. This supplements the complete parametrization and the printed 48-state family; it does not assert that 48 is minimal. The 153 Hesse states, their event-order witnesses, and all original printed profiles remain in the accompanying Hesse data.

## S2.30. States of the Cabello restrictions and completions

### *S2.30.1. States of the vertex-deleted 45-ray completion*

The nine-context $C_{18,9}$, its eighteen-context faithful completion, and both 68-ray parents have no context states by their retained parity proofs. For the 45-ray construction, use exactly the local labels in Tables S31 and S32. Table S84 lists all twelve admissible core supports, and Table S85 lists the fourteen bridge pairs with their fresh labels and zero-profiles. For a support $S_\alpha$, let $F_\alpha$ be the bridge indices missed by $S_\alpha$. On every hit bridge both fresh values are zero; on each missed bridge independently choose exactly one of its two fresh vertices. This enumerates every context state once.

TABLE S84: All twelve core supports; $m = |F_\alpha|$. The final column specifies extendibility under all geometric pair exclusions.

| $\alpha$ | $S_\alpha$ | $m$ | Geometric extension |
|---|---|---|---|
| 1 | $\{1,6,11,12\}$ | 8 | yes |
| 2 | $\{1,10,14,18\}$ | 8 | yes |
| 3 | $\{3,7,10,14\}$ | 8 | yes |
| 4 | $\{3,8,11,13\}$ | 8 | yes |
| 5 | $\{3,13,14,17\}$ | 8 | no |
| 6 | $\{2,6,10,15\}$ | 8 | yes |
| 7 | $\{6,12,15,16\}$ | 8 | no |
| 8 | $\{8,10,15,18\}$ | 8 | yes |
| 9 | $\{9,16,18,19\}$ | 9 | no |
| 10 | $\{9,17,18,19\}$ | 9 | no |
| 11 | $\{2,11,13,19\}$ | 8 | yes |
| 12 | $\{7,11,12,19\}$ | 8 | yes |

TABLE S85: The fourteen bridge pairs, fresh labels, and zero-profiles; each $Z_m$ is the set of missed-bridge branch indices.

| $m$ | Bridge | Fresh pair | $Z_m$ |
|---|---|---|---|
| 1 | $\{1,2\}$ | $\{21,22\}$ | $\{3,4,5,7,8,9,10,12\}$ |
| 2 | $\{1,13\}$ | $\{23,24\}$ | $\{3,6,7,8,9,10,12\}$ |
| 3 | $\{1,17\}$ | $\{25,26\}$ | $\{3,4,6,7,8,9,11,12\}$ |
| 4 | $\{2,14\}$ | $\{27,28\}$ | $\{1,4,7,8,9,10,12\}$ |
| 5 | $\{2,17\}$ | $\{29,30\}$ | $\{1,2,3,4,7,8,9,12\}$ |
| 6 | $\{3,16\}$ | $\{31,32\}$ | $\{1,2,6,8,10,11,12\}$ |
| 7 | $\{6,17\}$ | $\{33,34\}$ | $\{2,3,4,8,9,11,12\}$ |
| 8 | $\{7,8\}$ | $\{35,36\}$ | $\{1,2,5,6,7,9,10,11\}$ |
| 9 | $\{7,15\}$ | $\{37,38\}$ | $\{1,2,4,5,9,10,11\}$ |
| 10 | $\{7,16\}$ | $\{39,40\}$ | $\{1,2,4,5,6,8,10,11\}$ |
| 11 | $\{8,12\}$ | $\{41,42\}$ | $\{2,3,5,6,9,10,11\}$ |
| 12 | $\{8,16\}$ | $\{43,44\}$ | $\{1,2,3,5,6,10,11,12\}$ |
| 13 | $\{10,19\}$ | $\{45,46\}$ | $\{1,4,5,7\}$ |
| 14 | $\{11,18\}$ | $\{47,48\}$ | $\{3,5,6,7\}$ |

Taking the first fresh vertex on every missed bridge gives $s_{\alpha,0}$, and taking the second gives $s_{\alpha,1}$. Table S86 explicitly enumerates all 24 canonical context states. Their profiles are nonzero and pairwise distinct on all 45 vertices. The twelve branch supports and fourteen distinct nonempty zero-profiles verify this by Lemma 21(3).

TABLE S86: All 24 canonical context-state supports of the 45-ray example. Some fail geometric pair exclusions.

| $\alpha$ | $\varepsilon$ | Support |
|---|---|---|
| 1 | 0 | $\{1,6,11,12,27,29,31,35,37,39,43,45\}$ |
| 1 | 1 | $\{1,6,11,12,28,30,32,36,38,40,44,46\}$ |
| 2 | 0 | $\{1,10,14,18,29,31,33,35,37,39,41,43\}$ |
| 2 | 1 | $\{1,10,14,18,30,32,34,36,38,40,42,44\}$ |
| 3 | 0 | $\{3,7,10,14,21,23,25,29,33,41,43,47\}$ |
| 3 | 1 | $\{3,7,10,14,22,24,26,30,34,42,44,48\}$ |
| 4 | 0 | $\{3,8,11,13,21,25,27,29,33,37,39,45\}$ |
| 4 | 1 | $\{3,8,11,13,22,26,28,30,34,38,40,46\}$ |
| 5 | 0 | $\{3,13,14,17,21,35,37,39,41,43,45,47\}$ |
| 5 | 1 | $\{3,13,14,17,22,36,38,40,42,44,46,48\}$ |
| 6 | 0 | $\{2,6,10,15,23,25,31,35,39,41,43,47\}$ |
| 6 | 1 | $\{2,6,10,15,24,26,32,36,40,42,44,48\}$ |
| 7 | 0 | $\{6,12,15,16,21,23,25,27,29,35,45,47\}$ |
| 7 | 1 | $\{6,12,15,16,22,24,26,28,30,36,46,48\}$ |
| 8 | 0 | $\{8,10,15,18,21,23,25,27,29,31,33,39\}$ |
| 8 | 1 | $\{8,10,15,18,22,24,26,28,30,32,34,40\}$ |
| 9 | 0 | $\{9,16,18,19,21,23,25,27,29,33,35,37,41\}$ |
| 9 | 1 | $\{9,16,18,19,22,24,26,28,30,34,36,38,42\}$ |
| 10 | 0 | $\{9,17,18,19,21,23,27,31,35,37,39,41,43\}$ |
| 10 | 1 | $\{9,17,18,19,22,24,28,32,36,38,40,42,44\}$ |
| 11 | 0 | $\{2,11,13,19,25,31,33,35,37,39,41,43\}$ |
| 11 | 1 | $\{2,11,13,19,26,32,34,36,38,40,42,44\}$ |
| 12 | 0 | $\{7,11,12,19,21,23,25,27,29,31,33,43\}$ |
| 12 | 1 | $\{7,11,12,19,22,24,26,28,30,32,34,44\}$ |

Ten branches miss eight pairs and two miss nine. Therefore

$$|\mathcal{S}_{\mathrm{ctx}}| = 10\,2^8 + 2\,2^9 = 3{,}584, \qquad |\mathcal{S}_{\mathrm{geom}}| = 8\,2^8 = 2048. \tag{S22}$$

The geometric branches are $1, 2, 3, 4, 6, 8, 11, 12$; arbitrary fresh choices within each of these branches remain geometrically compatible. The other branches are excluded by the four forced pairs. Both full enumerations are supplied as compressed CSV data, in addition to this complete finite parametrization. Every geometric state vanishes on 9, 16, and 17, so this family is neither unital nor separating.

For an explicit counterexample, the context state supported at

$$\{6, 12, 15, 16, 21, 23, 25, 27, 29, 35, 45, 47\}$$

selects both 16 and 21, although $v_{16} \cdot v_{21} = 0$.

### *S2.30.2. Deleting one of the nine Cabello contexts*

Number the nine contexts as the first nine rows of Table S29. For every deletion there are 26 context states, six geometric states, and a proper four-coloring of the remaining context graph. The color strings in Table S87 use the ordered ray list $(1, 2, 3, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20)$. The geometric states vanish precisely on the omitted context.

TABLE S87: All single-context deletion counts and four-coloring witnesses for $C_{18,9}$.

| $e$ | Omitted context | $\lvert\mathcal{S}_{\mathrm{ctx}}\rvert$ | $\lvert\mathcal{S}_{\mathrm{geom}}\rvert$ | Four-color string |
|---|---|---|---|---|
| 1 | $\{9, 10, 12, 13\}$ | 26 | 6 | 012102203020112333 |
| 2 | $\{9, 11, 14, 15\}$ | 26 | 6 | 012021231012102333 |
| 3 | $\{10, 11, 16, 17\}$ | 26 | 6 | 012021320012102333 |
| 4 | $\{2, 3, 12, 18\}$ | 26 | 6 | 013103302120113322 |
| 5 | $\{1, 3, 15, 19\}$ | 26 | 6 | 103130320102113232 |
| 6 | $\{1, 2, 17, 20\}$ | 26 | 6 | 330130230103112222 |
| 7 | $\{6, 7, 13, 18\}$ | 26 | 6 | 012221213030102333 |
| 8 | $\{6, 8, 14, 19\}$ | 26 | 6 | 012202230013112333 |
| 9 | $\{7, 8, 16, 20\}$ | 26 | 6 | 012021320012113332 |

For deletion of context 9, $\{7, 8, 16, 20\}$, Table S88 explicitly enumerates all 26 supports; exactly the six marked rows satisfy every geometric pair exclusion. All nine full deletion-state lists accompany the data.

TABLE S88: All 26 states after deleting context 9; geometric compatibility is marked explicitly.

| $j$ | Support | Geometric |
|---|---|---|
| 1 | $\{1, 6, 11, 12\}$ | yes |
| 2 | $\{1, 7, 8, 11, 12\}$ | no |
| 3 | $\{1, 7, 12, 14, 16\}$ | no |
| 4 | $\{1, 8, 9, 16, 18\}$ | no |
| 5 | $\{1, 10, 14, 18\}$ | yes |
| 6 | $\{2, 6, 10, 15\}$ | yes |
| 7 | $\{2, 7, 8, 10, 15\}$ | no |
| 8 | $\{2, 8, 13, 15, 16\}$ | no |
| 9 | $\{2, 7, 9, 16, 19\}$ | no |
| 10 | $\{2, 11, 13, 19\}$ | yes |
| 11 | $\{3, 6, 9, 17\}$ | no |
| 12 | $\{3, 7, 8, 9, 17\}$ | no |
| 13 | $\{3, 13, 14, 17\}$ | yes |
| 14 | $\{6, 12, 15, 17\}$ | no |
| 15 | $\{7, 8, 12, 15, 17\}$ | no |
| 16 | $\{9, 17, 18, 19\}$ | yes |
| 17 | $\{3, 6, 9, 16, 20\}$ | no |
| 18 | $\{3, 7, 8, 9, 16, 20\}$ | no |
| 19 | $\{3, 7, 10, 14, 20\}$ | no |
| 20 | $\{3, 8, 11, 13, 20\}$ | no |
| 21 | $\{3, 13, 14, 16, 20\}$ | no |
| 22 | $\{6, 12, 15, 16, 20\}$ | no |
| 23 | $\{7, 11, 12, 19, 20\}$ | no |
| 24 | $\{7, 8, 12, 15, 16, 20\}$ | no |
| 25 | $\{9, 16, 18, 19, 20\}$ | no |
| 26 | $\{8, 10, 15, 18, 20\}$ | no |

### *S2.30.3. All single-ray deletions and the alternative parity subsystem*

The next table records the two span failures and all forced-zero core rays after geometric pair exclusion for every possible single-ray deletion. An entry $\{u, v\} : z$ means $v_z \in \mathrm{span}(v_u, v_v)$. Every row has twelve compatible core states, 3,584 abstract completion states, and 2,048 geometric completion states, with exactly four forced extra pairs and no other extra orthogonality in the supplied realization. The complete core supports and 24 canonical states for every row are also supplied in the data.

TABLE S89: All eighteen vertex-deletion checks; the completed geometric families fail unitality and separation in every case.

| Deleted ray | First span failure | Second span failure | Forced-zero core rays |
|---|---|---|---|
| 1 | $\{15, 19\} : 2$ | $\{17, 20\} : 3$ | $\{2, 3, 13\}$ |
| 2 | $\{12, 18\} : 1$ | $\{17, 20\} : 3$ | $\{1, 3, 14\}$ |
| 3 | $\{12, 18\} : 1$ | $\{15, 19\} : 2$ | $\{1, 2, 16\}$ |
| 6 | $\{13, 18\} : 8$ | $\{14, 19\} : 7$ | $\{7, 8, 17\}$ |
| 7 | $\{13, 18\} : 8$ | $\{16, 20\} : 6$ | $\{6, 8, 15\}$ |
| 8 | $\{14, 19\} : 7$ | $\{16, 20\} : 6$ | $\{6, 7, 12\}$ |
| 9 | $\{12, 13\} : 11$ | $\{14, 15\} : 10$ | $\{10, 11, 20\}$ |
| 10 | $\{12, 13\} : 11$ | $\{16, 17\} : 9$ | $\{9, 11, 19\}$ |
| 11 | $\{14, 15\} : 10$ | $\{16, 17\} : 9$ | $\{9, 10, 18\}$ |
| 12 | $\{2, 3\} : 13$ | $\{9, 10\} : 18$ | $\{8, 13, 18\}$ |
| 13 | $\{6, 7\} : 12$ | $\{9, 10\} : 18$ | $\{1, 12, 18\}$ |
| 14 | $\{6, 8\} : 15$ | $\{9, 11\} : 19$ | $\{2, 15, 19\}$ |
| 15 | $\{1, 3\} : 14$ | $\{9, 11\} : 19$ | $\{7, 14, 19\}$ |
| 16 | $\{7, 8\} : 17$ | $\{10, 11\} : 20$ | $\{3, 17, 20\}$ |
| 17 | $\{1, 2\} : 16$ | $\{10, 11\} : 20$ | $\{6, 16, 20\}$ |
| 18 | $\{2, 3\} : 13$ | $\{6, 7\} : 12$ | $\{11, 12, 13\}$ |
| 19 | $\{1, 3\} : 14$ | $\{6, 8\} : 15$ | $\{10, 14, 15\}$ |
| 20 | $\{1, 2\} : 16$ | $\{7, 8\} : 17$ | $\{9, 16, 17\}$ |

For the alternative subsystem, take parent variant 2 of Table S35 and retain vertices 1–17 and 66–68. It is stateless. Each ray deletion instead leaves the positive geometric state count below; restriction of any such state shows that no eighteen-ray subset can be KS.

TABLE S90: Geometric state counts after every single-ray deletion of the twenty-ray parity subsystem.

| Deleted ray | States | Deleted ray | States |
|---|---|---|---|
| 1 | 8 | 2 | 8 |
| 3 | 8 | 4 | 28 |
| 5 | 28 | 6 | 8 |
| 7 | 8 | 8 | 8 |
| 9 | 12 | 10 | 12 |
| 11 | 12 | 12 | 10 |
| 13 | 10 | 14 | 10 |
| 15 | 10 | 16 | 10 |
| 17 | 10 | 66 | 8 |
| 67 | 8 | 68 | 8 |

## S3. SUPPLEMENTARY PROOFS, REDUCTIONS, AND EXACT VERIFICATION CERTIFICATES

### S3.1. Formulas used by the finite certificates

The following formulas restore the explicit conventions used by the later certificates. State indices are local to the corresponding complete table. For Hesse, the six-state vertex-separating family, its seven-state event-separating extension, and the eighteen-state order-determining family are respectively

$$\Omega_6 = \{s_3, s_{31}, s_{75}, s_{89}, s_{113}, s_{151}\}. \tag{S23}$$

$$\Omega_7 = \Omega_6 \cup \{s_{84}\} \tag{S24}$$

$$\begin{aligned}\Omega_{18} = \{&s_2, s_8, s_{20}, s_{30}, s_{40}, s_{45},\\ &s_{54}, s_{64}, s_{73}, s_{79}, s_{83}, s_{93},\\ &s_{105}, s_{109}, s_{119}, s_{123}, s_{127}, s_{150}\}.\end{aligned} \tag{S25}$$

The members of $\Omega_6$ are denoted $\sigma_1, \dots, \sigma_6$ in the displayed order. Their atom profiles and the nontrivial atom–coatom order are

$$\kappa(i) = (\sigma_1(i), \dots, \sigma_6(i)). \tag{S26}$$

$$a_i \le a_j^{\perp} \quad \Longleftrightarrow \quad \{i, j\} \in E(\Gamma(H_{37})). \tag{S27}$$

For $H_{65}$, the following core coloring extends to the completed hypergraph. A base-compatible independent support $S_\alpha$ is extended canonically using $\varepsilon \in \{0, 1\}$. In the plane formula, $\eta_A, \eta_B$ are an integer orthogonal basis and $n_A = \|\eta_A\|^2$, $n_B = \|\eta_B\|^2$.

$$\begin{aligned}(c(1), \dots, c(17)) = (&0, 1, 2, 3, 0, 1, 2, 4, 0, 1, 3,\\ &3, 4, 2, 1, 0, 2).\end{aligned} \tag{S28}$$

$$\begin{aligned}&\big(s_{\alpha,\varepsilon}(w_{m,1}),\ s_{\alpha,\varepsilon}(w_{m,2})\big)\\ &\quad = \begin{cases}(0, 0), & |S_\alpha \cap \{u_m, v_m\}| = 1,\\ (1-\varepsilon,\ \varepsilon), & S_\alpha \cap \{u_m, v_m\} = \emptyset.\end{cases}\end{aligned} \tag{S29}$$

$$\begin{aligned}w_{m,1}(a, b) &= a\,\eta_A + b\,\eta_B,\\ w_{m,2}(a, b) &= -n_B b\,\eta_A + n_A a\,\eta_B\end{aligned} \tag{S30}$$

The eight rays adjoined to completed Yu–Oh and the nine uncovered pairs of the Cabello subsystem are fixed by the following local labels. The last display records the completed historical $\Gamma_1$ TITS incidence.

$$\begin{aligned}&(0, 1, 2), (0, 2, -1), (0, 1, -2), (0, 2, 1),\\ &(1, 2, 0), (2, -1, 0), (1, -2, 0), (2, 1, 0).\end{aligned} \tag{S31}$$

$$\begin{aligned}&\{1, 13\}, \{2, 14\}, \{3, 16\}, \{6, 17\}, \{7, 15\},\\ &\{8, 12\}, \{9, c\}, \{10, b\}, \{11, a\}.\end{aligned} \tag{S32}$$

$$E(I_{16,9}) = E(B_{13,7}) \cup \{\{4, 11, 24\}, \{4, 10, 18\}\}. \tag{S33}$$

*a. Fractional value of the faithful Cabello completion.* The geometric orthogonality graph of $C_{18,9}$ has independence number four and exactly 54 independent four-sets, with each of its 18 rays belonging to twelve of them. Assigning weight 1/12 to each set gives a fractional cover of total weight 9/2, matching 18/4. Fresh pair completion preserves this value by the main completion proposition, proving $\chi_f(C_{36,18}) = 9/2$ as recorded in the main comparison. The complete 54-set certificate is in the electronic audit data.

## S3.2. Snark obstructions and quantitative robustness bounds

### *S3.2.1. Cubic graphs, Blanuša snarks, and Isaacs dot products*

For a finite simple cubic graph $X$, let $H(X)$ be the context hypergraph whose vertices are the edges of $X$ and whose contexts are the three edges incident with each vertex of $X$. Then $\Gamma(H(X)) = L(X)$ and $\chi(H(X)) = \chi'(X)$. An orthogonal ray assignment for $H(X)$ in $\mathbb{F}^3$ is an assignment of nonzero vectors to the edges of $X$ such that incident edges are orthogonal.

Silva and Ibiapina[5] (Question 6) asked whether an orthogonal vector edge coloring in $\mathbb{R}^3$ can exist when $\chi'(X) = 4$. Theorem 11 excludes the Petersen graph $P$ even over $\mathbb{C}$. Using the multipole forcing law of Theorem 11(1), we can extend this exclusion to the two 18-vertex Blanuša snarks[6] and the infinite family of Isaacs dot-product snarks[7].

Recall that for a cubic graph $G$ with two nonadjacent edges $e_1 = \{x_1, y_1\}$ and $e_2 = \{x_2, y_2\}$, the Isaacs dot product $G \cdot P$ is formed by removing $e_1, e_2$ from $G$, removing two adjacent vertices $u, v$ from the Petersen graph $P$ (leaving the 4-pole $P \setminus \{u, v\}$ whose dual hypergraph is precisely $H_8$), and attaching the two dangling edges $\{2, 8\}$ of $u$ to $\{x_1, y_1\}$ and the two dangling edges $\{13, 14\}$ of $v$ to $\{x_2, y_2\}$; the two distinct choices of nonadjacent edges in $G = P$ produce the two Blanuša snarks $B_1$ and $B_2$[6–8].

**Corollary 1** (Obstruction for Blanuša and dot-product snarks)**.** *If a cubic graph $G$ admits no orthogonal vector edge coloring in $\mathbb{C}^3$, then neither does any Isaacs dot product $G \cdot P$. In particular, both 18-vertex Blanuša snarks $B_1, B_2 = P \cdot P$ and all iterated dot products $X = P \cdot P \cdots P$ satisfy $\chi(H(X)) = \chi'(X) = 4 > 3 = \omega(L(X))$ and admit no orthogonal vector edge coloring in $\mathbb{C}^3$.*

*Proof.* In any orthogonal vector edge coloring of $G \cdot P$ in $\mathbb{C}^3$, the restriction to the embedded copy of $H_8 = H(P \setminus \{u, v\})$ satisfies $[v_2] = [v_8]$ and $[v_{13}] = [v_{14}]$ by Theorem 11(1). Assigning the common ray $[v_2] = [v_8]$ to $e_1 = \{x_1, y_1\}$ and the common ray $[v_{13}] = [v_{14}]$ to $e_2 = \{x_2, y_2\}$ preserves orthogonality at $x_1, y_1, x_2, y_2$ and therefore yields a valid orthogonal vector edge coloring of $G$ in $\mathbb{C}^3$. Since $P$ admits no such coloring by Theorem 11, induction excludes $B_1, B_2$ and all iterated dot products with $P$. □

Generalized Petersen graphs[9] and flower snarks provide further families for comparing real, complex, injective, and edge-faithful realizability.

### *S3.2.2. Explicit analytic lower bound and tetrahedral tight-frame upper bounds for $\Delta_3$*

Let $A$ be the 30 edges of $\Gamma(G_{32})$ and define the minimum total squared orthogonality error in $\mathbb{C}^3$:

$$\Delta_3 = \min_{\substack{v_i \in \mathbb{C}^3 \\ \|v_i\|=1 \ (1 \le i \le 15)}} \sum_{\{i,j\} \in A} |\langle v_i, v_j \rangle|^2. \tag{S34}$$

By compactness of $(S^5)^{15}$ and Theorem 11, $\Delta_3 > 0$. We now establish both a certified rational analytic lower bound and explicit tetrahedral tight-frame upper bounds.

**Proposition 2** (Quantitative bounds on $\Delta_3$)**.** *Every assignment of unit vectors $v_1, \ldots, v_{15} \in \mathbb{C}^3$ satisfies*

$$\varepsilon := \max_{\{i,j\} \in A} |\langle v_i, v_j \rangle| > \frac{1}{120}, \tag{S35}$$

*and the minimum total squared error in Eq. (S34) satisfies*

$$\begin{aligned} \frac{1}{14400} < \Delta_3 &\le \min_{a^2+b^2+c^2=1} F(a, b, c) \\ &\le \frac{9275914990340646}{10001032226635921} < 0.927496 < \frac{18}{19}, \end{aligned} \tag{S36}$$

*where*

$$F(a,b,c) = 12a^2 + 12(ab+bc-ca)^2 + 6(b^2-a^2-c^2)^2. \tag{S37}$$

*Proof.* To prove the lower bound (S35), let $\varepsilon = \max_{\{i,j\}\in A} |\langle v_i, v_j\rangle|$. For any context $e = \{x,y,z\}$, the Gram matrix of $\{v_x, v_y, v_z\}$ has diagonal entries 1 and off-diagonal entries of modulus at most $\varepsilon$, so Gershgorin's circle theorem implies $\|I_3 - (v_x v_x^\dagger + v_y v_y^\dagger + v_z v_z^\dagger)\|_{\mathrm{op}} \le 2\varepsilon$. Hence for any unit vectors $u, w \in \mathbb{C}^3$,

$$\Big|1 - \big(|\langle u,x\rangle|^2 + |\langle u,y\rangle|^2 + |\langle u,z\rangle|^2\big)\Big| \le 2\varepsilon, \tag{S38}$$

and if $\{u,x\}, \{w,z\} \in A$, then

$$\big|\langle u,w\rangle - \langle u,y\rangle\langle y,w\rangle\big| \le 4\varepsilon. \tag{S39}$$

In $\{5,6,7\}$ and $\{9,10,11\}$, the vectors $w_7 = v_{12} - v_5\langle v_5, v_{12}\rangle$ and $w_9 = v_4 - v_{11}\langle v_{11}, v_4\rangle$ satisfy $\|w_7 - v_7\langle v_7, v_{12}\rangle\| \le 3\varepsilon$ and $\|w_9 - v_9\langle v_9, v_4\rangle\| \le 3\varepsilon$, so $|\langle w_7, w_9\rangle| \le 7\varepsilon$. Expanding,

$$\begin{aligned}\langle w_7, w_9\rangle = {} & \langle v_{12}, v_4\rangle - \langle v_{12}, v_{11}\rangle\langle v_{11}, v_4\rangle - \langle v_{12}, v_5\rangle\langle v_5, v_4\rangle \\ & + \langle v_{12}, v_5\rangle\langle v_5, v_{11}\rangle\langle v_{11}, v_4\rangle.\end{aligned}$$

The two middle terms are at most $\varepsilon$ in modulus, since $|\langle v_{12}, v_{11}\rangle| \le \varepsilon$ and $|\langle v_5, v_4\rangle| \le \varepsilon$. Eq. (S39) on $\{1,2,3\}$ replaces each of $\langle v_{12}, v_4\rangle$, $\langle v_{12}, v_5\rangle$, $\langle v_5, v_{11}\rangle$, and $\langle v_{11}, v_4\rangle$ by its product through $v_2$ with error at most $4\varepsilon$, and hence the triple product with error at most $12\varepsilon$, since all factors have modulus at most one. The main terms combine to

$$\big(1 + |\langle v_2, v_5\rangle|^2 |\langle v_2, v_{11}\rangle|^2\big)\langle v_{12}, v_2\rangle\langle v_2, v_4\rangle,$$

whose real prefactor is at least one, so

$$\big|\langle v_{12}, v_2\rangle\langle v_2, v_4\rangle\big| \le (7+2+4+12)\varepsilon = 25\varepsilon.$$

By reflection symmetry, we may assume $|\langle v_2, v_4\rangle| \le 5\sqrt{\varepsilon}$. Applying Eq. (S38) to $v_4$ in $\{1,2,3\}$ and to $v_2$ in $\{3,4,5\}$ gives $|\langle v_1, v_4\rangle|^2 \ge 1 - 27\varepsilon - \varepsilon^2$ and $|\langle v_2, v_5\rangle|^2 \ge 1 - 27\varepsilon - \varepsilon^2$, so $\min_\theta \|v_1 - e^{i\theta} v_4\| \le \sqrt{54\varepsilon + 2\varepsilon^2}$ and $\min_\theta \|v_2 - e^{i\theta} v_5\| \le \sqrt{54\varepsilon + 2\varepsilon^2}$. Since $|\langle v_4, v_{10}\rangle| \le \varepsilon$ and $|\langle v_5, v_7\rangle| \le \varepsilon$, Eq. (S38) applied to $v_1$ in $\{9,10,11\}$ and to $v_2$ in $\{7,8,9\}$ gives

$$|\langle v_i, v_9\rangle|^2 \ge 1 - 2\varepsilon - \varepsilon^2 - \big(\varepsilon + \sqrt{54\varepsilon + 2\varepsilon^2}\big)^2 \qquad (i = 1,2).$$

Comparing the sum of these two inequalities with $|\langle v_1, v_9\rangle|^2 + |\langle v_2, v_9\rangle|^2 \le 1 + 2\varepsilon$ from $\{1,2,3\}$ gives

$$1 \le 6\varepsilon + 2\varepsilon^2 + 2\big(\varepsilon + \sqrt{54\varepsilon + 2\varepsilon^2}\big)^2.$$

The right-hand side is strictly increasing for $\varepsilon > 0$, and at $\varepsilon = 1/120$ we have $54\varepsilon + 2\varepsilon^2 = 3241/7200 < (27/40)^2$, so the right-hand side is strictly bounded above by

$$\frac{6}{120} + \frac{2}{120^2} + 2\left(\frac{1}{120} + \frac{27}{40}\right)^2 = \frac{1417}{1440} < 1,$$

certifying $\varepsilon > 1/120$ and $\Delta_3 \ge \varepsilon^2 > 1/14400$.

For the upper bounds, set $v_4 = e_1$, $v_8 = e_2$, $v_{12} = e_3$, and for any unit vector $(a,b,c) \in S^2$, define the remaining 12 unit vectors as representatives of the tetrahedral $A_4$-orbit of rays:

$$\begin{aligned} v_1 &= (-b,-c,-a)^T, & v_2 &= (c,-a,b)^T, \\ v_3 &= (-a,-b,c)^T, & v_5 &= (-a,b,c)^T, \\ v_6 &= (-b,c,-a)^T, & v_7 &= (c,a,b)^T, \\ v_9 &= (-c,-a,b)^T, & v_{10} &= (-a,b,-c)^T, \\ v_{11} &= (b,-c,-a)^T, & v_{13} &= (a,b,c)^T, \\ v_{14} &= (b,c,-a)^T, & v_{15} &= (-c,a,b)^T. \end{aligned} \tag{S40}$$

These 15 vectors form a tight frame $\sum_{i=1}^{15} v_i v_i^T = 5I_3$, and the sum of squared overlaps on the 30 edges of $\Gamma(G_{32})$ equals $F(a,b,c)$ in Eq. (S37). Evaluating Eq. (S40) at the normalized integer direction $(a,b,c) = (1,3,-3)/\sqrt{19}$ yields rational squared overlaps with $\Delta_3 \le 342/361 = 18/19 \approx 0.947368$ and $\max_{\{i,j\}\in A} |\langle v_i, v_j\rangle|^2 = 1/19$; evaluating at the refined normalized integer direction $(a,b,c) = (2332, 7101, -6644)/\sqrt{100005161}$ yields the certified rational bound in Eq. (S36). □

Numerical minimization of $F$ on $S^2$ (as well as unrestricted local searches over $(S^5)^{15}$) finds an attained candidate minimum near 0.927495724691 at $(a,b,c) \approx (0.23319713, 0.71005263, -0.66441280)$; certifying whether the unrestricted global minimum $\Delta_3$ coincides with $\min_{S^2} F$ remains an open optimization problem. Choosing instead the rational unit vector $(a,b,c) = \frac{1}{3}(1,2,-2)$ in Eq. (S40) makes $ab + bc - ca = 0$, so four of the ten contexts of $G_{32}$ ($\{1,2,3\}, \{5,6,7\}, \{9,10,11\}, \{13,14,15\}$) become exact orthonormal bases of $\mathbb{R}^3$, all 15 rays are distinct, and $\Delta_3 \le 38/27$ with $\max_{\{i,j\}\in A} |\langle v_i, v_j\rangle| = 1/3$.

### S3.3. Hesse $H_{37}$: minimum state families

All state indices in this appendix use the canonical enumeration of Appendix S2.27. The minima for unital vertex separation and event separation have the counting lower bounds $2^5 - 1 < 37$ and $2^6 < 76$. The families in Eqs. (S23) and (S24) attain those bounds: their profiles give respectively 37 distinct nonzero atom profiles and 76 distinct event profiles. The seven-state family still has 462 ordered event pairs for which inclusion of the profiles does not imply the lattice order. The family in Eq. (S25) has no such pair and hence gives the upper bound eighteen for order determination. This appendix supplies a computer-assisted proof of the matching lower bound.

#### *S3.3.1. Reduction to families of seventeen states*

Let $D = \{P_0, P_1, P_2, Q_0, Q_1, Q_2\}$. For each unordered pair $\{u, v\} \subset D$, order determination requires a state with $s(u) = s(v) = 1$: the pair is nonorthogonal, so $u \not\le v^\perp$. Every state takes value one on at most two members of $D$. Consequently the fifteen pairs require fifteen different states. Table S91 gives their complete witness classes. These classes are disjoint, their union $T$ has 27 states, and a choice of one witness from each class has $1^3 2^{12} = 4{,}096$ possibilities.

TABLE S91: Complete joint-one witness classes for the fifteen pairs in $D$. Each entry lists all canonical indices $r$ for which $s_r(u) = s_r(v) = 1$.

| Pair $\{u, v\}$ | Witness indices $r$ |
|---|---|
| $\{P_0, P_1\}$ | 150 |
| $\{P_0, P_2\}$ | 73 |
| $\{P_0, Q_0\}$ | 93, 145 |
| $\{P_0, Q_1\}$ | 127, 133 |
| $\{P_0, Q_2\}$ | 79, 99 |
| $\{P_1, P_2\}$ | 2 |
| $\{P_1, Q_0\}$ | 105, 139 |
| $\{P_1, Q_1\}$ | 14, 119 |
| $\{P_1, Q_2\}$ | 40, 48 |
| $\{P_2, Q_0\}$ | 54, 60 |
| $\{P_2, Q_1\}$ | 8, 26 |
| $\{P_2, Q_2\}$ | 20, 68 |
| $\{Q_0, Q_1\}$ | 109, 113 |
| $\{Q_0, Q_2\}$ | 83, 87 |
| $\{Q_1, Q_2\}$ | 30, 34 |

The nine nonorthogonal pairs $\{z, C_{ak}\}$ require further joint-one witnesses. No state in $T$ witnesses any of them. Indeed, if a state activates a $P$ vertex then $s(z) = 0$. If it instead activates two $Q$ vertices, the state equations force the selected $M$ vertices into the remaining $Q$ column, and all nine $C$ values are zero. Any state outside $T$ witnesses at most six of the nine pairs: for $s(z) = 1$, the three selected $M$ vertices force at least three $C$ values to vanish. Thus at least two states outside $T$ are necessary.

It follows that an order-determining family of at most seventeen states would have exactly seventeen members: one from each of the fifteen classes in Table S91, and two outside $T$ that together witness all nine $\{z, C_{ak}\}$ pairs. Enumerating the $\binom{126}{2}$ unordered pairs outside $T$ leaves exactly 132 with this property. The reduction therefore leaves precisely

$$4{,}096 \cdot 132 = 540{,}672 \tag{S41}$$

candidate families. The nine pairs alone do not require three extra states; for example, $s_1$ and $s_{72}$ jointly witness all nine. The remaining order requirements must also be checked.

#### S3.3.2. *Exhaustive order certificate*

For each of the 5,407 ordered event pairs $x \not\leq y$, define its nonempty witness set

$$W_{xy} = \{r \in \{1, \ldots, 153\} : s_r(x) = 1,\ s_r(y) = 0\}. \tag{S42}$$

A family is order determining exactly when its index set meets every $W_{xy}$. Repeated sets and proper supersets can be discarded, because meeting an inclusion-minimal set also meets every set containing it. There are exactly 81 inclusion-minimal witness sets. The exhaustive check forms each of the 540,672 families in Eq. (S41) and counts the minimal witness sets that it meets. The largest count is 75, strictly below 81. Hence no family of seventeen states is order determining. Together with $\Omega_{18}$, this proves the minimum eighteen.

The accompanying Hesse extension checker, invoked by the joint entry point, reconstructs the contexts and all 153 states from their symbolic parametrization, checks their agreement with `hesse37-data.json`, and performs this finite search using integer bit sets and the Python standard library. No numerical tolerance, optimization solver, or unreported search cutoff is used. It also verifies the printed seven- and eighteen-state families and every entry of Table S91. From the extracted joint project directory, run the joint entry point, without Python's optimization option:

```
python verification/verify-integrated.py
```

The generated file `verification/_work/hesse/hesse37-extensions-certificate.json` records all 81 minimal witness sets, a representative event pair for each, all 132 possible extra pairs, the coverage-count distribution over the 540,672 candidates, and all event profiles for $\Omega_6$, $\Omega_7$, and $\Omega_{18}$. The generated file `verification/_work/hesse/hesse37-order18-witnesses.csv` records a witness in $\Omega_{18}$ for every $x \not\leq y$. These two files use the same numbered event labels as the full-family certificate in Appendix S2.27.6.

The program additionally checks Berge girth five from the incidence graph, computes every meet and join in the 76-event order, verifies the orthomodular law, and verifies the projector sum in Eq. (30) exactly in $\mathbb{Q}(\zeta)$. The nonexistence of an injective real representation is proved analytically in Sec. V A.

### S3.4. Four-dimensional $H_{\mathrm{YO}}^{(4)}$ and $H_{65}$: completion and verification

The coordinates in Table S17 can be reproduced by the following finite selection rule. Normalize an integer vector by dividing by the greatest common divisor of its entries and choosing the sign so that its first nonzero entry is positive. Form the set of normalized nonzero vectors obtained from $\{-3, -2, \ldots, 3\}^4$, ordered by squared norm, then maximum absolute entry, then lexicographically. Process bridge edges in the order $e_{13}, \ldots, e_{24}, e_1, \ldots, e_{12}$. For each pair, choose $\eta_A$ to be the first vector in this ordered pool orthogonal to its endpoints, and $\eta_B$ the first such vector also orthogonal to $\eta_A$.

For all $(a, b) \in \{-7, -6, \ldots, 7\}^2 \setminus \{(0, 0)\}$, form the pair in Eq. (S30) and normalize both vectors. Discard pairs that repeat an already placed ray or have an orthogonality to any already placed vertex other than the two endpoints. Among the remaining ordered pairs choose the lexicographically smallest score

$$\left( \max_{j,k} |(w_{m,j})_k|,\ \|w_{m,1}\|^2 + \|w_{m,2}\|^2,\ w_{m,1}, w_{m,2} \right). \tag{S43}$$

In Eq. (S43), the order on the last two entries is the usual lexicographic order on their integer coordinates. This rule yields exactly the 48 completion vectors in Table S17. It is a recipe for this specific example; the general existence assertion is established independently by Lemma 21.

The companion verifier parses the integrated coordinate and state tables, reproduces this completion rule and all 393,216 state extensions, and checks the maximal cliques, fractional-coloring certificate, atom-order failure, forced hub-completion zero, and warm-up data. It is included with the complete state file and invoked by the joint entry point in Sec. S3.7.

### S3.5. Exact Cabello completion and forced-zero certificates

For each of the nine uncovered pairs in Eq. (S32), rational row reduction verifies that its two endpoint vectors span no third core ray. If two endpoint planes were orthogonal, their four endpoints would form another full basis. Exact four-clique enumeration gives only the original nine bases, all of whose pairs are already covered. This excludes such

an orthogonal-plane pair. The integer vectors in Table S28 directly verify the resulting faithful completion: 36 distinct projective rays, eighteen full bases, and 108 orthogonal pairs, with every pair covered by exactly one listed basis. The nine-context graph and the full eighteen-ray orthogonality graph both fail four-colorability and admit five-colorings, as checked by exact finite search in the companion certificate. Lemma 21(2) transfers the chromatic value to the completion.

For the vertex-deleted core with $c$ removed, rational row reduction finds exactly the span failures

$$v_{16} \in \mathrm{span}(v_1, v_2), \qquad v_{17} \in \mathrm{span}(v_7, v_8).$$

There is no orthogonal-plane-pair obstruction. The finite-avoidance choice in Lemma 21 can therefore avoid all extra incidences except those forced by these two spans; the explicit 45-ray inventory has exactly the four pairs printed in Sec. S1.17.2. The two fresh vectors completing $\{1, 2\}$ must both be orthogonal to 16, and those completing $\{7, 8\}$ must both be orthogonal to 17.

The twelve rows of Table S84 give a short complete proof of the resulting obstruction. Whenever 16 is selected, neither 1 nor 2 is selected. The context completing $\{1, 2\}$ then needs one fresh value equal to one, while geometric exclusion with 16 requires both to be zero. Hence every geometric extension has $s(16) = 0$. The same argument with $\{7, 8\}$ gives $s(17) = 0$. These implications use the displayed core states in addition to the span incidences; the span incidences alone would not imply them. The retained basis $\{10, 11, 16, 17\}$ gives $s(10) + s(11) = 1$. If $s(9) = 1$, the other two hub bases give $s(10) = s(11) = 0$, a contradiction. Thus 9, 16, and 17 have identical zero profiles.

Conversely, each of the eight indicated core rows avoids 16 and 17, so all its fresh-pair choices satisfy the four extra exclusions. Each misses eight bridges. This proves the exact count $8\,2^8 = 2{,}048$, as well as failure of geometric unitality and separation. The two excluded core rays remain impossible for every fresh-pair completion of these fixed planes; additional accidental orthogonalities cannot restore them. Table S89 supplies the individual span checks for all eighteen deletions. Their full core tables, exact integer completions, and extension counts are included in `data/cabello18-review-checks.json`.

To audit the ancestry result, enumerate every four-clique after adjoining either three-ray choice to all 65 original rays. In both cases there are exactly 35 bases, 195 geometric pairs, and 192 context-covered pairs; the maximum basis intersection is two. The first union contains the displayed eighteen-ray parity proof. The second union contains the twenty-ray proof with multiplicities $4, 4, 2, \ldots, 2$ and eleven contexts. Each of its twenty ray deletions has a compatible assignment by Table S90. All twenty rays lie in Peres's 24-ray set. Enumerating the $\binom{24}{4} = 10{,}626$ twenty-ray subsets of that set, with all bases they contain, gives exactly 96 ray-critical KS subsets, each with eleven bases and all mutually isomorphic; the twenty-ray proof is one of them. Since Peres's set contains Kernaghan's critical twenty-ray set and 95 others[3], the twenty-ray proof is isomorphic to Kernaghan's[10]. All coordinates, full basis lists, and the preserved $H_{65}$ restriction are checked independently of the unrestricted ancestry existence proposition.

### S3.6. Transverse extension and criticality certificates

For $d = 3$, number the two copies of an $n$-vertex input as $i$ and $n + i$, and the fresh transverse vertex as $2n + i$. The transverse contexts are $\{i, n + i, 2n + i\}$ and the horizontal contexts are the two numbered copies of the original list. This explicitly specifies all 45 vertices and 33 contexts for $T(H_9)$, and all 75 vertices and 57 contexts for $T(H^+_{\mathrm{YO}})$.

For each deleted original context $e$, a strong three-coloring of $H \setminus \{e\}$ having two equal colors and a singleton color on $e$ gives a state after deletion of one horizontal copy of $e$. Use the repeated-color indicator on the deleted copy and the singleton-color indicator on the intact copy. Both have exactly one active vertex in each of their required contexts and their supports are disjoint; the remaining transverse values are forced. For deletion in the other copy, interchange the two indicators. Table S92 supplies an explicit coloring for every required context of both inputs. The color strings use original vertex order, and their labels 0,1,2 may be permuted.

TABLE S92: Explicit deletion colorings with pattern (1,1,2) and the number of states after deleting the corresponding first-copy horizontal context.

| Input | Deleted original context | Color string | Extension states |
|---|---|---|---|
| $H_9$ | $\{1,2,3\}$ | 212010201201122 | 20 |
| $H_9$ | $\{1,11,12\}$ | 201020120121221 | 20 |
| $H_9$ | $\{2,8,15\}$ | 021020120121220 | 12 |
| $H_9$ | $\{3,4,5\}$ | 012120102012212 | 20 |
| $H_9$ | $\{4,10,13\}$ | 012102120121000 | 12 |
| $H_9$ | $\{5,6,7\}$ | 012010120121220 | 20 |
| $H_9$ | $\{6,12,14\}$ | 012102102012202 | 12 |
| $H_9$ | $\{7,8,9\}$ | 012010200121222 | 20 |
| $H_9$ | $\{9,10,11\}$ | 012010201121222 | 20 |
| $H^+_{\mathrm{YO}}$ | $\{1,2,3\}$ | 200021210110202010201122221 | 16 |
| $H^+_{\mathrm{YO}}$ | $\{1,10,19\}$ | 210021201212020001011202l | 8 |
| $H^+_{\mathrm{YO}}$ | $\{1,13,20\}$ | 120210201012120011200220l | 8 |
| $H^+_{\mathrm{YO}}$ | $\{2,5,8\}$ | 021021012120201122211001 2 | 24 |
| $H^+_{\mathrm{YO}}$ | $\{3,11,14\}$ | 012021102120200122210001 1 | 8 |
| $H^+_{\mathrm{YO}}$ | $\{3,12,25\}$ | 021210102101222020210101 0 | 8 |
| $H^+_{\mathrm{YO}}$ | $\{4,5,6\}$ | 021200210102121110220221 0 | 16 |
| $H^+_{\mathrm{YO}}$ | $\{4,10,18\}$ | 021210201202121010120201 0 | 8 |
| $H^+_{\mathrm{YO}}$ | $\{4,11,24\}$ | 021201012120201120211000 2 | 8 |
| $H^+_{\mathrm{YO}}$ | $\{6,12,17\}$ | 021210102100222110210101 2 | 8 |
| $H^+_{\mathrm{YO}}$ | $\{6,13,22\}$ | 021012102120200112210001 2 | 8 |
| $H^+_{\mathrm{YO}}$ | $\{7,8,9\}$ | 021012002120101112222001 2 | 16 |
| $H^+_{\mathrm{YO}}$ | $\{7,11,15\}$ | 021201012100221120211001 2 | 8 |
| $H^+_{\mathrm{YO}}$ | $\{7,13,21\}$ | 021012102120100112220001 2 | 8 |
| $H^+_{\mathrm{YO}}$ | $\{9,10,23\}$ | 021210201102121010220201 0 | 8 |
| $H^+_{\mathrm{YO}}$ | $\{9,12,16\}$ | 021210102102222010210101 0 | 8 |

Deleting one transverse context $T_v$ instead permits an ordered pair of original state supports $S_1, S_2$ exactly when $S_1 \cap S_2 \subseteq \{v\}$. In these non-three-colorable inputs disjoint pairs do not exist, so the intersection must equal $\{v\}$; the isolated fresh vertex has two choices. For $H_9$ this gives four states for each $v = 1, \ldots, 12$ and twelve states for each $v = 13, 14, 15$. Together with Table S92, this proves criticality of all 33 contexts. For completed Yu–Oh it gives zero states at $v = 1, \ldots, 13$ and eight states at $v = 14, \ldots, 25$. Thus horizontal criticality does not imply criticality of this full transverse construction. The corresponding $G_{32}$ extension has 45 vertices and 35 contexts and is stateless. Its distinct six state supports have pairwise intersection size one; for $H_9$ the distinct eight supports have intersection size one in 21 pairs and size three in seven pairs.

### S3.7. Data and verification

The companion electronic package contains coordinate and context inventories, complete state and coloring enumerations, verification programs, exact parent inventories, and finite certificates. Vertex labels and state orderings agree with the tables in this supplement. The verification programs check the common Yu–Oh numbering, the published 49-ray parent map, the parent vectors and contexts, the full $H_{65}$ state inventory, and cross-reference targets. The finite lower-bound certificate for the eighteen-state Hesse minimum is the exhaustive 540,672-candidate argument of Sec. S3.3.

Run the companion verification entry point from the project directory:

```
python verification/verify-integrated.py
```

The exact checks use rational or $\mathbb{Q}(\zeta)$ arithmetic, with $\zeta^2 + \zeta + 1 = 0$. Reports and hashes accompany the source.

The finite checks certify the instances and certificates they explicitly test. The analytic realization and existence results also rely on their proofs in the text.

### S3.8. Historical Kochen–Specker gadget certificates (1965 Fig. 1 and 1967 $\Gamma_1$)

#### *S3.8.1. The 1965 triangles and the completed historical TITS*

Identifying the repeated occurrences of vertices 1 and 2 in Kochen–Specker's 1965 Fig. 1 gives

$$\{134, 325, 2\,9\,11, 11\,1\,10, 479, 5\,8\,10, 678\}. \tag{S44}$$

These are also the triples used in the displayed function definitions on p. 182 of Ref.[11]. Their four state supports are

$$\{1,5,6,9\}, \quad \{2,4,6,10\}, \quad \{3,6,9,10\}, \quad \{4,5,6,11\}.$$

The TIFS between 1 and 2 is nonvacuous. Directly, if both were true, contexts 134, 325, 2 9 11, and 11 1 10 would make 3,4,5,9,10,11 false; contexts 479 and 5 8 10 would then make both 7 and 8 true, contradicting 678.

Every state has 6 true and 7,8 false, so no three-coloring can exist: the three color-class states of any coloring would need to be disjoint. The graph is abstract rather than an injective ray example. In a purported injective realization in $\mathbb{C}^3$, vertices 1 and 9 are both orthogonal to 4 and 11. Distinct rays 4 and 11 span a plane, whose orthogonal complement is one-dimensional; thus rays 1 and 9 coincide. This proves the obstruction even over the complex field. No claim that the 1965 authors presented this graph as a Hilbert-space realization is intended.

For the 1967 $\Gamma_1$[12] (pp. 68,70), the correspondence between its original vertices and $I_{16,9}$ is

| $\Gamma_1$ | $a_0$ | $a_1$ | $a_2$ | $a_3$ | $a_4$ | $a_5$ | $a_6$ | $a_7$ | $a_8$ | $a_9$ |
|---|---|---|---|---|---|---|---|---|---|---|
| $I_{16,9}$ | 10 | 9 | 1 | 7 | 3 | 8 | 2 | 11 | 4 | 24 |

The already complete triangles are $\{a_1, a_3, a_5\}$, $\{a_2, a_4, a_6\}$, and $\{a_7, a_8, a_9\}$. The six remaining edges

$$a_0a_1, \ a_0a_2, \ a_0a_8, \ a_3a_7, \ a_4a_7, \ a_5a_6$$

are completed by fresh vertices $23, 19, 18, 15, 14, 5$, respectively. The resulting nine contexts are exactly Eq. (S33). This also identifies its directed TITS $a_0 \to a_9$ with $10 \to 24$.

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